Chapter 1 — *Bioenergetics and Biologic oxidation*

Bioenergetics

I. *DEFINITIONS:*

A. Energy is the capacity to do work. It may be mechanical, chemical or electrical energy.

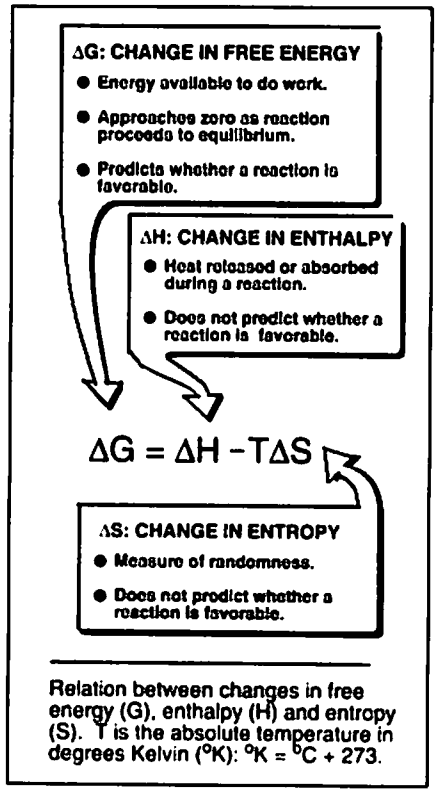
B. Bioenergetics or thermodynamics is the study of the energy changes accompanying biochemical reactions.

II. *TYPES OF ENERGY:*

A. **Heat energy:** is used to maintain body temperature.

B. **Free energy, = Delta G = (Δ G):**

1. Is useful energy for body activities as muscle contraction, nerve impulse etc.
2. The direction of chemical reaction is determined by 2 factors:
   a) Heat content (ΔH) of both reactant and product. It is called **enthalpy.**
   b) Randomness (Δ S) or disorder of both reactant and product. It is called: **entropy.**
   c) Combination of the 2 factors enthalpy (ΔH) - entropy (Δ S) will make a third factor called free energy (Δ G). So every substance has free energy, which determines the direction of reaction.



Relation between changes in free energy (G), enthalpy (H) and entropy (S). T is the absolute temperature in degrees Kelvin (°K): °K = °C + 273.

C. **<u>Forms of free energy (Free energy change, Δ G):</u>**

If we consider a reaction as follows: A ⇌ B

1. The direction of reaction may be A → B or B → A.
2. **Negative exergonic -ΔG: (A → B):** if ΔG is a negative number, this means that the free energy content of product (B) is less than that of product (A).

a) The reaction proceeds spontaneously in the cell from A →B. It is irreversible.

b) The reaction is said to be **exergonic.**

c) In these reactions there is a net loss of energy. This energy is used to form ATP. The compounds derived from food stuff intermediates (e.g. phosphoenol pyruvate, creatine phosphate etc) lose energy (~P) to form ATP.

d) Examples:

(i) Complete oxidation of creatine phosphate gives rise 10.3 Kcal/mol. In vitro this energy is lost in the form of heat. In vivo most of energy (-ΔG 7 Kcal/mol is used to form ATP). The remaining energy (-ΔG 3 Kcal/mol) is used to maintain body temperature.

Creatine~P + ADP → Creatine + ATP + -ΔG 3 Kcal/mol

(ii) Complete oxidation of phosphoenol pyruvate gives rise 14.8 Kcal/mol. In vitro this energy is lost in the form of heat. In vivo most of energy (-ΔG 7 Kcal/mol is used to form ATP). The remaining energy (-ΔG 7.5 Kcal/mol) is used to maintain body temperature.

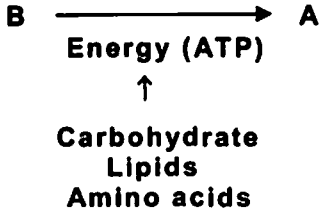
Phosphenol pyruvate + ADP → Pyruvate + ATP + -ΔG 7.5 Kcal/mol

3. **Positive endergonic +ΔG:** (B → A) if ΔG is positive number, this means that the free energy content of (B) is more than that of product (A).

a) There is net gain of energy.

b) The reaction does not go spontaneously from B to A except after addition of energy (ATP) from another source. This source is the catabolism of carbohydrate, lipids (fatty acids) and amino acids.

c) The reaction is said to be **endergonic**



d) Examples:

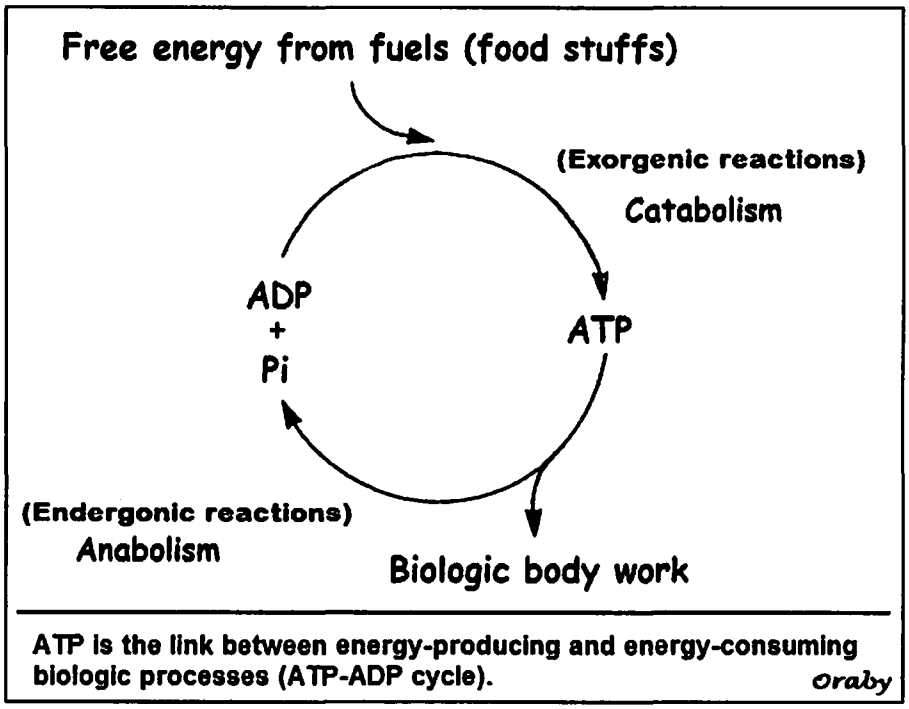
Creatine + ATP → Creatine~P + ADP + +ΔG 1.5 Kcal/mol

4. **ΔG is zero:** if ΔG = 0, the reactants, are in equilibrium. [Note: when a reaction is proceeding spontaneously - that is free energy is being lost, then the reaction continues until ΔG reaches zero and equilibrium is established].
5. The **exergonic** reactions (energy producing reactions) are called: **catabolism** e.g. glycolysis and fatty acids oxidation. The **endergonic** reactions (energy utilizing reactions) are called: **anabolism** e.g. synthesis of glycogen and fatty acids. Both catabolism and anabolism constitute metabolism.

III. *ATP ACTS AS AN ENERGY CARRIER:*

A. Introduction:

1. The free energy made available through the catabolism of fuels (carbohydrates, lipids, and amino acids) is not transmitted directly to the reactions requiring energy. Instead it is used to synthesize a compound that acts as a carrier of free energy, which is adenosine triphosphate (ATP).
2. **ATP-ADP cycle:** It is the relationship between ATP generation and its utilization into ADP. ATP molecules are generated by exorgenic reactions. They are utilized (endergonic reactions) for different body works.

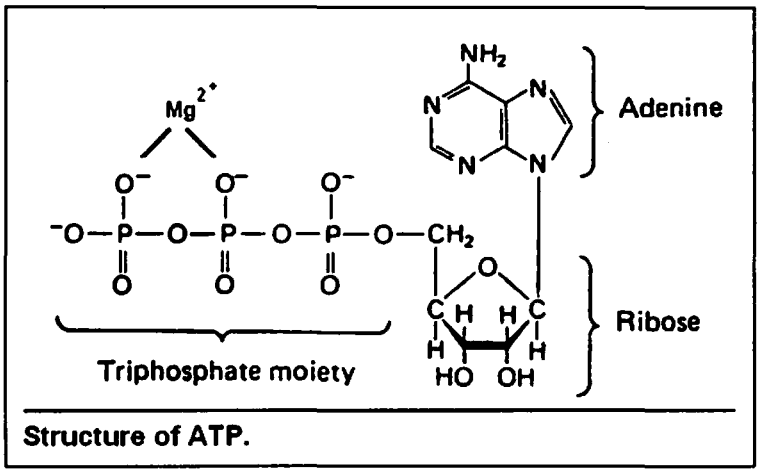


ATP is the link between energy-producing and energy-consuming biologic processes (ATP-ADP cycle). *Oraby*

B. Structure:

1. It is a member of a class of compounds called nucleotide triphosphates (ATP, GTP, UTP, CTP and TTP).

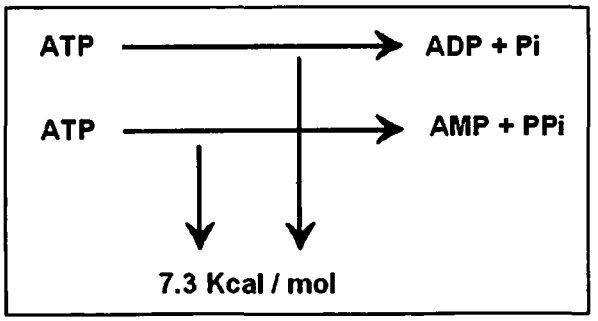
2. It has 3 component parts: a nitrogenous base (adenine), a sugar (ribose) and 3 phosphoryl groups that are joined to ribose by a phosphate ester bond and to each other by phospho-anhydride bonds.



Structure of ATP.

C. Importance of ATP:

1. The most important parts of the ATP molecule are its **2-phosphoanhydride bonds** (bonds between phosphate molecules number 1&2 and 2&3). Breakdown of either of these bonds is accompanied by a large decrease in free energy i.e. release energy.



2. Any bond whose breakdown is accompanied by a large decrease in free energy (≥ 5 Kcal/mol) is termed a **high energy bond** and represented by the symbol (~).
3. Breakdown of one high energy bond of ATP gives 7.3 Kcal.
4. Because ATP has two high energy bonds, it can serve as a link between energy yielding processes or **exergonic reactions** (e.g. catabolism of glucose and fatty acids) and energy requiring processes or **endergonic reactions** (e.g. anabolic pathways).
   a) Catabolic reactions give energy, which can be stored in the form of ATP.
   b) Anabolic reactions can utilize energy through hydrolysis of ATP.

IV. *UNITS OF FREE ENERGY:*

A. **Traditional units *(calorie)*:** energy measured in calories (one calorie is the amount of energy required to raise one gram of water by one degree (15°C-16°C).

B. **SI units *(joule)*:** energy measured in joules (one calorie = 4.128 joule).

V. *CALCULATION OF CALORIES:*

By using Bomb calorimeter (complete oxidation of foodstuff by burning in the presence of oxygen).

A. One gram of carbohydrate → 4 Kcal

B. One gram of fat → 9 Kcal

C. One gram of protein → 41 Kcal.

VI. *DIFFERENCE BETWEEN IN VITRO AND IN VIVO OXIDATION OF FOODSTUFF:*

A. **In vitro:** all free energy is released as heat.

B. **In vivo:** part is conserved in the form ATP and the remaining is released as heat.

VII. *PHOSPHATE BONDS:*

A. **High energy phosphate bonds:** Give rise to 7.3 Kcal/mol. Examples:

1. **ATP:** Its free energy is about -7.3Kcal/mol **(see before).**
2. **Creatine phosphate (C~P):** Excess ATP is stored in the form of C~P in muscles. Its free energy is about -10.3 Kcal/mol. (see protein metabolism)
3. **Carbomyl phosphate:** Its free energy is about -12.3 Kcal/mol. (see protein metabolism)

B. **Low energy phosphate bonds:** Give rise **less than** 7.3 Kcal/mol. Examples:

1. **Glucose-1-phosphate:** -5 Kcal/mol.
2. **Glucose-6-phosphate:** -3.3 Kcal/mol.
3. **Fructose-6-phosphate:** -3.8 Kcal/mol.
4. **AMP:** -3.4 Kcal/mol.
5. **Glyceraldehyde-3-phosphate:** -2.2 Kcal/mol.

Biologic oxidation

I. *Introduction:*

A. Energy is required to maintain the structure and function of the living Cells. This energy is derived from oxidation of carbohydrates, lipids and proteins of diet.

B. The energy liberated is partly converted into useful form ATP, which is known as ***energy currency*** *of the living cells.*

C. Each gram of carbohydrate and protein gives about 4 Kcal on oxidation, while each gram of fat gives about 9 Kcal.

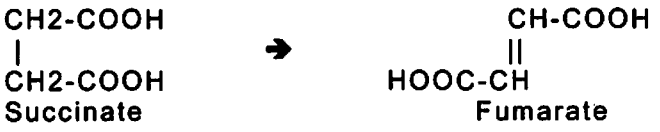
II. *Oxidation and reduction:*

A. **Oxidation means:**

1. **Addition of oxygen to a compound:**

$$2\ Cu + O_2 \rightarrow 2\ CuO \text{ (cupric oxide).}$$

2. **Removal of hydrogen from a compound:**



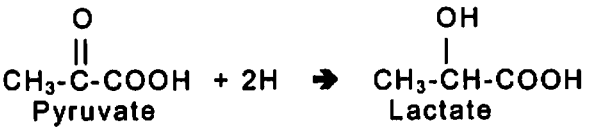
3. **Removal of electron from an ion or atom** i.e. increases in positive charges of an atom or an ion:

$$Fe^{++} \rightarrow Fe^{+++} + e$$

B. **Reduction means:**

1. **Removal of oxygen from a compound.**

2. **Addition of hydrogen to a compound:**



3. **Addition of electron to an ion or atom** i.e. decrease in positive charges of an atom or an ion:

$$Fe^{+++} + e \rightarrow Fe^{++}$$

C. Commonly the oxidation reactions are accompanied by reduction reactions and they are called **redox reactions**. Redox reactions are accompanied by energy liberation, necessary for the cells.

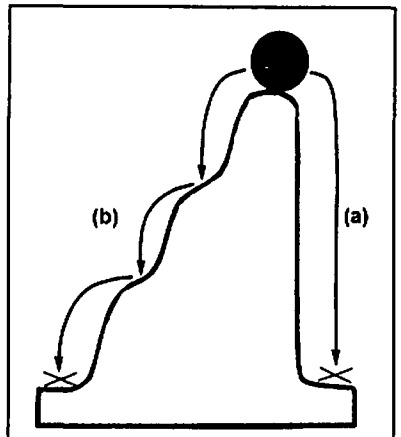
1. **Hydrogen** plays an important role in liberation of energy as represented by the following example:

$$2\ H_2 + O_2 \rightarrow 2\ H_2O + \text{energy}$$

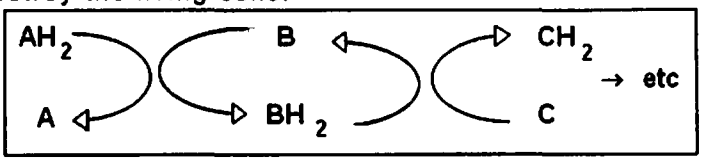
2. In this redox reaction, H_2 is oxidized while O_2 is reduced, and if occurs it will be accompanied by a massive energy explosion.
3. This simple example is similar to the fundamental reactions which provide energy in the living cells, and instead of massive energy is liberated, hydrogen must be transferred to oxygen in gradual steps. Thus, small fractions of energy are liberated and stored for further use.
4. Note that hydrogen atom is formed of one electron (e) and one proton (H^+). The removal of hydrogen atom or electron from a compound is always accompanied by a release of energy.

C. **Redox potential (= electron affinity):**

1. Oxygen has the highest electron affinity i.e. highest redox potential.
2. Hydrogen has the lowest electron affinity i.e. lowest redox potential.
3. **Redox chain:**
   a) **It is a chain of different compounds of increasing redox potentials between hydrogen and oxygen**
   b) **The living cells depend on redox reactions for their energy requirements. The reactions start by removal of H_2 from the substrate, which is then transferred to different components of redox chain, and finally to oxygen to form water. Components of redox chains have a redox potential higher than hydrogen and lower than oxygen.**
   c) During hydrogen (H^+ and electron) transfer through different components of the redox chain, energy is liberated in steps and in small utilizable amounts instead of a massive energy production, in the form of heat, which if happens may destroy the living cells.



Model of energy loss by big rock rolling down hill using two different pathways:
a. Direct with massive energy production.
b. Indirect in steps with release of small amounts of energy.



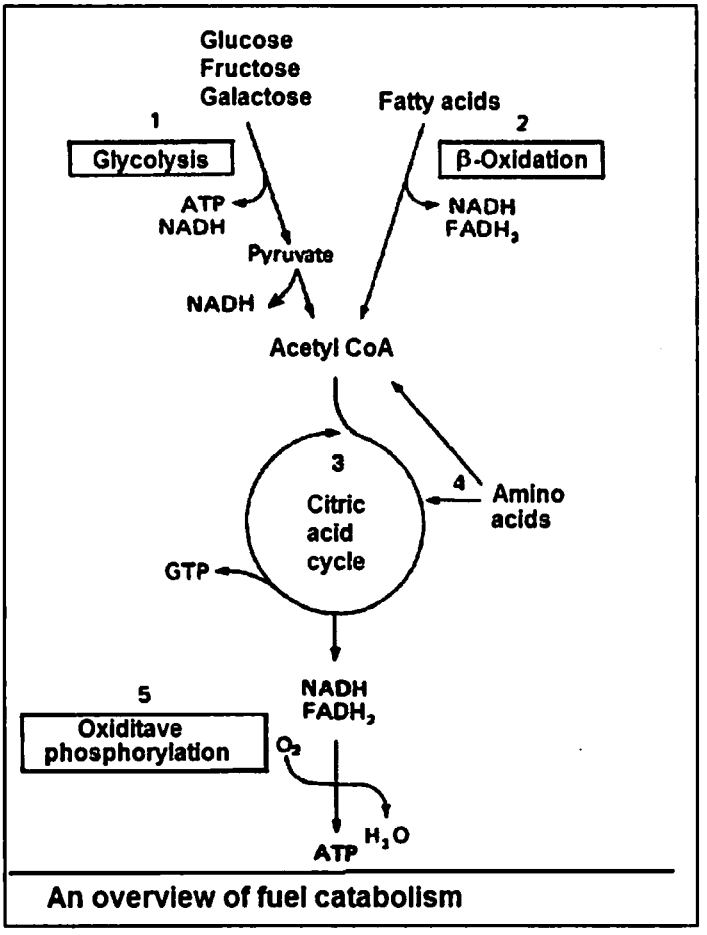
Hydrogen is transferred from A to B and from B to C because B has a higher redox potential (electron affinity) than A, while C has a higher redox potential than B... etc. In this way hydrogen (electron + H^+) moves from compounds of lower redox potential to compounds of higher redox potential.

III. *Respiratory chain (also called electron transport chain):*

A. Definition:

It is the final common pathway in aerobic cells by which electrons derived from various substances are transferred to oxygen to form water.

1. A variety of substances (carbohydrate, amino acids and fatty acids) can use respiratory chain as a final pathway as they give electrons to oxidized coenzymes NAD^+ and FAD^+ to form the energy rich reduced coenzymes, $NADH^+$, $FADH_2$.



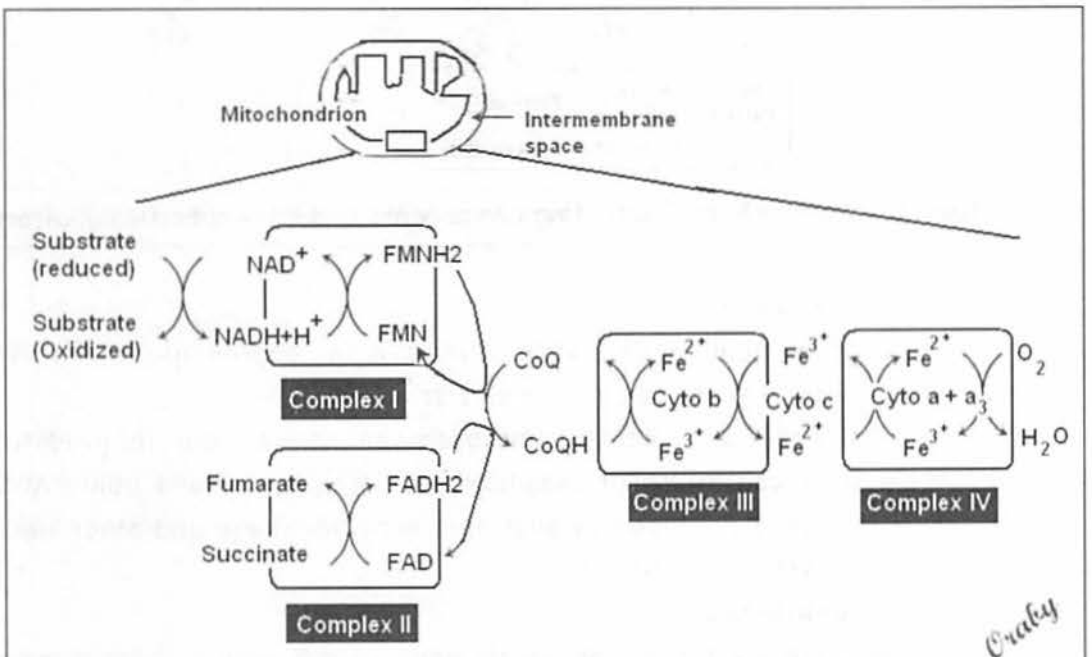
An overview of fuel catabolism

2. $NADH^+$, $FADH_2$ give hydrogen and a pair of electrons to electron carriers collectively called the **respiratory chain components**.
3. Hydrogen and electrons flow through respiratory chain in steps from the more electronegative component (low redox potential) to the more electropositive substance. **Oxygen** is the most electropositive component i.e. has the highest electron affinity (=highest redox potential). So oxygen is the final acceptor of electrons and protons in respiratory chain.
4. As electrons are passed down the respiratory chain, they lose much of their free energy. Part of this energy can be captured

and stored by the production of ATP from ADP and inorganic phosphate(Pi). This process is called **oxidative phosphorylation**. The remainder of the free energy not trapped as ATP is released as heat.

B. **Organization of the respiratory chain :**

1. The inner mitochondrial membrane contains 5 separate enzyme complexes, called complex I, II, III, IV and V. Complex V catalyses ATP synthesis.
   a) **Each complex** accepts or donates electrons to relatively mobile electron carriers such as coenzyme Q and cytochrome C.
   b) **Each carrier** of electron transport chain can receive electrons from the more electronegative donor and can subsequently donate electrons to the next more electropositive carrier in the chain. Finally electrons combine with oxygen and protons to form water.



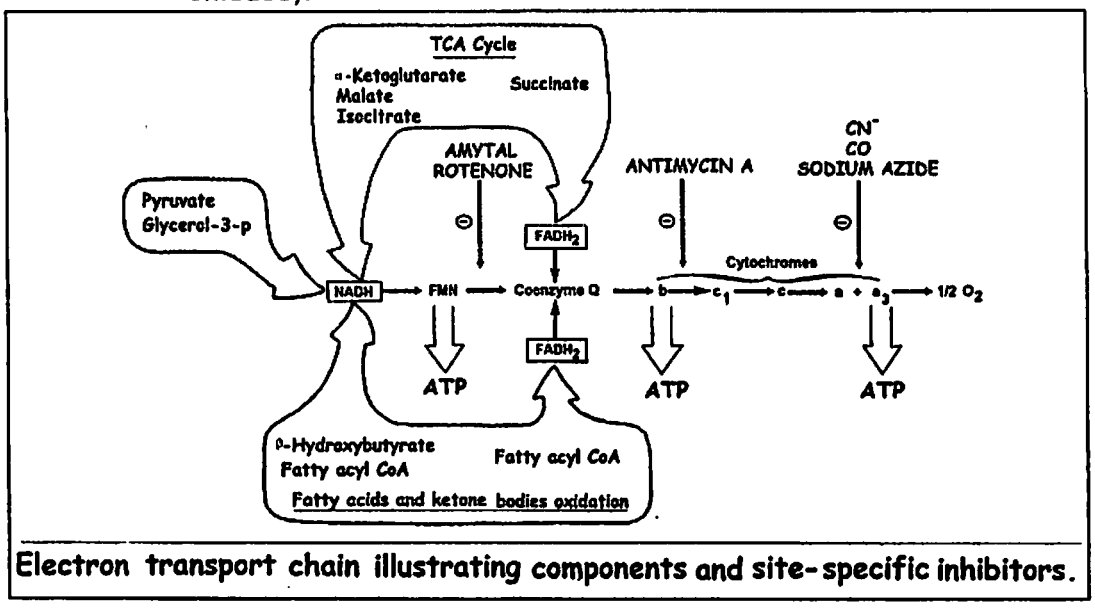
Electron transport chain. [Note: Coplex V not shown].

2. **Components of the respiratory chain:** All members of the respiratory chain are protein except coenzyme Q. All are embedded in the inner mitochondrial membrane.
   a) **Complex I:** Contains an enzyme called NADH dehydrogenase
      (i) Its coenzyme is FMN.
      (ii) It contains several iron and sulfur atoms.
      (iii) It oxidizes $NADH+H^+$ into NAD. At the same time converts its coenzyme FMN into $FMNH_2$.

b) **Complex II:** Contains an enzyme called: flavoprotein dehydrogenase e.g. succinate dehydrogenase of TCA and acyl CoA dehydrogenase of fatty acid oxidation.
   (i) Its coenzyme is FAD.
   (ii) It contains iron and sulfur atoms.

c) **Complex III**: contains cytochrome b enzyme and cytochrome c_1.

d) **Complex IV**: contains cytochromes a + a_3 (=cytochrome c oxidase).



Electron transport chain illustrating components and site-specific inhibitors.

3. **Coenzyme Q:**
   a) It is quinine derivative with a long isoprenoid tail. It is a relatively mobile electron carrier.
   b) Coenzyme Q can accept hydrogen atoms both from $FMNH_2$, produced by NADH dehydrogenase (complex I) and from $FADH_2$, which is produced by succinate dehydrogenase and other similar enzymes (complex II).
4. **Cytochromes:**.
   a) There are 5 types of cytochromes; cyto b, cyto c_1, cyto c, cyto a and cyto a_3.
   b) All cytochromes are conjugated proteins formed of protein conjugated with heme ring. The heme ring contains iron (Fe). This iron oscillates between ferric ions (Fe^{3+}) when it loses an electron, and ferrous (Fe^{2+}) when it accepts electrons.
   c) Cytochrome b is associated with sulfur (S) in addition to iron (Fe).

d) **Cytochrome c:** is a relatively mobile carrier. It receives electrons from complex III (cyto b and cyto c_1) and transfer them to cytochrome c oxidase (cyto a and cyto a_3).

e) **Cytochrome c oxidase (a and a_3):** it is a complex that contains two heme, cytochrome a, cytochrome a_3, iron and copper. It is the last enzyme in the respiratory chain. It is the only electron carrier that can react directly with molecular oxygen. It receives electrons from cyto c and transfer them to one oxygen molecule.

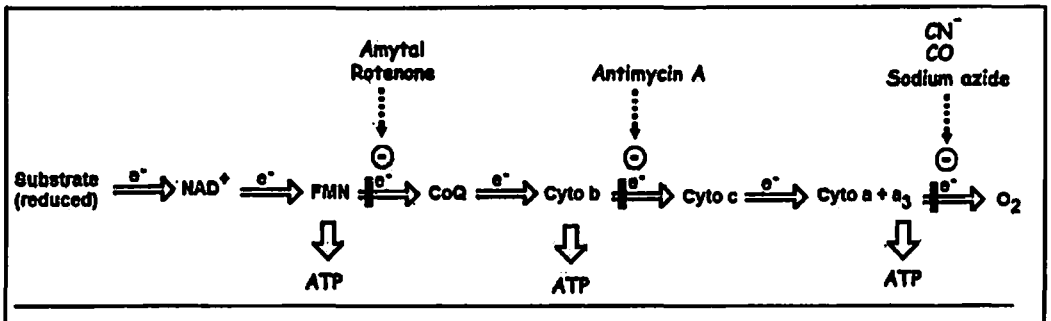
C. Reactions of respiratory chain:

1. **Entry via NADH + H^+:** NADH + H^+ obtained from reactions catalyzed by dehydrogenase enzymes e.g. dehydrogenase of TCA can join the chain giving electrons to FMN of complex I to coenzyme Q to cytochrome b, cytochrome c to cytochrome a + a_3 to the final acceptor O_2.
2. **Entry via $FADH_2$:** $FADH_2$ obtained from reactions catalyzed by flavoprotein dehydrogenase e.g. succinate dehydrogenase can join the chain directly giving electrons to coenzyme Q, then to cytochrome b, c, a + a_3 to the final acceptor oxygen.

D. Inhibitors of respiratory chain:

Are compounds preventing the passage of electrons by binding to a component of the chain, blocking the oxidation - reduction reaction.

1. There are specific sites for binding inhibitors.
   a) **Site I:** binding with complex I, preventing passage of electrons from FMN to coenzyme Q.
      (i) Example of inhibitors: **Barbiturates, "piericidin A"** antibiotic and by the insecticide and fish poison"rotenone".
   b) **Site II:** binding with complex III, preventing passage of electrons from cytochrome b to cytochrome c.
      (i) Example: Antimycin A and dimercaprol.
   c) **Site III:** binding with complex IV, preventing passage of electrons from cytochrome a + a_3 to O_2.
      (i) Example of inhibitors: H_2S, cyanide (CN^-), carbon monoxide and sodium azide.

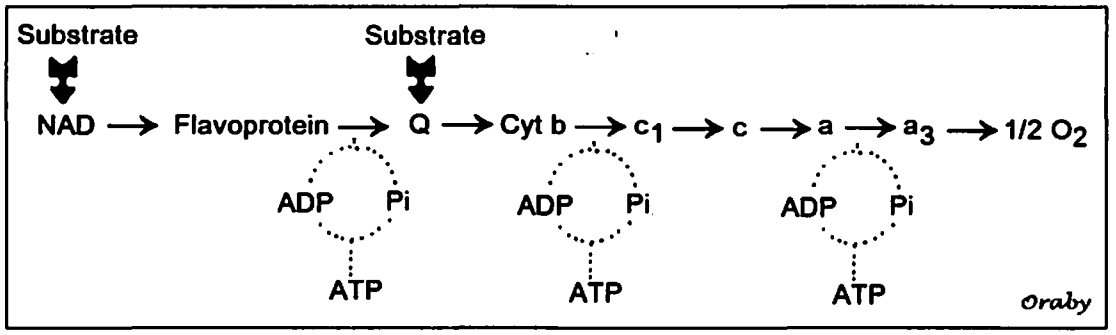


Site of respiratory chain inhibitors.

2. Because electron transport and oxidative phosphorylation are tightly coupled, inhibition of the respiratory chain also inhibits ATP synthase.

E. Release of free energy during electron transport:

1. **Free energy** is released as electrons are transferred along the electron transport chain from electron donor (reducing agent or reductant) to an electron acceptor (oxidizing agent or oxidant).
2. The electrons can be transferred in different forms, for example:
   a) As hydride ion (H) to NAD^+.
   b) As hydrogen atoms (H) to FAD.
   c) As electrons (e) to cytochromes.
3. **Redox pairs** (NAD^+ / $NADH+H^+$), (FAD / $FADH_2$) differ in their tendency to lose electrons. Each redox pair has a characteristic tendency (E_o) and it is constant to that pair. Its units are measured in volts. E_o is called standard reduction potential.
4. **E_o (standard reduction potential):** The standard reduction potentials of various redox pairs can be listed to range from the most negative E_o **to the most positive one.**
   a) The more negative E_o of a redox pair, the greater the tendency of that pair to lose electrons.
   b) The more positive E_o, the greater the tendency of that pair to accept electrons.
   c) Therefore, electrons flow from the pair with the more negative E_o to that with the more positive E_o.
5. **At three sites** (see the figure), the free energy released per electron pair transferred is sufficient to support the phosphorylation of ADP to ATP, which required about 7 Kcal/mol.



6. Electrons that enter the respiratory chain through the NAD-Q reductase complex support the synthesis of 3 mol of ATP. By contrast, electrons join the chain directly at the level of coenzyme Q (as in case of $FADH_2$ of succinate

dehydrogenase) will only support the synthesis of 2 mol of ATP.

Oxidation-reduction potentials and free-energy changes at sites in the electron transport chain that can support ATP formation:

Site	Oxidation-reduction potential E_0 (volts)	Free energy released (Kcal/mol)
At NAD^+ - Q	0.27	12.2
At cytochrome b →cytochrome c_1	0.22	09.9
At cytochrome a_3 → O_2	0.53	23.8

7. **P/O ratio:** It is the ratio between numbers of ADP mol changed into ATP to the number of oxygen atom (1/2 O_2) utilized by respiratory chain. It is 3:1 if electrons enter through NAD-coenzyme Q and it is 2:1 if electrons join the chain directly at the level of coenzyme Q.

d) *Oxidative phosphorylation:*

A. Introduction:

1. **Electrons are transferred down the respiratory chain from $NADH^+$ to oxygen. This is because $NADH^+$ is a strong electron donor, while oxygen is a strong electron acceptor.**
2. **The flow of electrons from $NADH^+$ to oxygen (oxidation) results in ATP synthesis by phosphorylation of ADP by inorganic phosphate, Pi (phosphorylation). Therefore, there is a coupling between oxidation and phosphorylation. Two theories explain the ATP synthesis, chemiosmotic hypothesis and membrane transport system.**

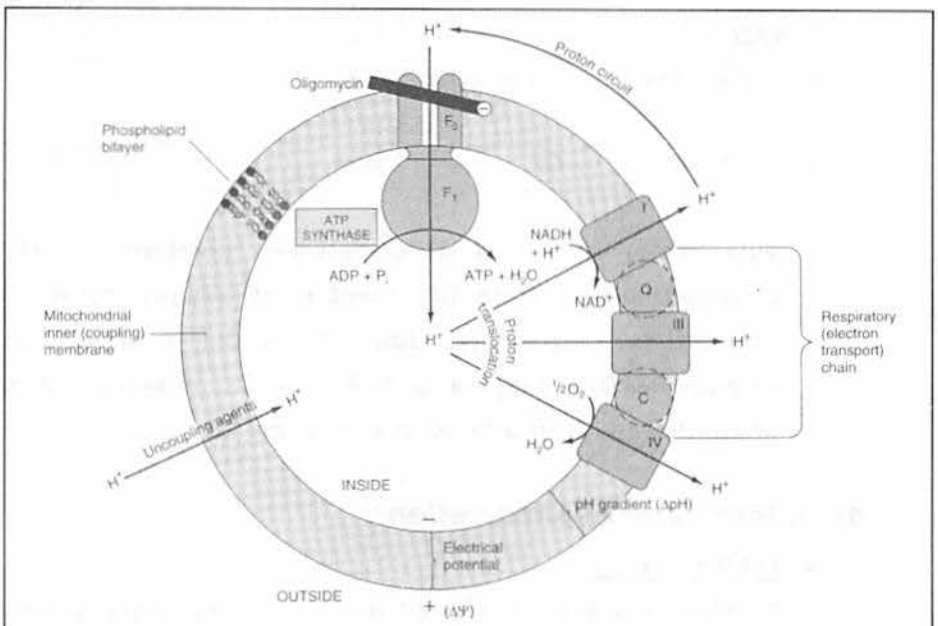
B. Chemiosmotic hypothesis:

Also called Mitchell hypothesis. This hypothesis is one form of oxidative phosphorylation. It can be summarized as follows:

1. **Proton pump:**
   a. The transport of electrons down the respiratory chain → Gives energy.
   b. This energy is used to transport H+ from the mitochondrial matrix → from inside to outside the membrane.
   c. This is done by complexes I, III and IV.
   d. This process creates across the inner mitochondrial membrane:
      i. **An electrical gradient**: (with more positive charges on the outside of the membrane than on the inside).

ii. **A pH gradient**: (the outside of the membrane is at lower pH than the inside).

e. The energy generated by this proton gradient is sufficient for ATP synthesis.



Principles of the chemiosmotic theory of oxidative phosphorylation. The main proton circuit is created by the coupling of oxidation in the respiratory chain to proton translocation from the inside to the outside of the membrane, driven by the respiratory chain complexes I, III, and IV, each of which acts as a *proton pump*. Q, ubiquinone; C, cytochrome *c*; F_1, F_o, protein subunits which utilize energy from the proton gradient to promote phosphorylation. Uncoupling agents such as dinitrophenol allow leakage of H^+ across the membrane, thus collapsing the electrochemical proton gradient. Oligomycin specifically blocks conduction of H^+ through F_o.

2. **ATP synthase (complex V):**

In the inner mitochondrial membrane, there is a phosphorylating enzyme complex: **ATP synthase** (or complex V).

a) It is formed of 2 subunits:

(i) F_1 subunit which protrudes into the matrix.

(ii) F_o subunit which is present in the membrane.

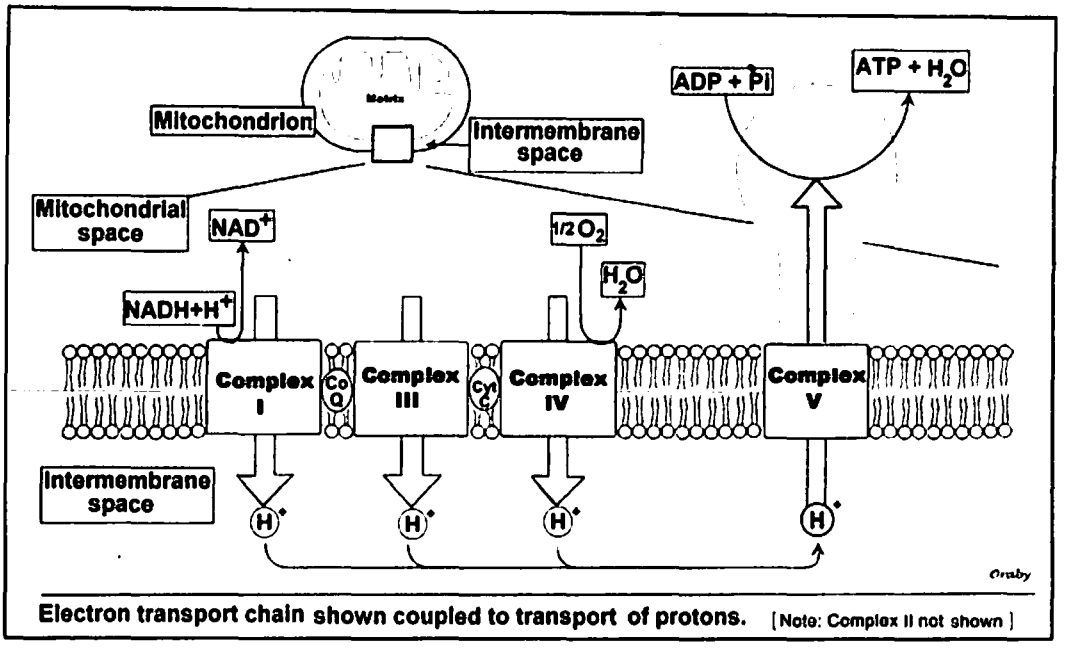
b) The protons outside the inner mitochondrial membrane can re-enter the mitochondrial matrix by passing through channel (F_o- F_1 complex) to pass by ATP synthase enzyme which is present in F_1 subunit. This results in the synthesis of ATP from ADP + Pi. At the same time decrease the pH and electrical gradients.

3. **Evidences support chemiosmotic theory:**

a) Addition of protons (acid) to the external medium of intact mitochondria leads to the generation of ATP.

b) ATP synthesis does not occur in soluble cytosol system where there is no ATP synthase. A closed membrane as mitochondria must be present in order to obtain oxidative phosphorylation.

c) The component of respiratory chain is organized in a sided manner as required by chemiosmotic theory.



Electron transport chain shown coupled to transport of protons. [Note: Complex II not shown]

4. **Uncouplers:**

These are substances that allow oxidation to proceed but prevent phosphorylation. So energy released by electron transport will be lost in the form of heat. This explains the cause of hotness after intake of these substances. Examples:

a) **Oligomycin:** This drug binds to the stalk of the ATP synthase , closes the H^+ channel, and prevent re-entry of protons to the mitochondrial matrix.

b) **2,4 Dinitrophenol:** It increases the permeability of the inner mitochondrial membrane to proton causing decrease proton gradient.

c) **Calcium and high doses of aspirin:** this explains the fever that accompanies toxic overdoses of these drugs.

d) **Ionophores:** e.g. antibiotics "valinomycin" and Nigericin. They are lipophilic substances. They have the ability to make a complex with cations as potassium "K^+" and facilitate their transport into mitochondria and other biological membranes. They inhibit phosphorylation because they decrease both electrical and pH gradient.

C. **Membrane transport systems:**

The inner mitochondrial membrane is impermeable to most charged or hydrophilic substances. However it contains numerous transport proteins (carrier) that permit passage of specific molecules from the

cytosol to the mitochondrial matrix e.g. ADP - ATP carrier which carriers ADP from cytosol into mitochondria, while carrying ATP from the matrix back to cytosol.

V. *Oxidation of extramitochondrial NADH +H⁺ :*

It is mediated by substrate shuttles (glycerophosphate shuttle and malate-aspartate shuttle).

- The 2 molecules of cytosolic NADH+H⁺ produced by glycolysis cannot penetrate mitochondrial membrane, however, they can be used to produce energy (4 or 6 ATP) by respiratory chain phosphorylation in the mitochondria.
- This can be done by using special carriers for hydrogen of NADH+H⁺. These carriers are either dihydroxyacetone phosphate (glycerophosphate shuttle) or oxaloacetate (aspartate - malate shuttle).

A. Glycerophosphate shuttle:

1) It is important in certain muscles and nerve cells.
2) The final energy produced is 2 x 2 ATP ➔ 4 ATP.
3) **Mechanism** (see the diagram):
   - ➢ The coenzyme of cytosolic glycerol -3- phosphate dehydrogenase is NAD⁺.
   - ➢ The coenzyme of mitochodrial glycerol-3-phosphate dehydrogenase is FAD.
   - ➢ Oxidation of $FADH_2$ in respiratory chain gives 2 ATP.

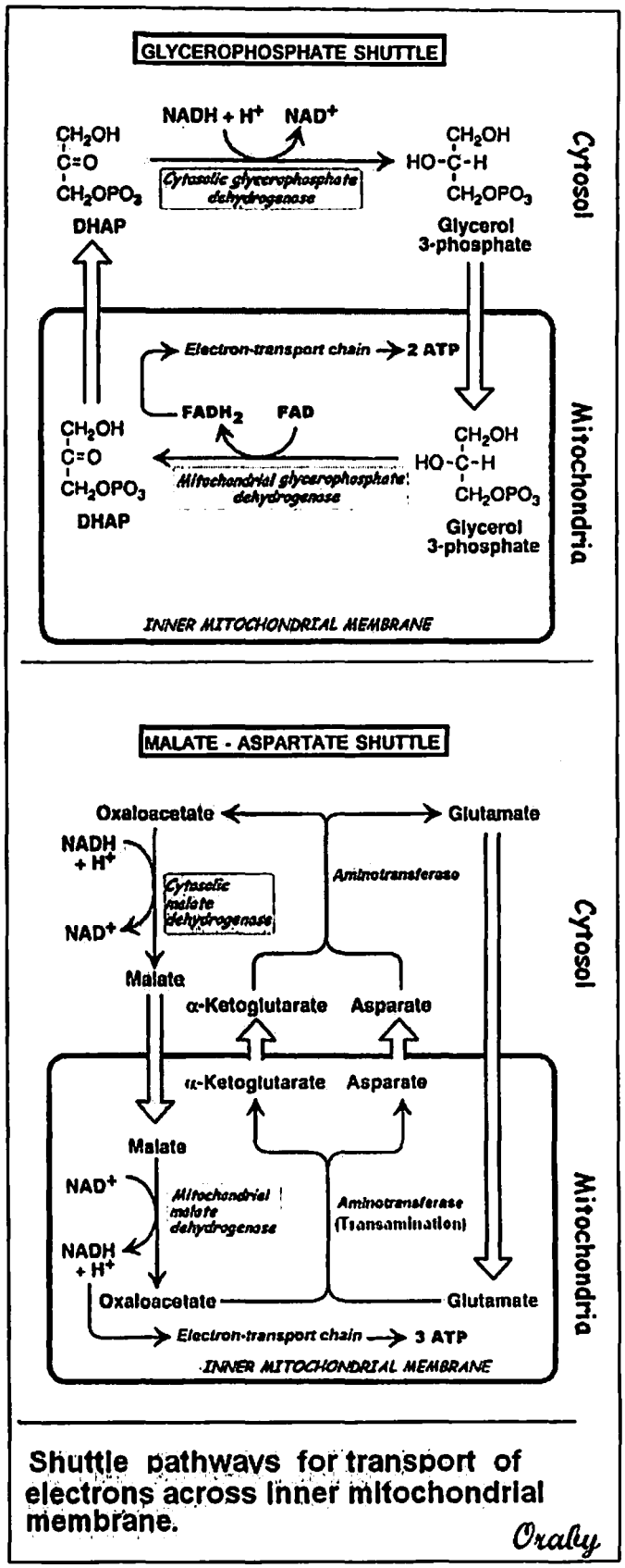
   As glycolysis gives 2 cytosolic NADH + H⁺ → 2 mitochondrial $FADH_2$ → 2 x 2 ATP = 4 ATP.

B. Malate - aspartate shuttle:

1) It is important in other tissues particularly liver and heart.
2) The final energy produced is 2x3 ATP ➔ 6 ATP.
3) **Mechanism:** see diagram
   - ➢ The coenzyme of both cytosolic and mitochondrial malate dehydrogenase is **NAD⁺.**
   - ➢ Oxidation of NADH+H⁺ in respiratory chain gives 3 ATP.

   As glycolysis gives 2 cytosolic NADH+H⁺ → 2 mitochondrial NADH+H⁺ → 2x 3 ATP = 6 ATP.

➢ **Note:** Mitochondrial membrane is impermeable to oxaloacetate. Therefore, oxaloacetate produced inside the mitochondrial cannot be transported to the cytosol again except by a special pathway called: transamination (see protein metabolism).



Shuttle pathways for transport of electrons across inner mitochondrial membrane.

Oraby

VI. *Mitochondrial structure:*

A. **The outer mitochondrial membrane:** is permeable to most small molecules.

B. **The inter-membrane space:** shows no barrier to the substances entering or leaving the mitochondrial matrix.

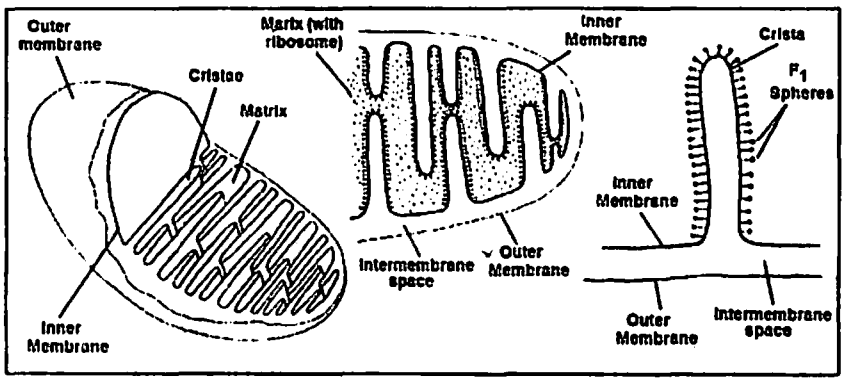
C. **The inner membrane:**

1. The inner mitochondrial membrane is impermeable to most small ions including H^+, Na^+ and K^+, small and large molecules as ATP, ADP, pyruvate and other metabolites important to mitochondrial function. Specialized carriers or transport systems are required to move ions or molecules across this membrane.

2. The inner mitochondrial membrane is highly convoluted. The convolutions are called **cristae** and serve to increase greatly the surface area of the membrane.
3. **ATP synthase complexes:** These complexes of proteins are considered as **inner membrane particles** and are attached to the inner surface of the inner mitochondrial membrane. They include the enzymes of respiratory (electron transport) chain.

D. **Matrix of mitochondrion:** It is a solution like a gel. It is bounded by the inner mitochondrial membrane and contains:

1. The enzymes of tricarboxylic acid cycle (TCA) with exception of succinate dehydrogenase, which is embedded in the inner membrane.
2. The enzymes of B-oxidation of fatty acids.
3. Miscellaneous enzyme systems.



VII) *Mitochondrial functions:*

The following **reactions** occur in the mitochondria:

A. **Carbohydrate metabolism:**

1. Oxidative decarboxylation of pyruvate and α ketoglutarate.
2. Tricarboxylic acid cycle.
3. Part of gluconeogenesis.

B. **Respiratory chain.**

1. Oxidative phosphorylation.
2. Most of ATP formation in the cells (cell battery).

C. **Lipids metabolism:**

1. β-Oxidation of Fatty acids.
2. Mitochondrial synthesis of fatty acids.
3. Ketogenesis.

D. **Protein metabolism:**

1. Transamination.
2. Part of heme synthesis.
3. Part of urea synthesis

Chapter 2 *Carbohydrate Metabolism*

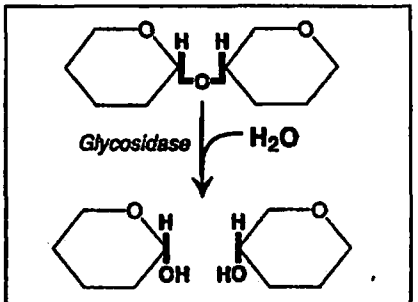
Digestion

I. ***Introduction:*** More than 60% of our foods are carbohydrates.

A. Starch, glycogen, sucrose, lactose and cellulose are the chief carbohydrates in our food.

B. Before intestinal absorption, they are hydrolyzed to monosaccharides (glucose, galactose and fructose).

C. A family of glycosidases that hydrolyzes carbohydrate into their monosaccharide components catalyzes hydrolysis of glycosidic bonds.



Hydrolysis of a glycosidic bond.

II. ***Digestion of carbohydrate by salivary α -amylase (ptylin) in the mouth:***

A. This enzyme is **produced by salivary glands.** Its optimum pH is **6.7.**

B. It is activated by chloride ions (Cl^-).

C. It acts on **cooked starch** and glycogen breaking α 1-4 bonds, converting them into maltose [a disaccharide containing two glucose molecules attached by α 1-4 linkage]. The bond of maltose is not attacked by α- amylase.

D. Because both starch and glycogen also contain α 1-6 bonds, the resulting digest contains **isomaltose** [a disaccharide in which two glucose molecules are attached by α 1-6 linkage].

E. **Because food remains for a short time in the mouth,** digestion of starch and glycogen may be incomplete and gives a partial digestion products called: **starch dextrins** (amylodextrin, erythrodextrin and achrodextrin).

F. Therefore, digestion of starch and glycogen **in the mouth** gives maltose, isomaltose and starch dextrins.

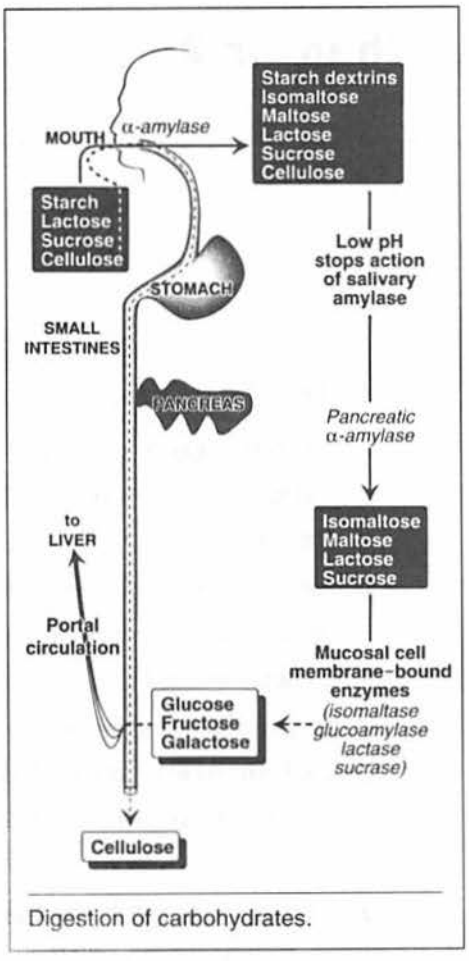
III. ***In the stomach:*** carbohydrate digestion stops temporarily due to the high acidity, which inactivates the salivary α-amylase.

IV. ***Digestion of carbohydrate by pancreatic α- amylase in the small intestine:***

A. α-amylase enzyme is produced by pancreas and acts in small intestine. Its optimum pH is **7.1**.

B. It is also activated by chloride ions.

C. It acts on cooked and uncooked starch, hydrolyzing them into maltose and isomaltose.



Digestion of carbohydrates.

V. ***Final carbohydrate digestion by intestinal enzymes:***

A. The final digestive processes occur at the small intestine and include the action of several disaccharidases. These enzymes are secreted through and remain associated with the brush border of the intestinal mucosal cells.

B. **<u>The disaccharidases include:</u>**

1- **Lactase** (β-galactosidase) which hydrolyses **lactose** into two molecules of glucose and galactose:

Lactose $\xrightarrow{Lactase}$ Glucose + Galactose

2- **Maltase** (α-glucosidase), which hydrolyses **maltose** into two molecules of glucose:

Maltose $\xrightarrow{Maltase}$ Glucose + Glucose

3- **Sucrase** (α-fructofuranosidase), which hydrolyses **sucrose** into two molecules of glucose and fructose:

Sucrose $\xrightarrow{Sucrase}$ Glucose + Fructose

4. **α-dextrinase** which hydrolyzes the (1,6) linkage of **isomaltose.**

Isomaltose —Dextrinase→ Glucose + Glucose

VI. *Digestion of cellulose:*

A. Cellulose contains β(1-4) bonds between glucose molecules.

B. In humans, there is no β (1-4) glucosidase that can digest such bonds. So cellulose passes as such in stool.

C. Cellulose helps water retention during the passage of food along the intestine → producing larger and softer feces → preventing constipation.

Absorption

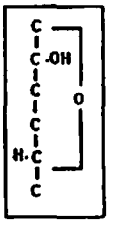
I. *Introduction:*

A. The end products of carbohydrate digestion are **monosaccharides**: glucose, galactose and fructose. They are absorbed from the jejunum to portal veins to the liver, where fructose and galactose are transformed into glucose.

B. Two mechanisms are responsible for absorption of monosaccharides: **active transport** (against concentration gradient i.e. from low to high concentration) and **passive transport (by facilitated diffusion).**

C. For active transport to take place, the structure of sugar should have:

1. **Hexose ring.**
2. **OH group at position 2** at the right side. Both of which are present in glucose and galactose. Fructose, which does not contain -OH group to the right at position 2 is absorbed more slowly than glucose and galactose by passive diffusion (slow process).



3. **A methyl or a substituted methyl group** should be present at carbon 5.

II. *Mechanisms of absorption:*

A. <u>Active transport:</u>

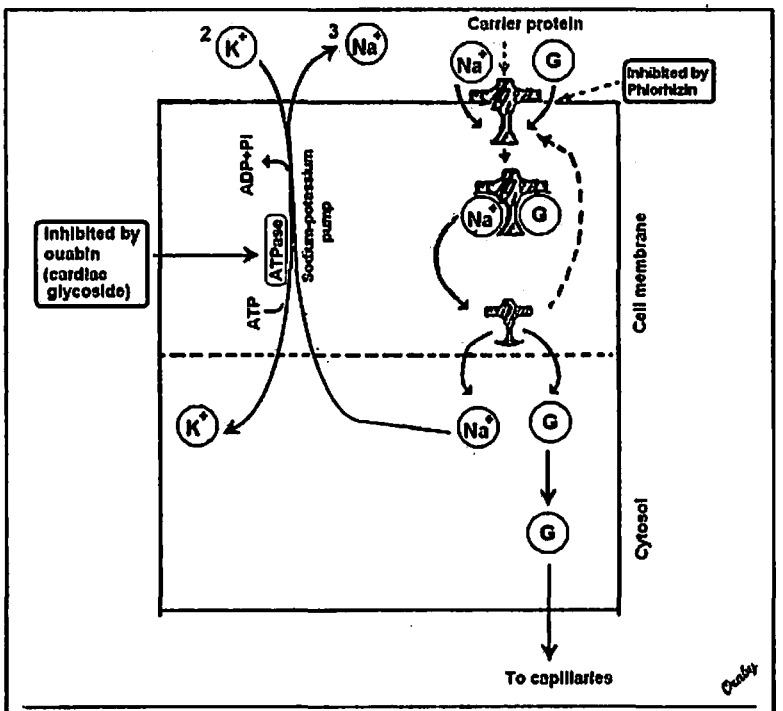
1. Mechanism of active transport:

a) In the cell membrane of the intestinal cells, there is a **mobile carrier protein** called **sodium – dependant glucose**

transporter (SGLT-1). It transports glucose to inside the cell using energy. The energy is derived from sodium-potassium pump. The transporter has 2 separate sites, one for sodium and the other for glucose. It transports them from the intestinal lumen across cell membrane to the cytosol. Then both glucose and sodium are released into the cytosol allowing the carrier to return for more transport of glucose and sodium.

b) The sodium is transported from high to low concentration (with concentration gradient) and at the same time causes the carrier to transport glucose against its concentration gradient. The Na^+ is expelled outside the cell by **sodium – potassium pump,** which needs ATP as a source of energy. The reaction is catalyzed by an enzyme called "**Adenosine triphosphatase** (ATPase)". Active transport is much more faster than passive transport.

c) **Insulin** increases the number of glucose transporters in tissues containing **insulin receptors** e.g. muscles and adipose tissue.



Transport the glucose across the intestinal epithelium. Active glucose transport is coupled to the Na^+ - K^+ pump.

2. **Inhibitors of active transport :**

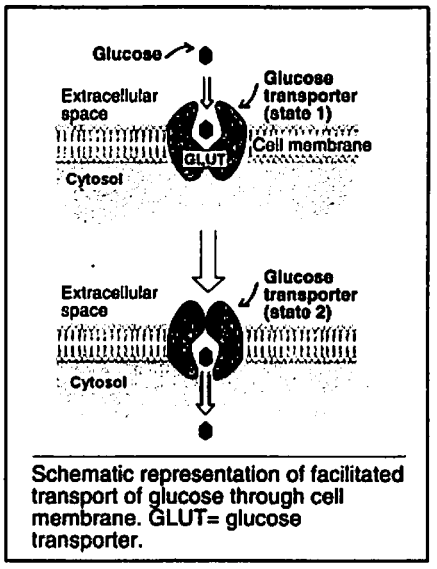
a) **Ouabin (cardiac glycoside):** Inhibits adenosine triphosphatase (ATPase) necessary for hydrolysis of ATP that produces energy of sodium-potassium pump.

b) **Phlorhizin:** Inhibits the binding of sodium in the carrier protein.

B. **Passive transport (facilitated diffusion):**

Sugars pass with concentration gradient i.e. from high to low concentration. It needs **no energy**. It occurs by means of sodium – independent facilitative transporter (GLUT-5). Fructose and pentoses are absorbed by this mechanism. Glucose and galactose can also use the same transporter if the concentration gradient is favorable.

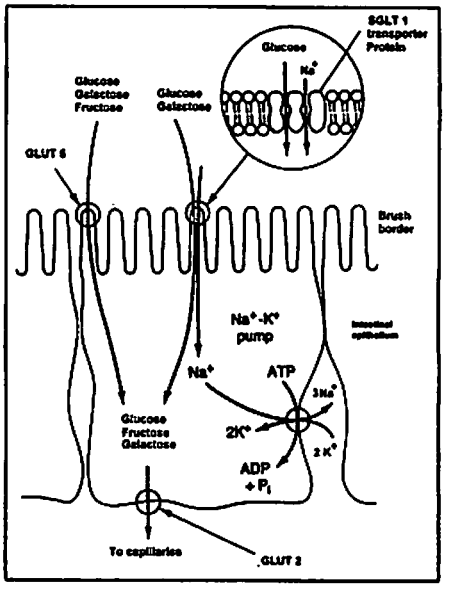
C. There is also **sodium – independent transporter (GLUT-2)**, that facilitates transport of sugars out of the cell i.e. to circulation.



Schematic representation of facilitated transport of glucose through cell membrane. GLUT= glucose transporter.

Summary of types and functions of most important glucose transporters:

	Function	*Site*
SGLT-1	Absorption of glucose by active transport (energy is derived from Na^+-K^+ pump.	Intestine and renal tubules
GLUT-5	Fructose transport · and to a lesser extent glucose and galactose.	Intestine and sperm
GLUT-2	Transport glucose out of intestinal and renal cells → circulation.	-Intestine and renal tubules -β cells of islets-liver



III. *Defects of carbohydrate digestion and absorption:*

A. Lactase deficiency = lactose intolerance:

1. Definition:

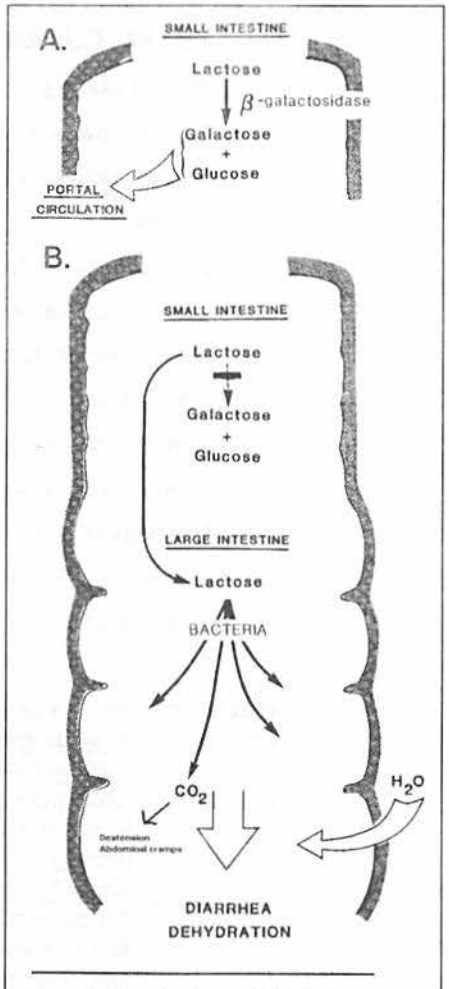
a) This is a deficiency of lactase enzyme which digests lactose into glucose and galactose

b) It may be:

(i) **Congenital:** This occurs very soon after birth (rare).

(ii) **Acquired:** This occurs later on, in life (common).

2. **Effect:** The presence of lactose in intestine causes:
   a) **Increased osmotic pressure:** So water will be drawn from the tissue (causing dehydration) into the large intestine (causing diarrhea).
   b) **Increased fermentation of lactose by bacteria:** Intestinal bacteria ferment lactose with subsequent production of CO_2 gas. This causes distention and abdominal cramps.
3. **Treatment:** Treatment of this disorder is simply by removing lactose (milk) from diet.



Normal **(A)** and abnormal **(B)** lactose metabolism.

B. **Sucrase deficiency:**
It is a rare condition, showing a similar signs and symptoms as lactase deficiency. It occurs early in childhood.

C. **Monosaccharides malabsorption:**
This is a congenital condition in which glucose and galactose are absorbed only slowly due to defect in the GLUT-1. Because fructose is not absorbed by the GLUT-1, its absorption is normal.

IV. *Fate of absorbed sugars:*

Monosaccharides (glucose, galactose and fructose) resulting from carbohydrate digestion are absorbed and undergo the following:

A. **Uptake by liver:**

After absorption the liver takes up sugars, where galactose and fructose are converted into glucose.

B. **Glucose utilization by tissues:** Glucose may undergo one of the following fate:

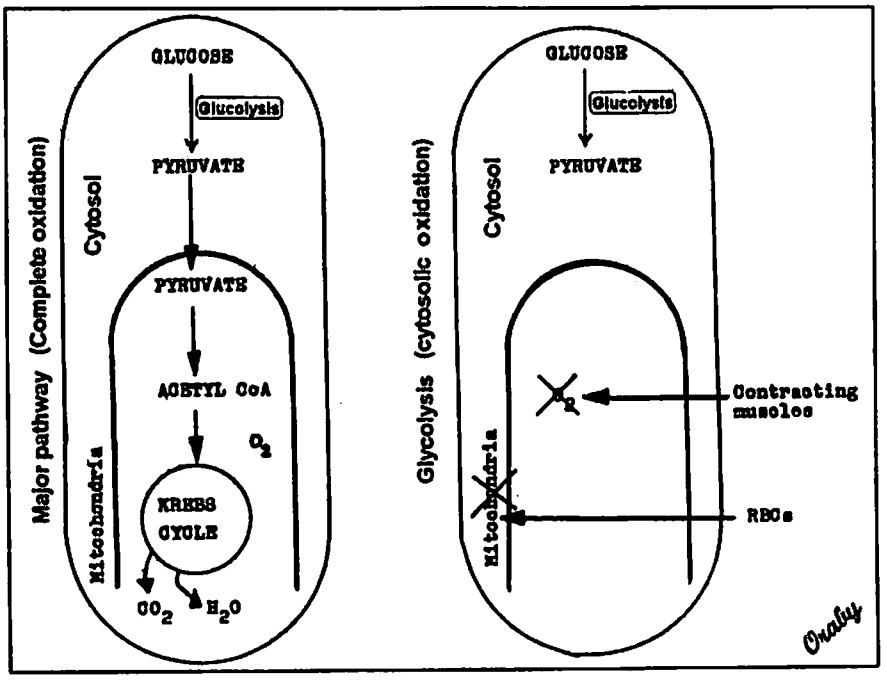
1. **Oxidation:** through
   a) **Major pathways** (glycolysis and Krebs` cycle) mainly for production of energy.
   b) **Pentose phosphate pathway:** for production of pentoses and NADPH + H^+.

c) **Uronic acid pathway:** for production of glucuronic acid. This sugar derivative is used in detoxication and enters in the structure of glycosaminoglycans.

2. **Storage in the form of:**
   a) **Glycogen:** glycogenesis.
   b) **Fat:** lipogenesis.
3. **Conversion:** to substances of biological importance:
   a) Ribose , deoxyribose ➔ RNA and DNA.
   b) Lactose ➔ milk.
   c) Glucosamine and galactosamine ➔ mucopolysaccharides.
   d) Glucuronic acid ➔ glycosaminoglycans and mucopolysaccharides.
   e) Fructose ➔ in semen.

GLUCOSE OXIDATION

Major Pathway



I. *Glycolysis (Embden Meyerhof Pathway):*

A. Definition:

1. Glycolysis means oxidation of glucose to give pyruvate (in the presence of oxygen) or lactate (in the absence of oxygen).

B. Site:

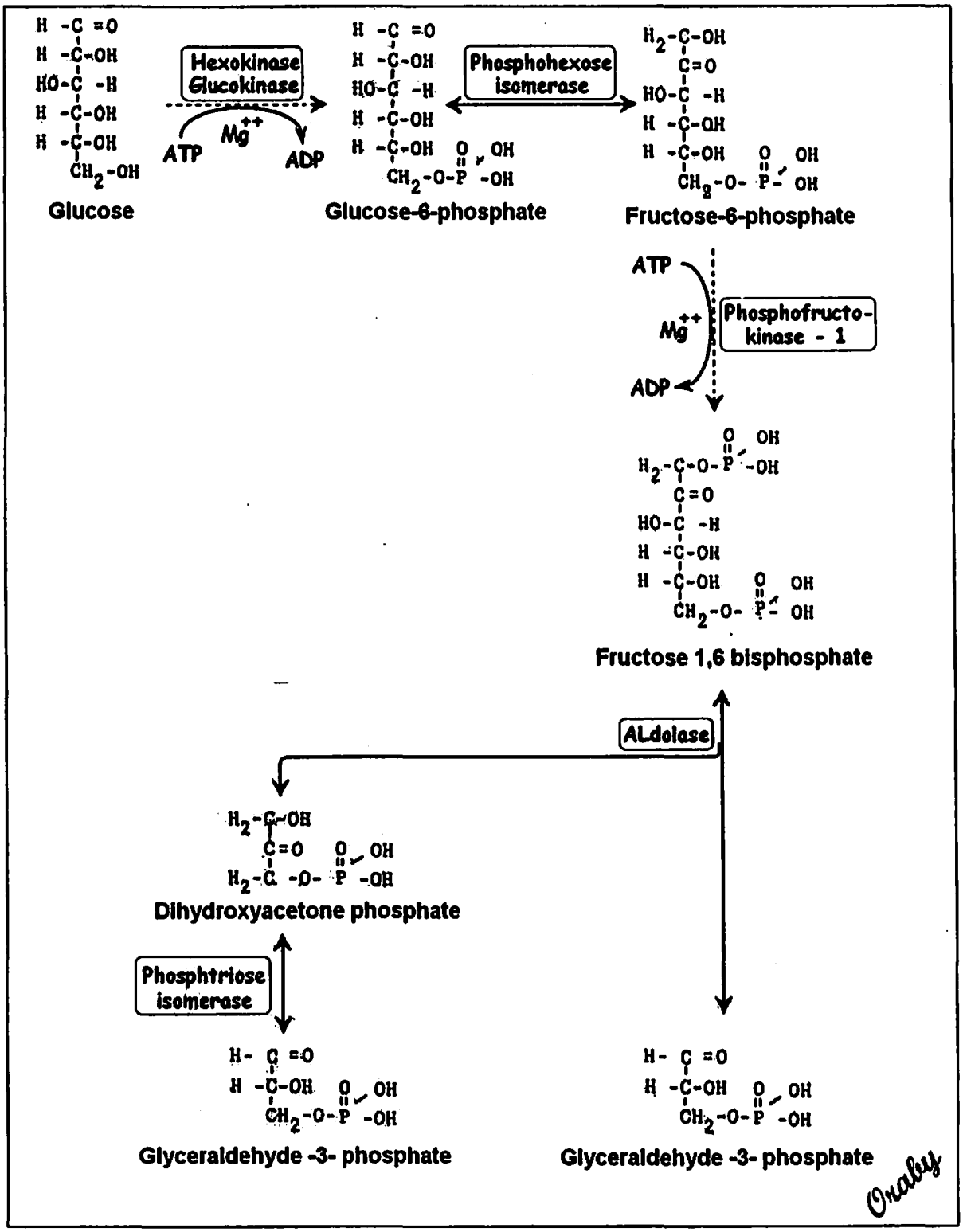
1. **Intracellular location:** cytosol.
2. **Organ location:** all tissue cells, but it is of physiological importance in:

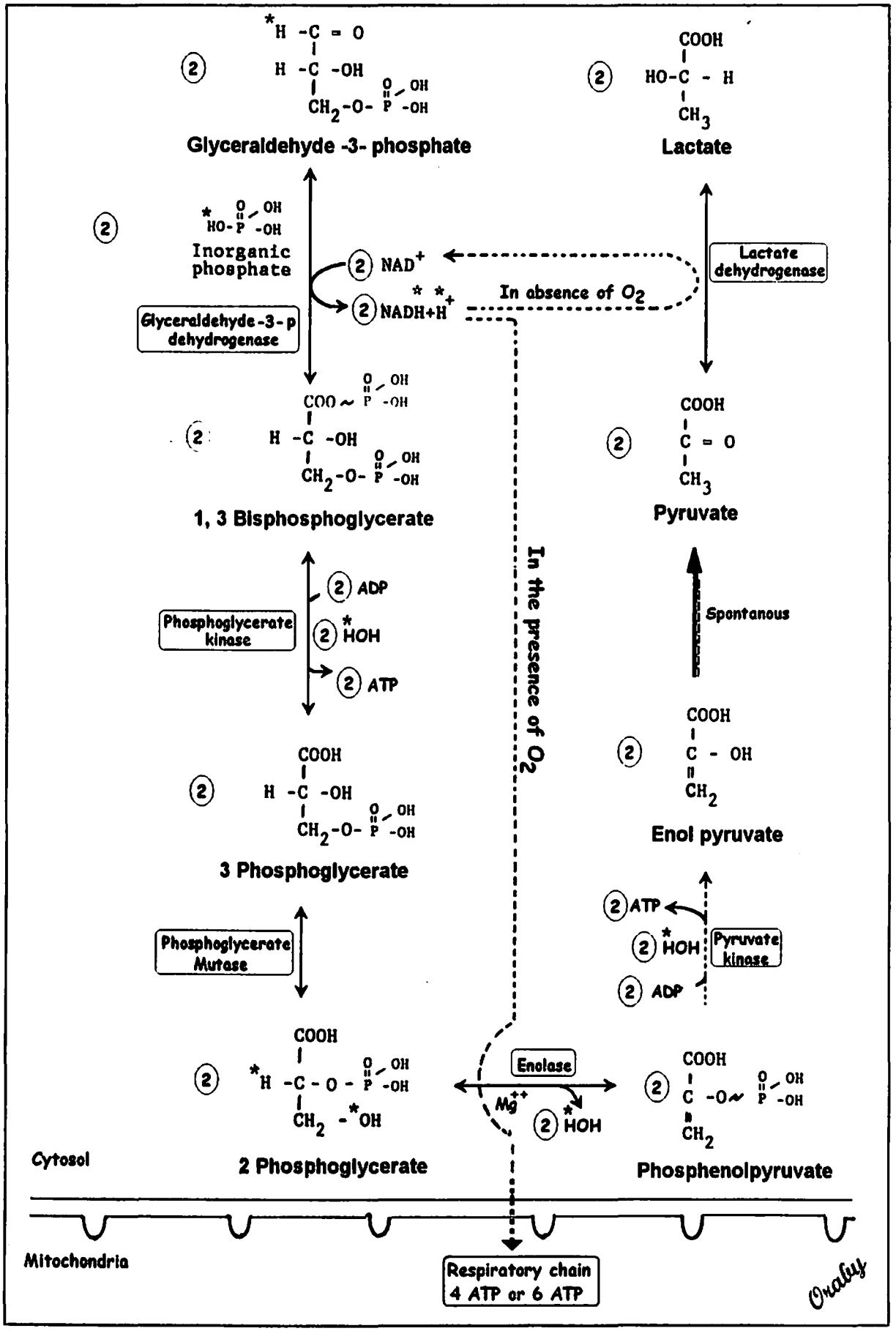
a) **Tissues with no mitochondria:** e.g. mature RBCs.

b) **Tissues with few mitochondria:** e.g. Testes and leucocytes.

c) **Tissues undergo frequent oxygen lack:** skeletal muscles especially during exercise.

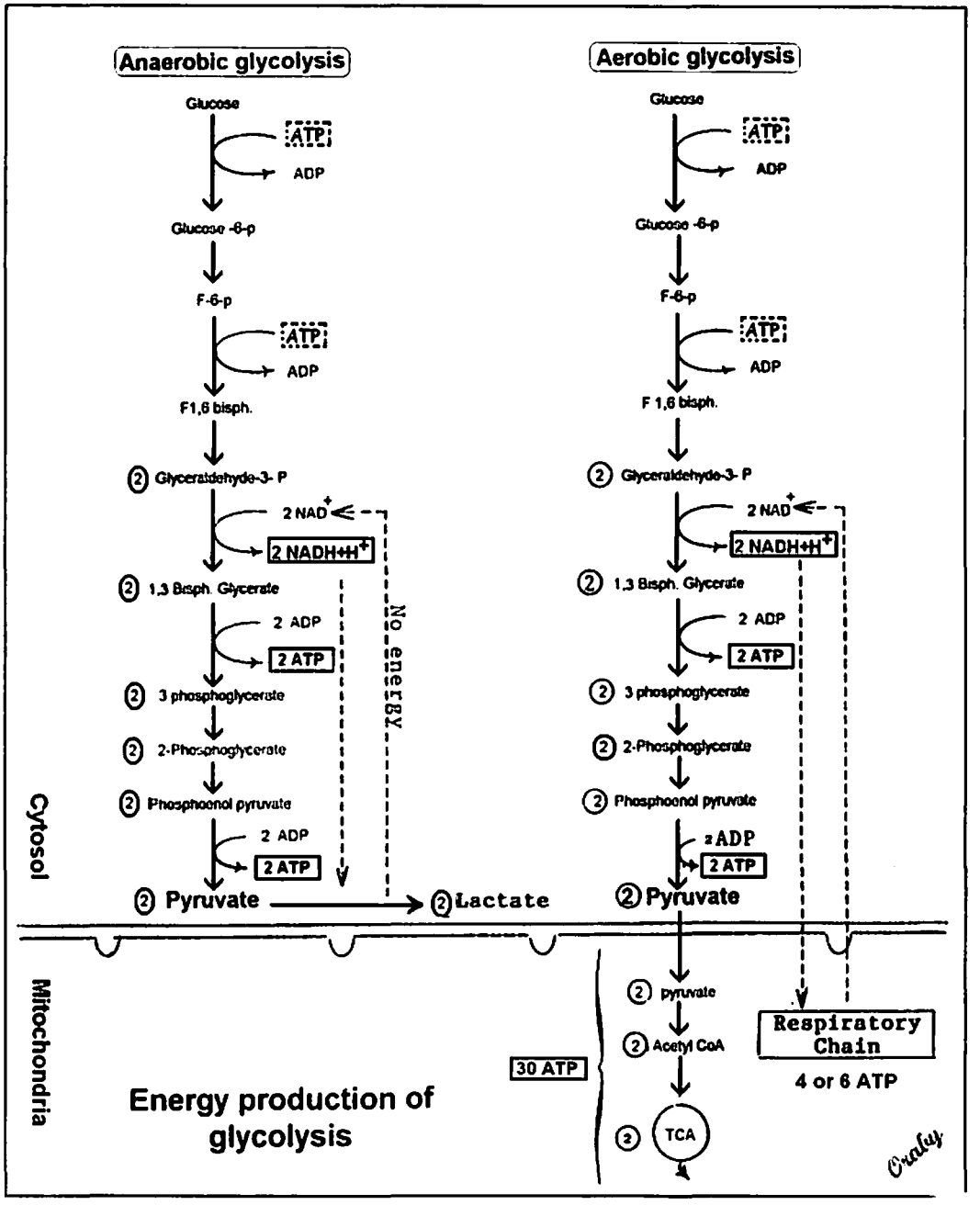
C. **Steps:**




Glyceraldehyde -3- phosphate
Lactate
Inorganic phosphate
2 NAD+
2 NADH+H+
In absence of O2
Lactate dehydrogenase
Glyceraldehyde-3-p dehydrogenase
1, 3 Bisphosphoglycerate
Pyruvate
2 ADP
2 HOH
2 ATP
Phosphoglycerate kinase
In the presence of O2
Spontanous
3 Phosphoglycerate
Enol pyruvate
Phosphoglycerate Mutase
2 ATP
2 HOH
2 ADP
Pyruvate kinase
Enolase
Mg++
2 HOH
2 Phosphoglycerate
Phosphenolpyruvate
Cytosol
Mitochondria
Respiratory chain
4 ATP or 6 ATP

Stages of glycolysis:

1. **Stage one (the energy requiring stage):**
   a) One molecule of **glucose** is converted into two molecules of **glyceraldehyde-3-phosphate.**
   b) These steps requires **2 molecules of ATP** (energy loss)
2. **Stage two (the energy producing stage):**
   a) The 2 molecules of **glyceraldehyde-3-phosphate** are converted into 2 **pyruvate** molecules (aerobic glycolysis) or 2 **lactate** molecules (anaerobic glycolysis).
   b) These steps produce ATP molecules (energy production).



D. Energy (ATP) production of glycolysis:

ATP production = ATP produced - ATP utilized

	ATP produced	ATP utilized	Net energy
In absence of oxygen (anaerobic glycolysis)	**4 ATP** (Substrate level phosphorylation) • 2ATP from 1,3 BPG. • 2ATP from phosphoenol pyruvate	**2 ATP** • From glucose to glucose-6-p. • From fructose-6-p to fructose 1,6 bisp.	**2 ATP**
In presence of oxygen (aerobic glycolysis)	**4 ATP** (Substrate level phosphorylation) • 2ATP from 1,3 BPG. • 2ATP from phosphoenol pyruvate + **4 ATP or 6 ATP** *(From oxidation of 2 NADH + H in mitochondria).	**2 ATP** • From glucose to glucose-6-p. • From fructose-6-p to fructose 1,6 bisphosphate.	**6ATP or 8ATP**

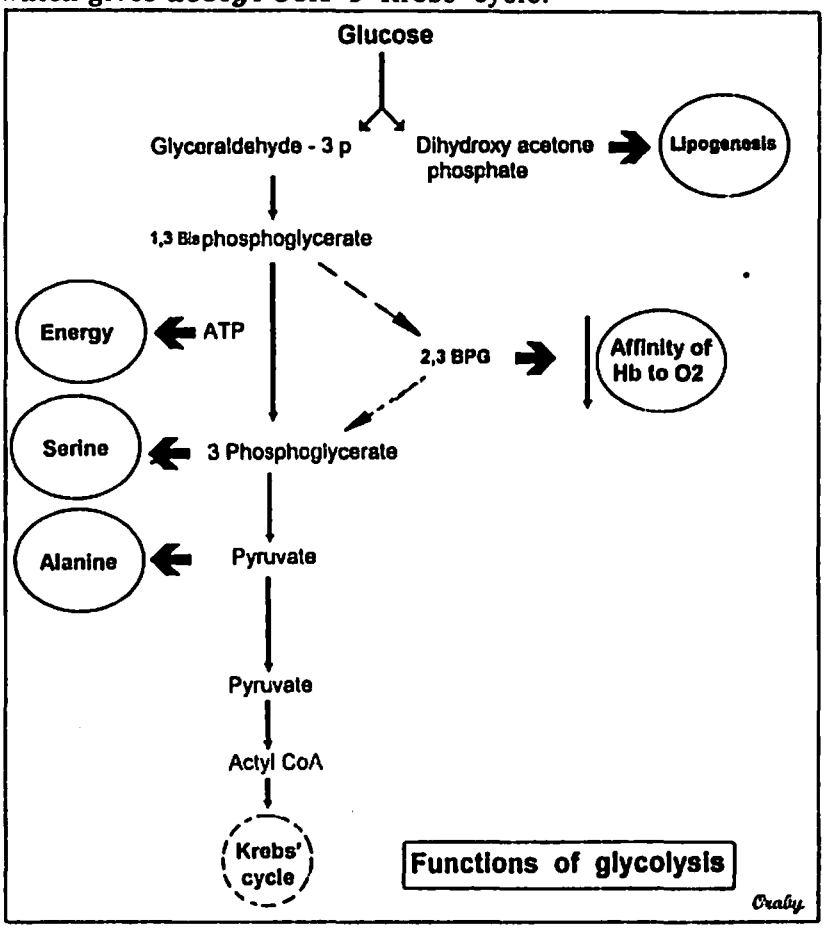
E. Oxidation of extramitochondrial NADH+H$^+$:

1. The 2 molecules of cytosolic NADH+H$^+$ cannot penetrate mitochondrial membrane, however, they can be used to produce energy (4 or 6 ATP) by respiratory chain phosphorylation in the mitochondria (see chapter I).

F. Biological importance (functions) of glycolysis:

1. **Energy production :**
   a) Anaerobic glycolysis gives 2 ATP.
   b) Acrobic glycolysis gives 8 ATP.
2. **Oxygenation of tissues:**
   Through formation of 2,3 bisphosphoglycerate, which decreases the affinity of Hemoglobin to O_2.
3. **Provides important intermediates:**
   a) Dihydroxyacetone phosphate: may give glycerol-3-phosphate, which is used for ***synthesis of triacylglycerols and phospholipids*** (lipogenesis).
   b) 3 Phosphoglycerate: which may be used for ***synthesis of amino acid serine.***
   c) Pyruvate: which may be used in ***synthesis of amino acid alanine.***

4. Aerobic glycolysis provides the mitochondria with ***pyruvate***, which gives ***acetyl CoA*** → Krebs' cycle.

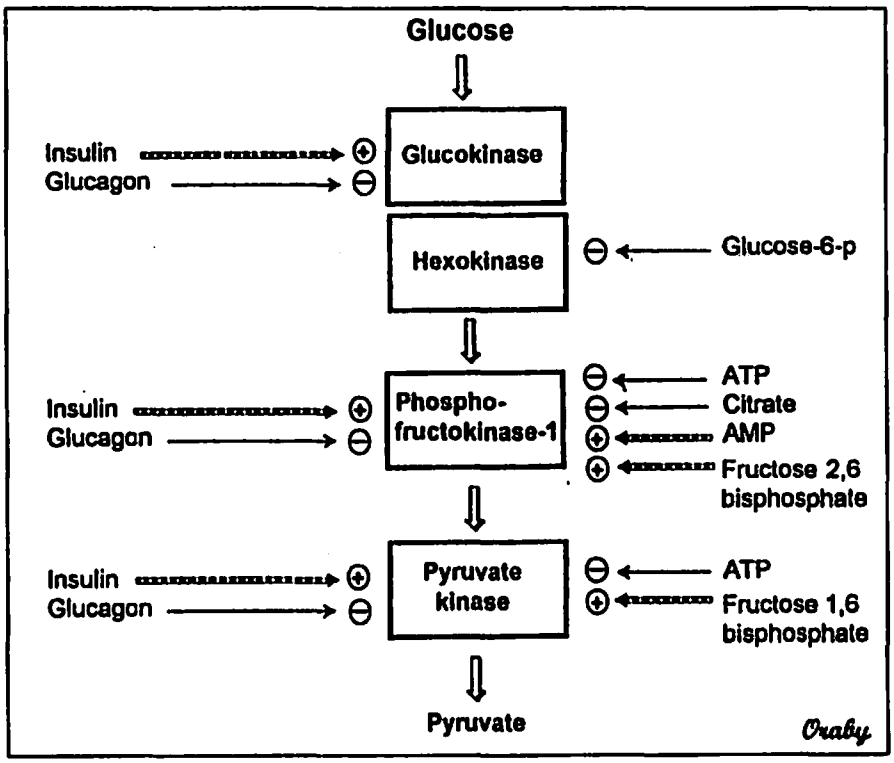


G. Regulation of glycolysis:

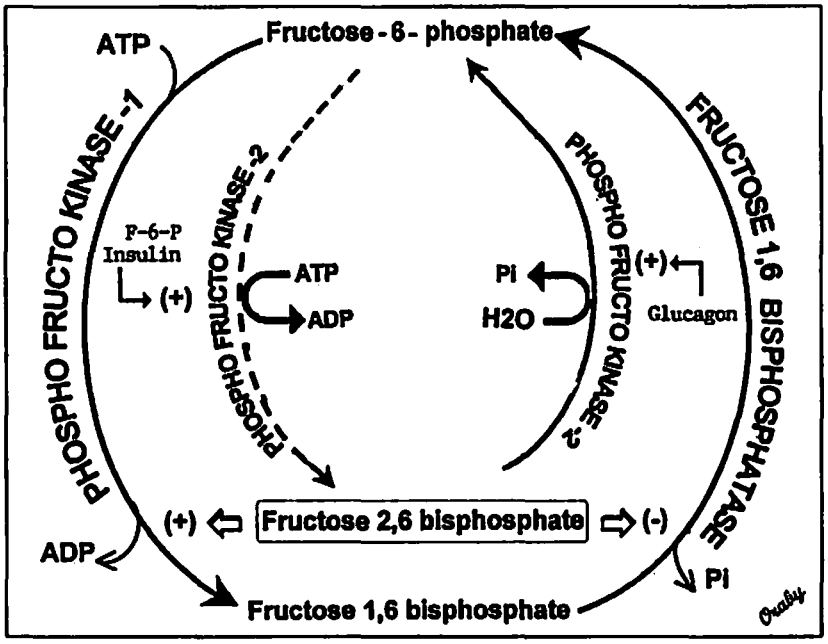
The rate of glycolysis is regulated by controlling of the three irreversible enzymes (key enzymes). These enzymes catalyze what is called committed reactions of the pathway. These enzymes are; glucokinase (hexokinase), phosphofructokinase-1 and pyruvate kinase.

1. **Hormonal regulation:**
   a) **Insulin:** ***Stimulates*** synthesis of all key enzymes of glycolysis. It is secreted after meal (in response to high blood glucose level).
   b) **Glucagon:** ***Inhibits*** the activity of all key enzymes of glycolysis. It is secreted in response to low blood glucose level.
2. **Energy regulation:**
   a) **High level of ATP** inhibits PFK-1 and pyruvate kinase.
   b) **High level of ADP and AMP** stimulate PFK-1.

3. **Substrate regulation:**
   a) Glucose-6-phosphate inhibits hexokinase (and not glucokinase).
   b) Fructose 2,6 bisphosphate stimulates phosphofructokinase-1.
   c) Citrate inhibits phosphofructokinase-1.
   d) Fructose 1,6 bisphosphate stimulates pyruvate kinase.
4. **Fructose 2,6 bisphosphate:**
   a) This substrate is produced from fructose-6-phosphate by reaction catalyzed by: phosphofructokinase-2 (PFK-2) enzyme.
   b) Fructose 2,6 bisphosphate stimulates glycolysis by allosteric stimulation of phosphofructokinase-1. It also inhibits gluconeogenesis by inhibiting fructose 1,6 bisphosphatase enzyme.



   c) PFK-2 (present mainly in liver) has bi-functional role:
      1) **Conversion of fructose-6-phosphate into fructose 2,6 bisphosphate.** This reaction is stimulated **after meal** by fructose-6-phosphate and insulin.
      2) **Conversion of fructose 2,6 bisphosphate into fructose-6-phosphate** (exactly as phosphatase). This reaction is stimulated **during fasting** by glucagon hormone.



H. Differences between aerobic and anaerobic glycolysis:

	Aerobic	Anaerobic
1-End product	Pyruvate	Lactate
2-Energy	6 or 8 ATP	2 ATP
3-Regeneration of NAD^+	Through respiratory chain in mitochondria	Through lactate formation
4-Availability to TCA in mitochondria	Available and 2 pyruvate can be oxidized to give 30 ATP	Not available as lactate is a cytosolic substrate

I. Substrate level phosphorylation in glycolysis:

This means phosphorylation of ADP to ATP at the reaction itself.

In glycolysis there are 2 examples:

- 1,3 Bisphosphoglycerate + ADP → 3 Phosphoglycerate + ATP
- Phospho-enolpyruvate + ADP → Enolpyruvate + ATP

J. Importance of lactate production in anaerobic glycolysis:

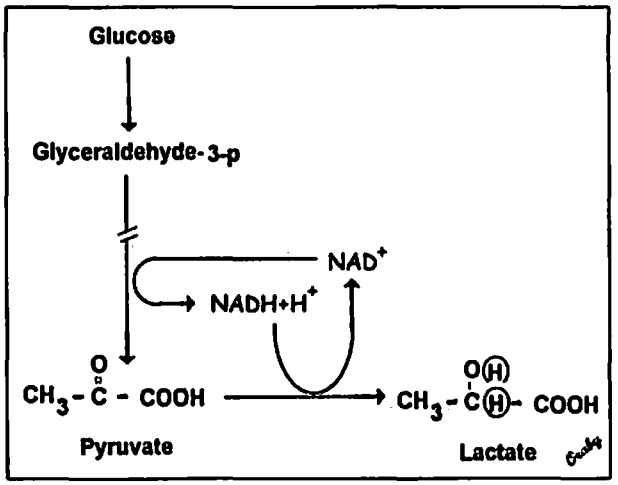
1. In absence of oxygen, lactate is the end product of glycolysis:

   Glucose → Pyruvate → Lactate

2. In absence of oxygen, $NADH+H^+$ is not oxidized by the respiratory chain, thus:

   The conversion of pyruvate to lactate is the mechanism for regeneration of NAD^+.

3. This helps continuity of glycolysis, as the generated NAD^+ will be used once more for oxidation of another glucose molecule.



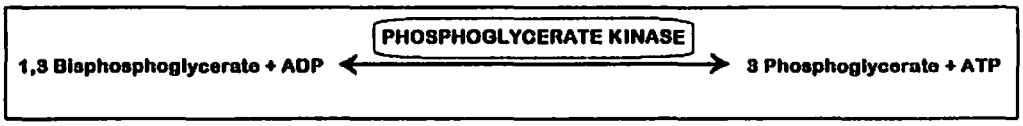
K. Special features of glycolysis in RBCs:

1. Mature RBCs contain **no mitochondria**, thus:
   a) They depend *only* upon **glycolysis** for energy production (=2 ATP).
   b) **Lactate** is always the end product.
2. Glucose uptake by RBCs is **independent on insulin** hormone.
3. **Reduction of met-hemoglobin:** Met-hemoglobin binds oxygen irreversibly. Glycolysis produces $NADH+H^+$, which used for reduction of met-hemoglobin in red cells, into hemoglobin. This reaction is catalyzed by **cytochrome b_5-met-haemoglobin reductase** system (cyt b_5):

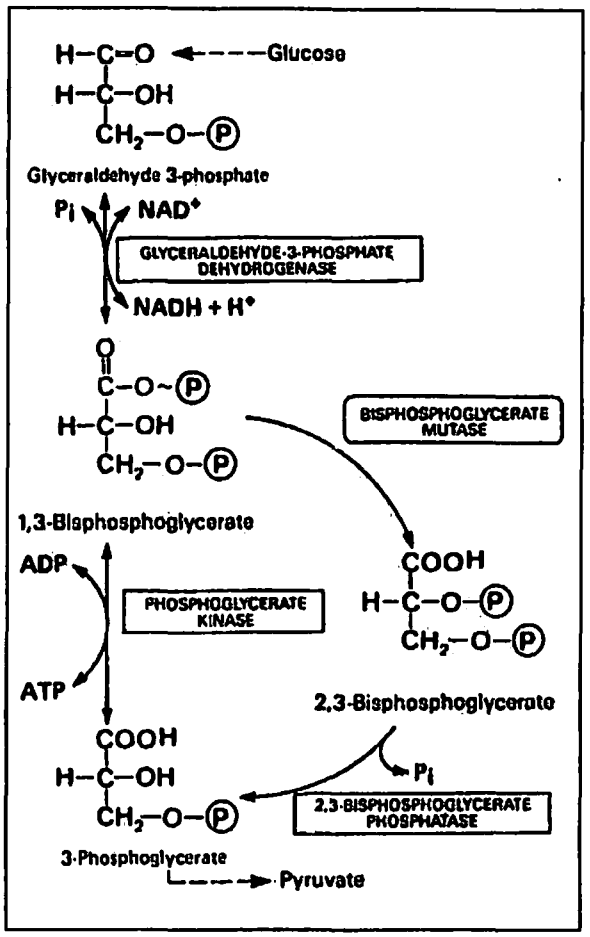
Met-Hemoglobin (Fe^{+++}) + $NADH+H^+$ → Hemoglobin (Fe^{++}) + NAD^+

4. **2,3 Bisphosphoglycerate:**
   a) The RBCs have the ability to form 2,3 bisphosphoglycerate (2,3 BPG) through what is called: Rapoport-Luebering cycle or 2,3 bisphosphoglycerate cycle. **2,3 BPG ↓ affinity of hemoglobin to O_2** → Good oxygenation of tissues.
   b) **Mechanism:**

In the erythrocytes of many mammalian species the reaction

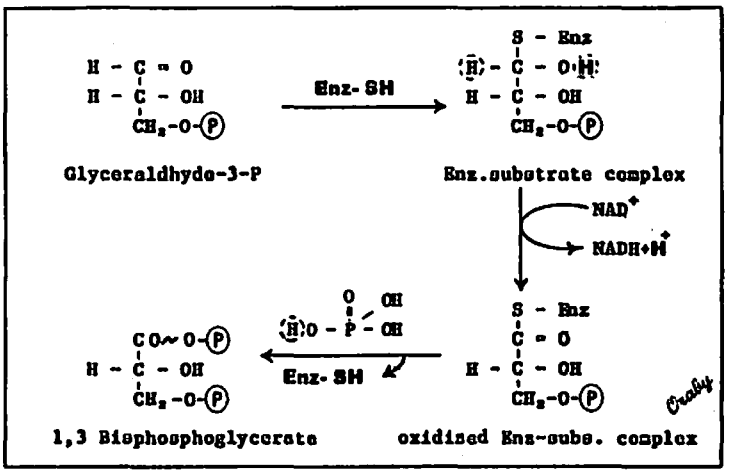


Is bypassed by other 2 reactions without producing energy (ATP) as follows:



L. Mechanism of oxidation of glyceraldhyde-3-phosphate:

1. Glyceraldhyde-3-phosphate is oxidized by glyceraldhyde-3-p dehydrogenase enzyme.
2. This enzyme contains -SH group in its active center:



M. Reversibility of glycolysis: Reversible reaction means that the same enzyme can catalyze the reaction in both directions.

1. All reactions of glycolysis –except 3- are reversible.

2. The 3 irreversible reactions (those are catalyzed by kinase enzymes) are:
   a) Glucose-6-phosphate → Glucose
   b) F 1,6 Bisphosphate → Fructose-6-phosphate
   c) Pyruvate → Phosphoenol pyruvate.
3. During fasting, glycolysis is reversed for synthesis of glucose from non-carbohydrate sources as lactate. This mechanism is called: ***gluconeogenesis.***

N. **Comparison between glucokinase and hexokinase enzymes:**

	Glucokinase	Hexokinase
1.Site	Liver only	All tissue cells
2.Affinity to glucose	Low affinity(high km) i.e. it acts only in the presence of high blood glucose concentration.	High affinity (low km) i.e. it acts even in the presence of low blood glucose concentration.
3.Substrate	Glucose only	Glucose, galactose and fructose
4.Effect of insulin	Induces synthesis of glucokinase	No effect
5.Effect of glucose-6-phosphate	No effect	Allosterically inhibits hexokinase
6.Function	Acts in liver after meals. It removes glucose coming in portal circulation, converting it into glucose-6-phosphate.	It phosphorylates glucose inside the body cells . This makes glucose concentration more in blood than inside the cells . This leads to continuous supply of glucose for the tissues even in the presence of low blood glucose concentration.

O. **<u>Lactate dehydrogenase</u>:**

1. It is an enzyme which catalyzes the reaction:

   Lactate ⇆ Pyruvate

2. ***This reaction helps the re-oxidation of NADH, H^+ into NAD^+.***
3. It has 5 isoenzymes: LD_1, LD_2, LD_3, LD_4 and LD_5.
4. **Medical importance :**

   Estimation of the activity of lactate dehydrogenase enzyme in plasma helps the diagnosis of heart and liver diseases:

   a) **LD_1:** Elevated in some heart diseases e.g. **myocardial infraction.**
   b) **LD_5:** Elevated in some liver diseases as **acute viral hepatitis.**

P. **In vitro inhibition of glycolysis:**

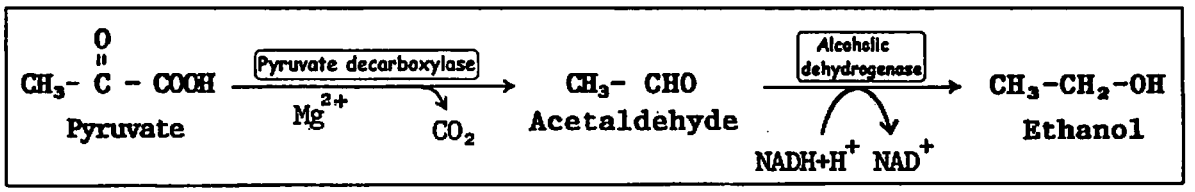
1. **Arsenate:** by competing with inorganic phosphate in the reaction:

   Glyceraldhyde-3-p → 1,3 bisphosphoglycerate

2. **Iodoacetate:** by inhibiting glyceraldhyde-3-p dehydrogenase.
3. **Fluoride:** Inhibits **enolase** enzyme. Clinical laboratories use fluoride to inhibit glycolysis by adding it to the blood before measuring blood glucose.

Q. **Fermentation:**

1. **Definition:** This is conversion of glucose into ethanol by yeast enzymes.
2. Pyruvate is formed by the same series of reactions of glycolysis. Then pyruvate is converted into acetaldehyde, then ethanol as follows:
3. Thus the end product of fermentation is CO_2 and ethanol.



R. **Sources and fate of lactate:**

1. **Sources:**

   From glycolysis especially in RBCs due to absence of mitochondria and muscle during exercises due to oxygen lack.

2. **Fate:**

   a) **Glucose formation:** [through lactic acid cycle (**Cori cycle**)]:

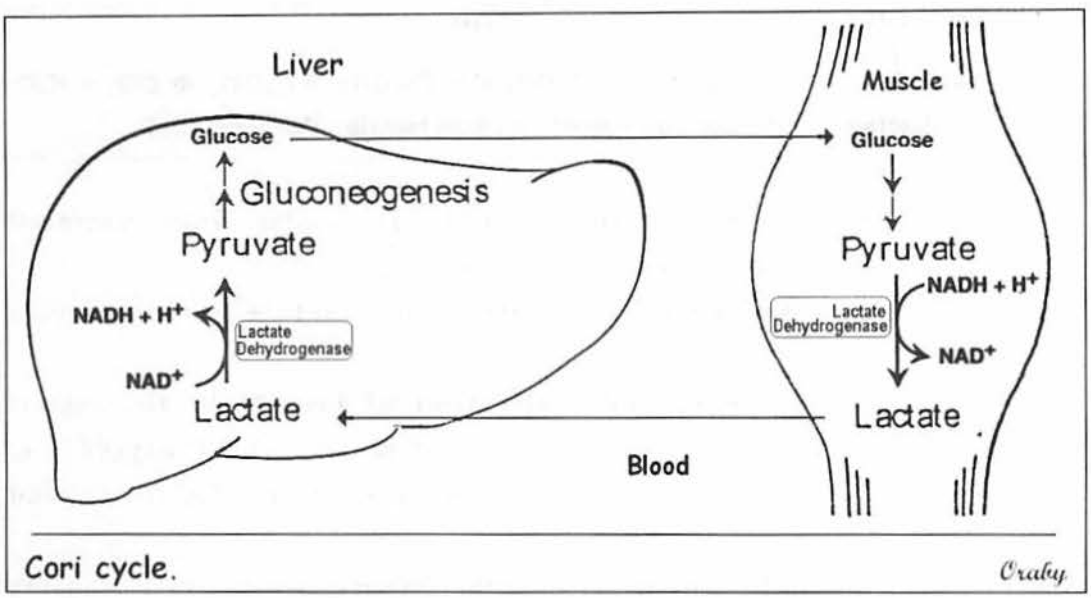
   1) Lactate formed in muscles and RBCs may diffuse to the blood then to the liver.
   2) In the liver, lactate is converted to glucose by gluconeogenesis. Glucose may diffuse back to the blood, then to red cells or muscles to be used for production of energy. This cycle is called: **Lactic acid cycle or Cori cycle.**

Definition of Cori cycle: It is the conversion of glucose into lactate in peripheral tissues, followed by conversion of lactate into glucose in liver.

b) **Conversion into pyruvate:** If oxygen gets available, lactate is converted into pyruvate, which proceeds into Krebs cycle.

c) Lactate may be accumulated in muscles causing **muscle fatigue.**

d) Lactate is **excreted** in urine and sweat.



Cori cycle.

S. **Clinical aspects of glycolysis:**

There are many diseases associated with impaired glycolysis. They include:

1. ***Pyruvate kinase deficiency.***
2. ***Hexokinase deficiency.***
3. ***Lactic acidosis.***

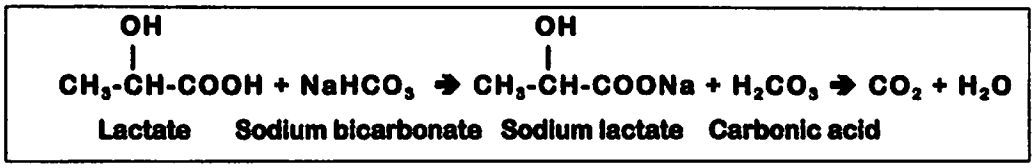
1. Pyruvate kinase (PK) deficiency:

a) This leads to excessive hemolysis of RBCs ➔ leading to hemolytic anemia.

b) Genetic deficiency of PK enzyme causes decrease in the rate of glycolysis and decreased production of ATP.

c) ATP is required for Na^+ -K^+ ATPase, which is important for stability of RBCs.

2. **Hexokinase deficiency:**
   It leads to hemolytic anemia due to decrease ATP production. The mechanism is similar to that of PK deficiency.
3. **Lactic acidosis:**
   a) **Definition and mechanism of lactic acidosis :**
      1) It is the lowered blood pH and bicarbonate levels due to increased blood lactate above normal level.
      2) This depletes bicarbonate ➔ ↓ pH of blood ➔ Lactic acidosis ➔ may lead to coma.



   b) **Causes of lactic acidosis:** It results from increased formation or decreased utilization of lactate.
      1) **Increased formation of lactate** as in severe muscular exercises.
         ➢ **Decreased utilization of lactate in tissues:** it occurs in cases of anoxia or lack of oxygen e.g. myocardial infarction, respiratory disorders, and anemia.
      2) **Phenformin:** is oral hypoglycemic drug, causing excessive anaerobic oxidation of glucose and excess lactate production.

II. *Mitochondrial pathway for glucose oxidation:*

Complete oxidation of glucose occurs in both cytosol (glycolysis) and mitochondria (Krebs` cycle). In the presence of O_2, pyruvate (the end product of glycolysis) passes by special pyruvate transporter into mitochondria which proceeds as follows:

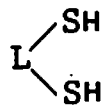
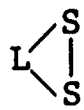
- Oxidative decarboxylation of pyruvate to acetyl CoA.
- Acetyl CoA is then oxidized completely to CO_2, H_2O through Krebs` cycle.

A. Oxidative decarboxylation of pyruvate to acetyl coenzyme A (= acetyl CoA):

1. **Enzyme:** ***Pyruvate dehydrogenase (PDH) complex:***
   a) This enzyme complex contains 3 subunits, which catalyzes the reaction in 4 steps. **These subunits are:** pyruvate decarboxylase, dihydrolipoyl transacetylase and dihydrolipoyl dehydrogenase.

b) This enzyme needs 5 coenzymes (all are vitamin B complex derivatives):

1) Vitamin B_1 = Thiamin pyrophosphate = TPP.

2) Lipoic acid = (reduced)  . (oxidized) 

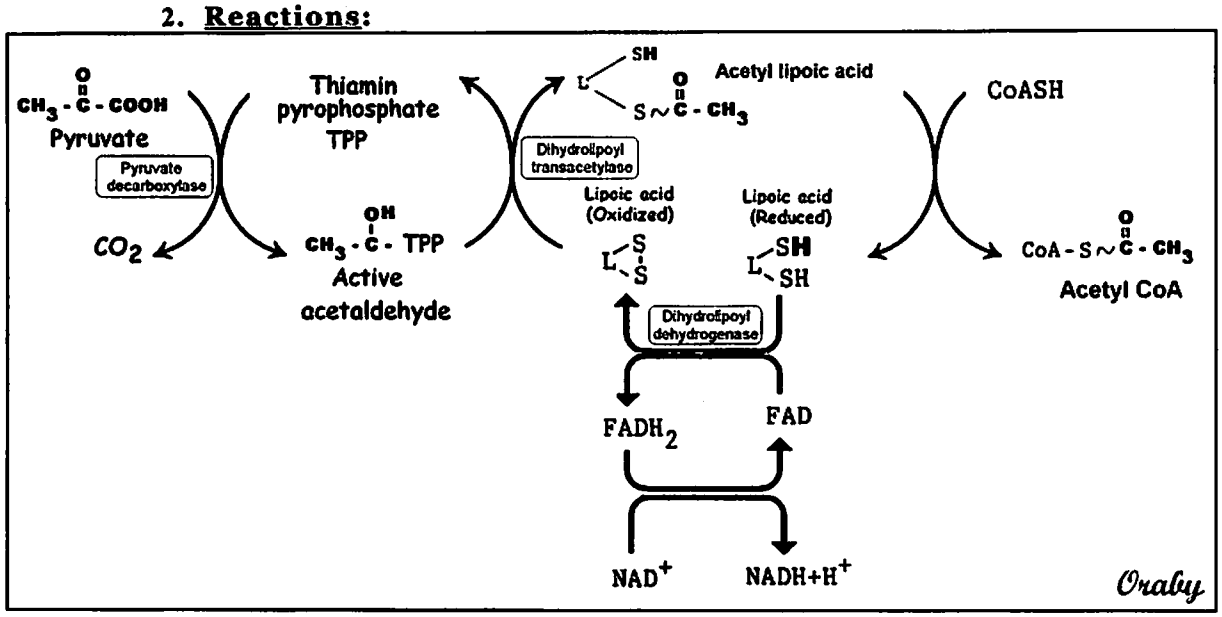
3) Coenzyme A = CoASH.

4) Flavin adenine dinucleotide = FAD.

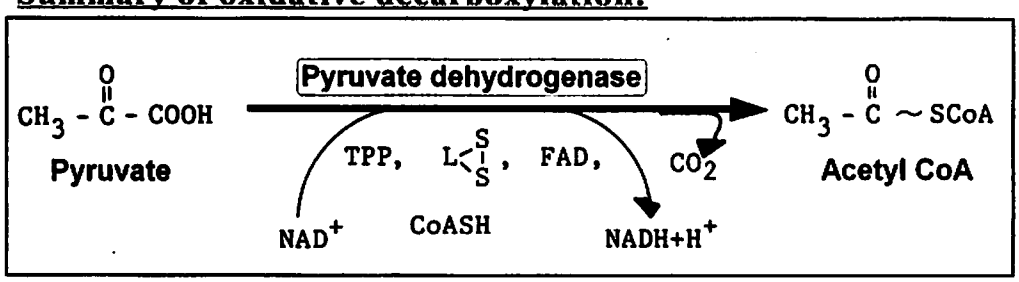
5) Nicotinamide adenine dinucleotide = NAD^+.

c) **Location: PDH** is located within the mitochondrial matrix.

2. **Reactions:**



Summary of oxidative decarboxylation:



3. **Energy production: ➔ (3 ATP)**
Oxidative decarboxylation of pyruvate to acetyl CoA produces one molecule of $NADH,H^+$. This produces 3 ATP molecules through respiratory chain phosphorylation.

4. **Regulation of oxidative decarboxylation (PDH):**

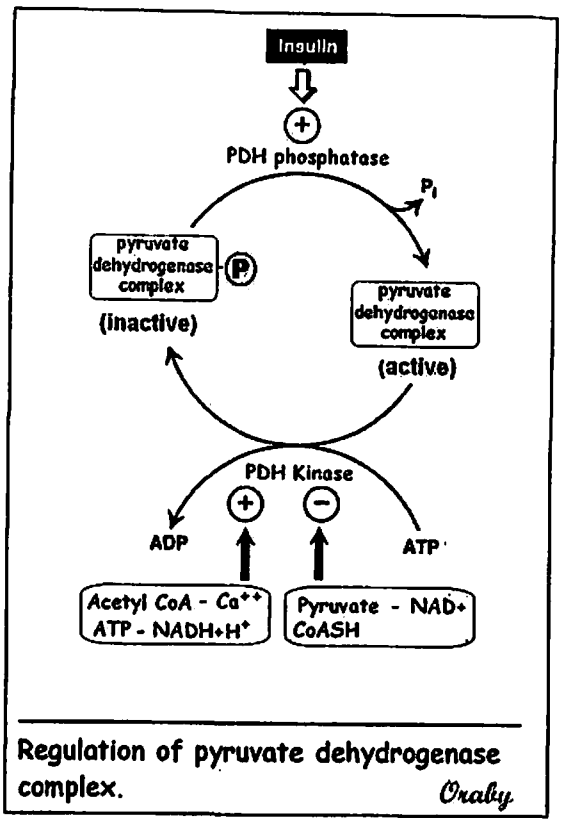
a) PDH exists in two forms: Phosphorylated (inactive) and dephosphorylated (active).

b) Pyruvate dehydrogenase (PDH) kinase enzyme converts active into inactive PDH enzyme.

c) PDH Phosphatase enzyme converts inactive into active PDH enzyme.

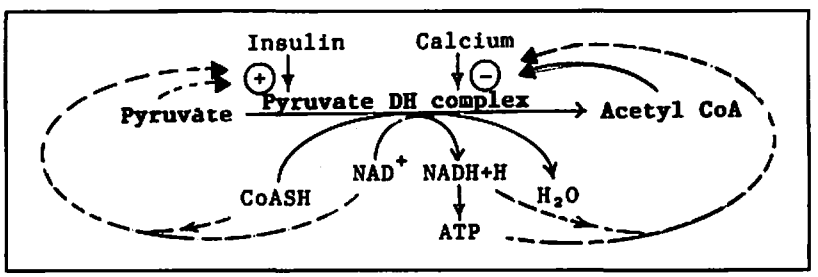
1) **Factors stimulating [+] PDH:**
   - ➢ Pyruvate.
   - ➢ CoASH.
   - ➢ NAD^+.
   - ➢ Insulin hormone.

2) **Factors inhibiting [-] PDH:**
   - ➢ $NADH,H^+$.
   - ➢ ATP.
   - ➢ Acetyl CoA.
   - ➢ Calcium ions.



Regulation of pyruvate dehydrogenase complex. Oraby

Summary of regulation:

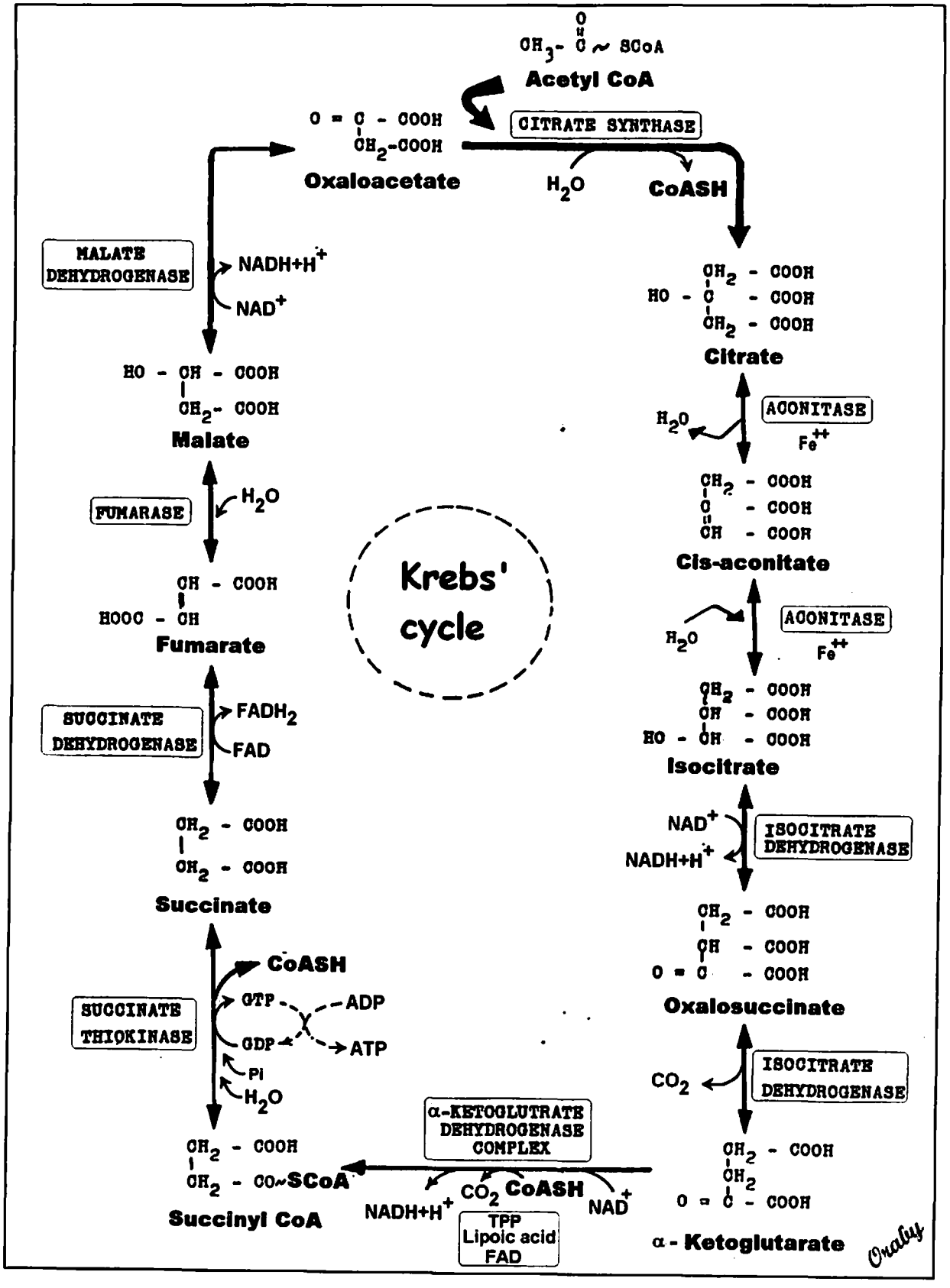


5. **In vitro inhibition of PDH:**
   a) Arsenic.
   b) Thiamin (B_1) deficiency.

B. **Krebs` cycle** [also known as citric acid cycle **(CAC)** or tricarboxylic acid cycle **(TCA) or** catabolism of acetyl CoA]:

1. **Definition:** TCA is a series of reactions in which acetyl CoA is oxidized into CO_2, H_2O and energy.
2. **Location:** Mitochondria.

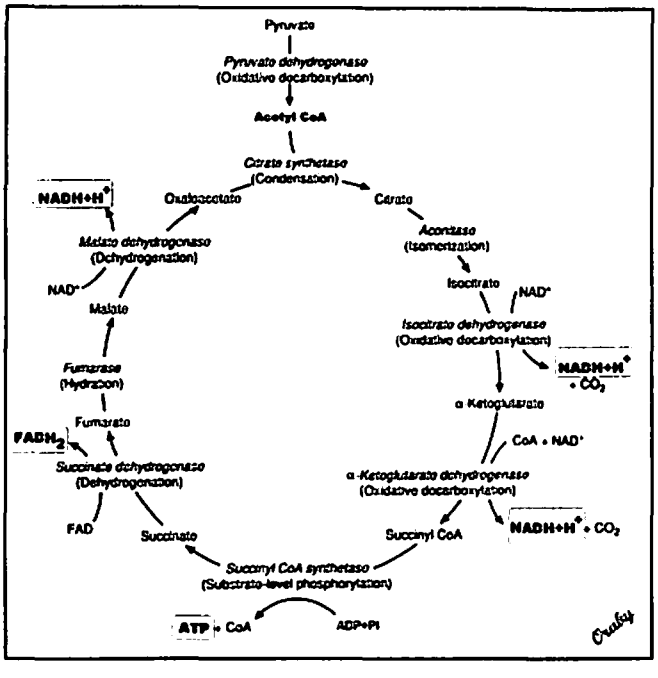
3. **Steps:**


Acetyl CoA
CITRATE SYNTHASE
Oxaloacetate
CoASH
MALATE DEHYDROGENASE
NADH+H+
NAD+
Malate
Citrate
FUMARASE
ACONITASE
Fe++
Cis-aconitate
Krebs' cycle
Fumarate
Isocitrate
SUCCINATE DEHYDROGENASE
FADH2
FAD
ISOCITRATE DEHYDROGENASE
Succinate
Oxalosuccinate
SUCCINATE THIOKINASE
GTP
GDP
ADP
ATP
Pi
ISOCITRATE DEHYDROGENASE
α-KETOGLUTRATE DEHYDROGENASE COMPLEX
TPP
Lipoic acid
FAD
Succinyl CoA
α - Ketoglutarate

3. **Steps:**
   a) The enzymes of TCA cycle are present in the mitochondrial matrix either free or attached to the inner surface of the mitochondrial membrane.
   b) The cycle is started by acetyl CoA (2 carbons) and oxaloacetate (4 carbons) to form citrate (6 carbons). It ends by oxaloacetate (4 carbons).

> **The difference between the starting compound (citrate, 6 carbons) and the ending compound (oxaloacetate, 4 carbons) is 2 carbons that are removed in the form of 2 CO_2. These 2 carbons are derived from acetyl CoA. For this reason acetyl CoA is completely catabolized in TCA and never give glucose.**

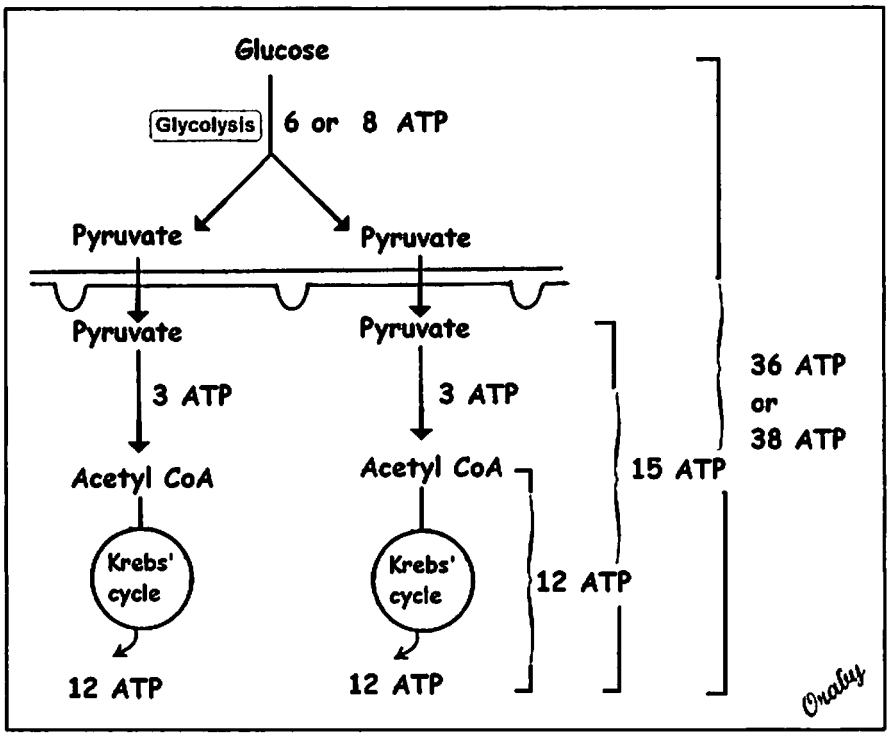
4. **Energy production of TCA:** (Energy produced by catabolism of acetyl CoA):
   a) Oxidation of one molecule of acetyl CoA in TCA produces 12 ATP molecules, 11 by respiratory chain phosphorylation and 1 by substrate level phosphorylation as follows :



Enzyme	Method of ATP production	No. of ATP
Isocitrate dehydrogenase	Oxidation of NADH+H+ by respiratory chain phosphorylation	**3 ATP**
α-Ketoglutarate dehydrogenase	Oxidation of NADH+H+ by respiratory chain phosphorylation	**3 ATP**
Succinyl CoA thiokinase	Substrate level phosphorylation	**1 ATP**
Succinate dehydrogenase	Oxidation of FADH by respiratory chain phosphorylation	**2 ATP**
Malate dehydrogenase	Oxidation of NADH+H+ by respiratory chain phosphorylation	**3 ATP**
Total =		**12 ATP**

b) Energy production of complete oxidation of one molecule of glucose:

- Glucose oxidation ➔ 36 or 38 ATP.
- Pyruvate oxidation ➔ 15 ATP.
- Acetyl CoA ➔ 12 ATP



5. **Oxidative decarboxylation of α-ketoglutarate to succinyl CoA:**

It is similar to the conversion of pyruvate to acetyl CoA.

a) **Enzymes:** α-ketoglutarate dehydrogenase complex.

b) **Coenzymes:** TPP, Lipoic acid, CoASH, FAD and NAD^+.

6. **Functions (significance) of TCA:**

TCA cycle is **amphibolic** i.e. it has catabolic (breakdown) and anabolic (formation) functions.

Energy: 12 ATP

Catabolic functions: Oxidation of carbohydrate, lipids and proteins.

Anabolic functions: Formation of:

* Amino acids
* Glucose
* Heme
* Fatty acid and cholesterol
* CO_2

a) **Production of energy** (12 ATP).

b) **Catabolic functions:** TCA is the final common pathway for oxidation of carbohydrates, fats and proteins (amino acids).

c) **Anabolic functions:** Formation of:

1) **Amino acids :**

α-Ketoglutarate —*Transamination*→ Glutamate.

Oxaloacetate —*Transamination*→ Aspratate.

2) **Glucose:** e.g.

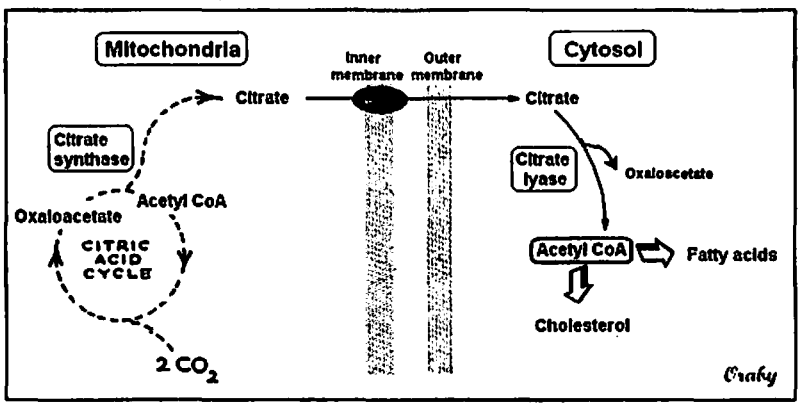
α-ketoglutarate —*Gluconeogenesis*→ Glucose.

3) **Heme synthesis :**

Succinyl CoA ➔ Heme.

4) **Fatty acid and cholesterol: from acetyl CoA (synthesized in cytosol):** (see the figure).

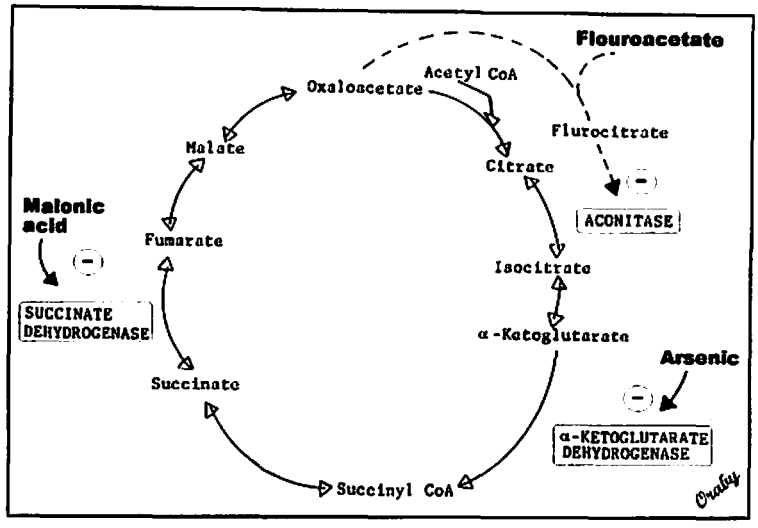
(Note that acetyl CoA is produced in mitochondria. The inner mitochondrial membrane is impermeable to acetyl CoA. To overcome this impermeability, acetyl CoA combines with oxaloacetate to form citrate. Citrate (diffuses to cytosol) ➔ Oxaloacetate + Acetyl CoA ➔ Fatty acid and cholesterol.



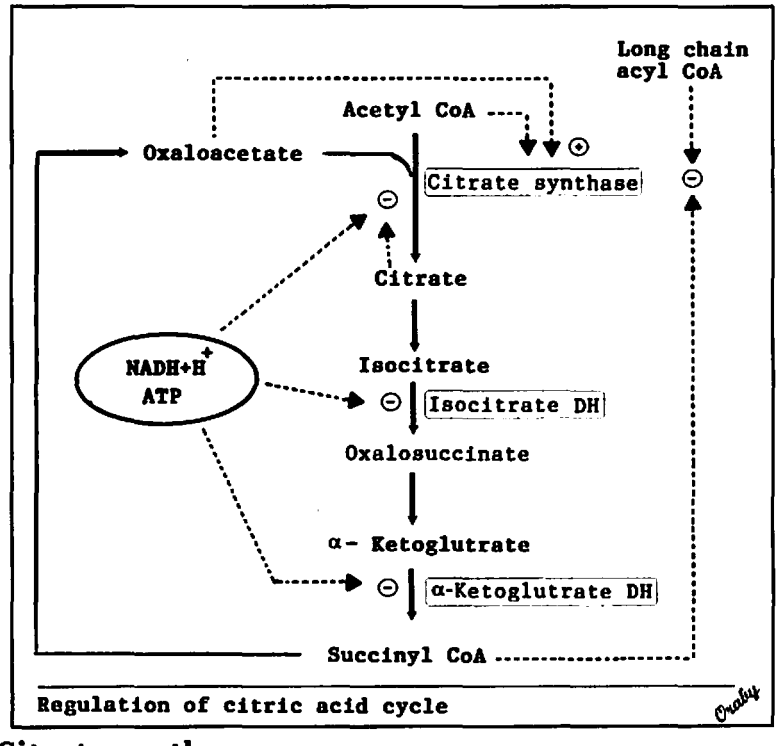
5) CO_2 produced in TCA is used in the following (CO_2 **fixation**) reactions:

- Pyruvate + CO_2 ➔ Oxaloacetate —Gluconeogenesis→ Glucose.
- Acetyl COA + CO_2 ➔ Malonyl CoA ➔ Fatty acids.
- Ammonia + ATP + CO_2 ➔ Carbamoyl phosphate ➔ Urea and pyrimidine.
- Propionyl CoA + CO_2 ➔ Methyl malonyl CoA ➔ Succinyl CoA ➔ Intermediate of citric acid cycle.
- Formation of C_6 of purine.
- Synthesis of H_2CO_3 / HCO_3 buffer.

7. **In vitro inhibition of TCA cycle :**
   a) **Flouroacetate** (F_1-CH_2-COSCoA): inhibits aconitase enzyme.
   b) **Arsenate:** inhibits α-ketoglutarate dehydrogenase enzyme.
   c) **Malonic acid**: inhibits succinate dehydrogenase enzyme (competitive inhibition).



8. **Regulation of citric acid cycle:**
   TCA is regulated through the **key enzymes** (citrate synthase, isocitrate dehydrogenase and α-ketoglutarate dehydrogenase) and the **availability of O_2:**



a) Citrate synthase :
   1) Stimulated by acetyl CoA, oxaloacetate, ADP and NAD^+.
   2) Inhibited by long chain acyl CoA, citrate, succinyl CoA, ATP and NADH, H^+.

b) Isocitrate dehydrogenase and α-ketoglutarate dehydrogenase:

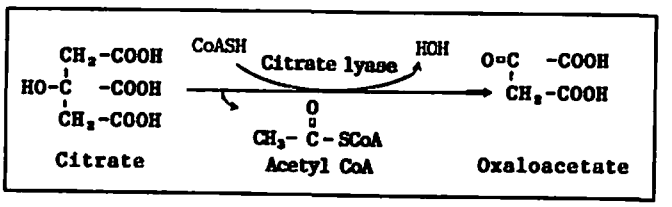
1) Stimulated by NAD^+, ADP.
2) Inhibited by NADH, H^+ and ATP.

c) **Availability of Oxygen:** Citric acid cycle needs oxygen to proceed (i.e. aerobic pathway). This is because in absence of oxygen, respiratory chain is inhibited leading to increase $NADH,H^+$ ➔ inhibition of TCA cycle.

9. Sources and fate of oxaloacetate:

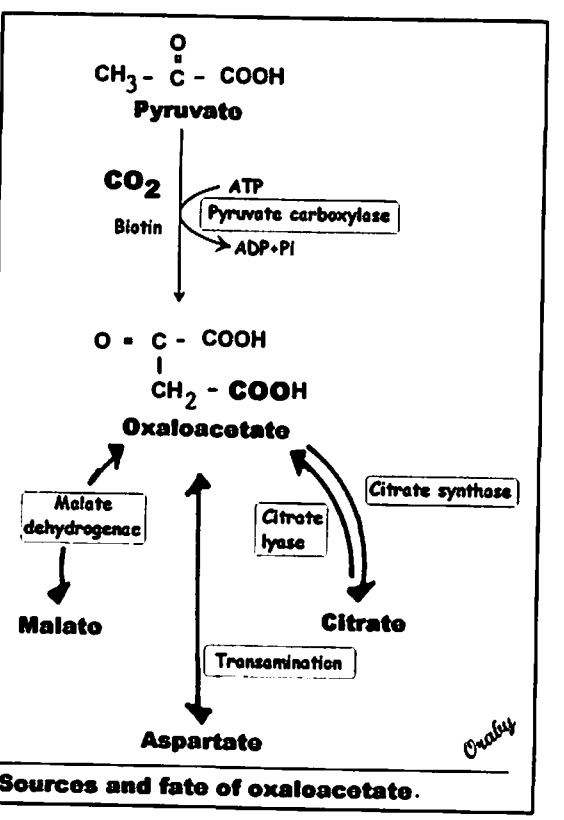
a) Sources of oxaloacetate:

1) **Oxidation of malate:** Final step in TCA cycle.
2) **Transamination of aspartate:** See protein metabolism.
3) **Carboxylation of pyruvate:** By pyruvate carboxylase and biotin (see gluconeogenesis).
4) **Cleavage of citrate:**



b) Fate of oxaloacetate:

1) **Formation of citrate:** By citrate synthase (first step in TCA cycle).
2) **Reduction to malate.**
3) **Transamination into aspartic acid.**



Sources and fate of oxaloacetate.

10. Energy production at substrate level in glucose oxidation:

a) The removal of hydrogen atoms from a compound is accompanied by a release of energy. If this energy is captured in phosphate or sulfate bonds, it will produce high-energy compounds.

b) The high energy compounds formed by glucose oxidation are:

1) Glyceraldhyde-3-p → 1,3 BPG (phosphate bond).
2) Phosphoglycerate → phosphoenol pyruvate (phosphate bond).
3) Pyruvate → Acetyl CoA (sulfate bond).
4) α-Ketoglutarate → Succinyl CoA (sulfate bond).

11. Pasteur effect:

a) It is the inhibition of glycolysis (anaerobic oxidation) by the presence of oxygen.

b) **Explanation:** Aerobic oxidation of glucose produces increased amount of ATP and citrate → those inhibit phosphofructokinase-1 (one of the key enzymes of glycolysis) → Inhibition of glycolysis.

III. *Pentose Phosphate Pathway (Hexose Phosphate Pathway):*

A. **Definition:** It is an alternative pathway for glucose oxidation where:

1. ATP (**energy**) is neither produced nor utilized.
2. Its main function is to produce **NADPH ,H^+** and **pentoses.**

B. **Location:**

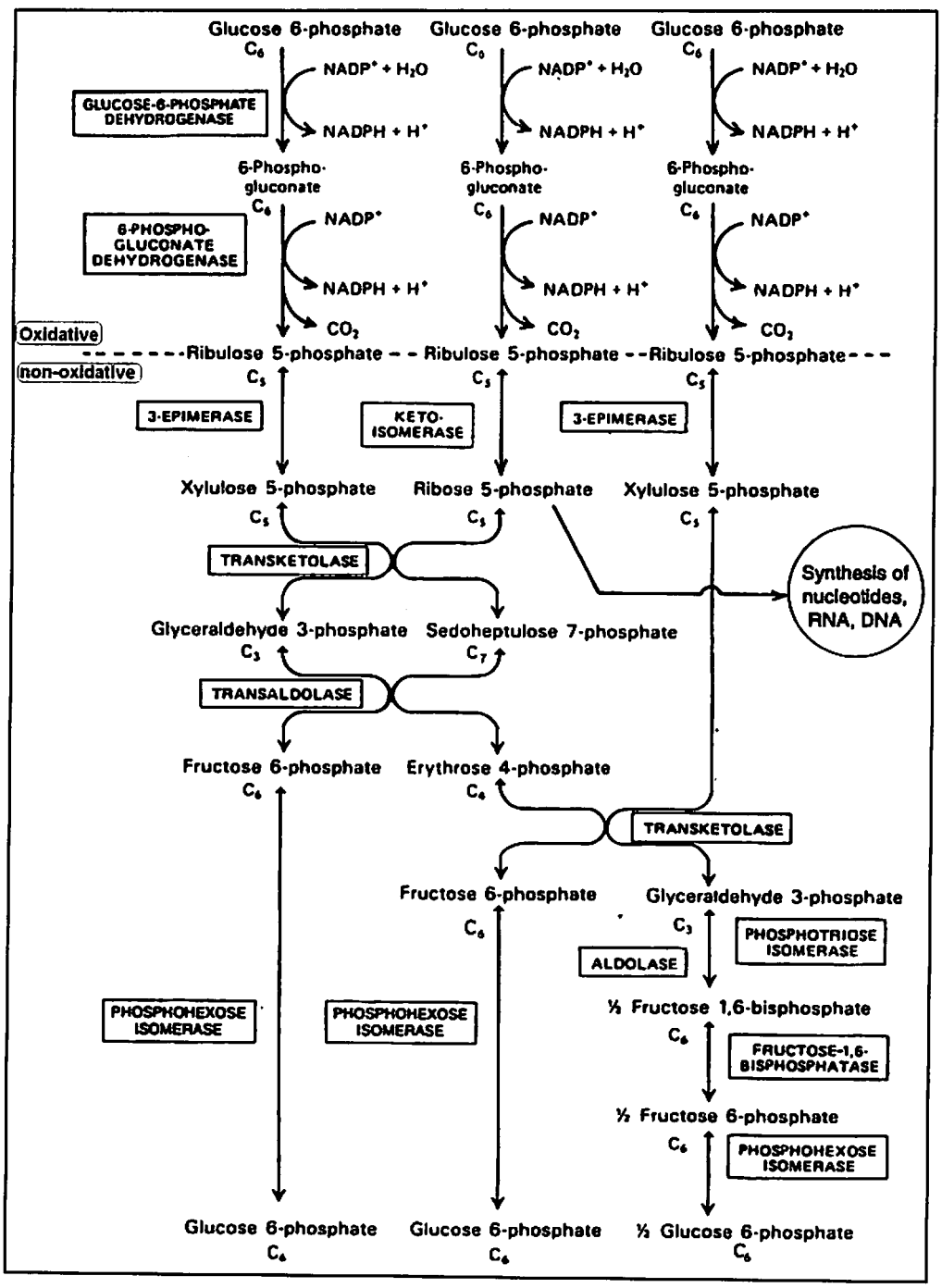
1. **Intracellular location:** cytosol.
2. **Organ location:**
   a) It is active in tissues where **NADPH,H^+** is needed for fatty acids and steroids synthesis.
      1) **Adipose tissue and liver:** It supplies NADPH,H^+ for Lipogenesis.
      2) **Adrenal cortex, ovaries and testes: It** supplies NADPH, H^+ for steroid synthesis.
      3) **Red cells: It** supplies NADPH,H^+ for production of reduced glutathione.
      4) **Retina:** It supplies NADPH,H^+ for reduction of retinal into retinol.

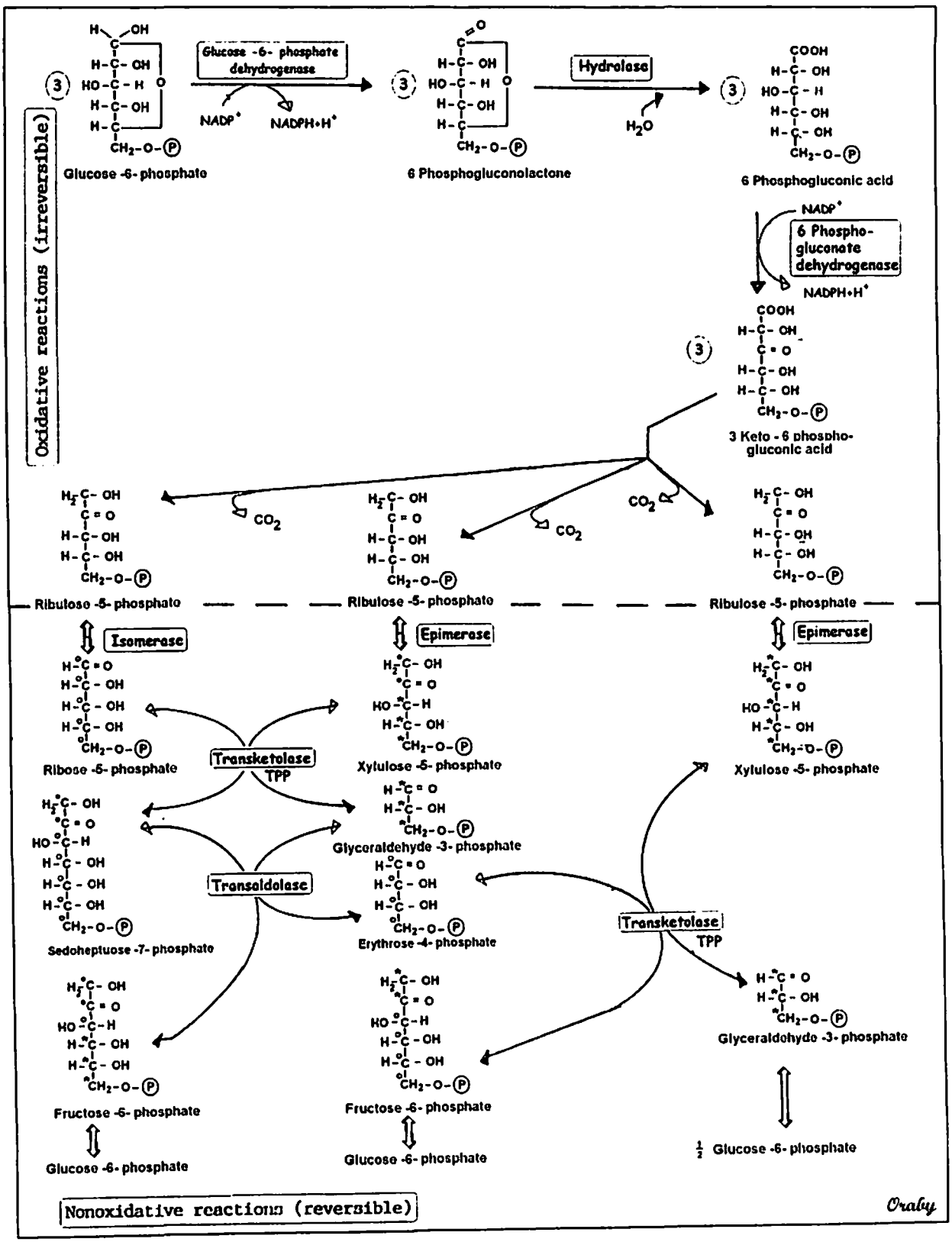
b) **In many tissues:** It supplies **pentoses** for synthesis of nucleotides.

C. Reactions (steps):

This pathway occurs in two phases; oxidative and non-oxidative:

1. **Oxidative (irreversible) phase:** where 3 molecules of glucose-6-phosphate are converted into 3 molecules of ribulsose-5-phosphate with production of $NADPH,H^+$ and CO_2 .
2. **Non-oxidative (reversible) phase:** Where the 3 molecules of ribulose-5-phosphate are interacted and converted into 2 molecules of glucose-6-phosphate and one molecule of glyceraldhyde-3-phosphate.




Oxidative reactions (irreversible)
Glucose -6- phosphate
Glucose -6- phosphate dehydrogenase
NADP+
NADPH+H+
6 Phosphogluconolactone
Hydrolase
H2O
6 Phosphogluconic acid
6 Phospho-gluconate dehydrogenase
3 Keto - 6 phospho-gluconic acid
CO2
Ribulose -5- phosphate
Isomerase
Epimerase
Ribose -5- phosphate
Xylulose -5- phosphate
Transketolase
TPP
Transaldolase
Sedoheptuose -7- phosphate
Glyceraldehyde -3- phosphate
Erythrose -4- phosphate
Fructose -6- phosphate
Glucose -6- phosphate
½ Glucose -6- phosphate
Nonoxidative reactions (reversible)
Oraby

D. Functions of pentose phosphate pathway:

Production of pentoses: ***(which are constituent of)***

- *RNA ,DNA .*
- *ATP ,GTP……..etc.*
- *NAD^+ ,FAD……etc.*

Production of NADPH + H^+: ***(which is necessary for)***

- *synthesis of substrates e.g. FA , cholesterol…….etc.*
- *Reduction of glutathione.*
- *Hydroxylation of aromatic compounds.*
- *Phagocytosis and respiratory burst.*

1. Production of pentoses:

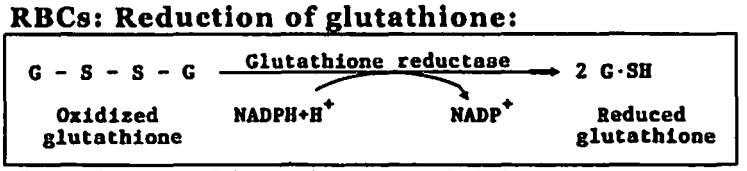
Which are essential for synthesis of nucleic acids (RNA and DNA), nucleotides (as ATP, GTP) and coenzymes (as NAD^+, NADP, FAD).

2. Production of NADPH, H^+: It is important for:

a) Synthesis of many substrates:

1) Synthesis of **fatty acids** (lipogenesis) cholesterol and other steroid hormones.
2) Synthesis of **sphingosine** and **galactolipids.**
3) Essential for **glucuronic acid** metabolism.
4) Synthesis of **non essential amino acids.**
5) Synthesis of **malate** from pyruvate by malic enzyme.

b) In RBCs: Reduction of glutathione:



Reduced glutathione is essential for:

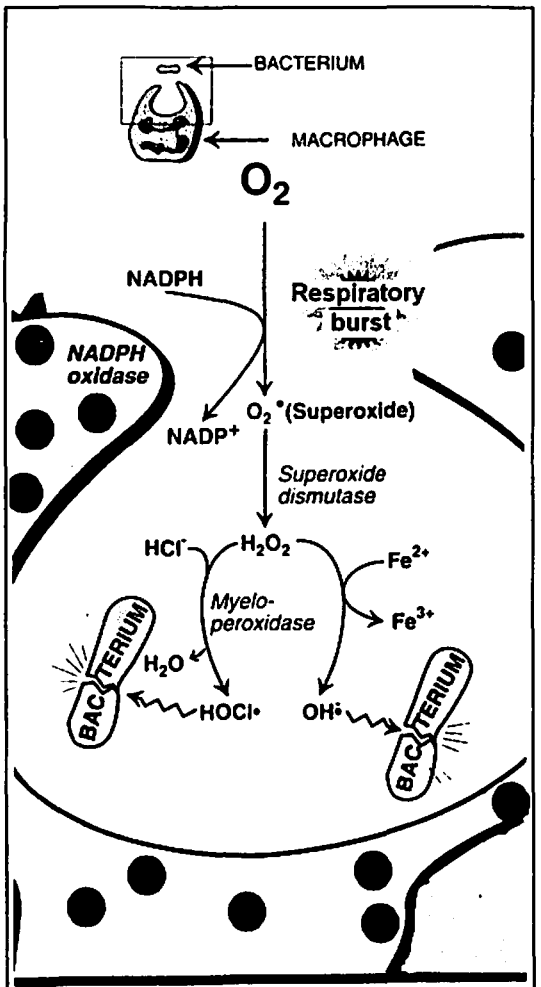
1) Normal **integrity of RBCs.**
2) Maintenance of **SH group** of RBCs enzymes.
3) Removal of **hydrogen peroxide (H_2O_2),** which is a toxic compound that causes cell membrane fragility.
4) Inactivation of insulin
5) Detoxication of many drugs and carcinogens

c) In liver: Hydroxylation of aromatic and aliphatic compounds:

NADPH,H^+ acts as coenzyme for **liver microsomal P-450 mono-oxygenase system (enzyme).** This is the major pathway for the **hydroxylation** of toxic aromatic and aliphatic compounds such as steroids, alcohols and many drugs converting them into non-toxic compounds (see chapter of xenobiotics, part I).

d) Phagocytosis by white blood cells (respiratory burst):

1) Phagocytosis is the engulfment of microorganisms and foreign bodies by white blood cells.
2) White blood cells contain an enzyme called: **NADPH+H⁺ oxidase enzyme** present in cell membrane.
3) After phagocytosis has occurred, NADPH+H⁺ oxidase converts oxygen, O_2 (derived from surrounding tissues) into super oxide ions (O_2^-).



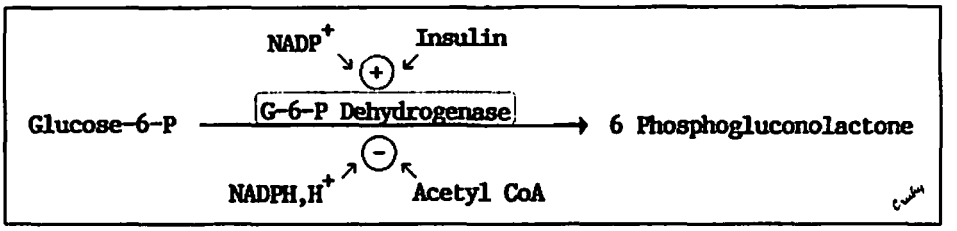
Neutrophil phagocytosis and the oxygen-dependent *myeloperoxidase* system for killing bacteria in phagolysosomes.

4) **Definition of respiratory burst:**
   It is the rapid consumption of molecular oxygen that accompanies the formation of superoxid.
5) Superoxide is then converted into H_2O_2 by **superoxide dismutase** enzyme.
6) H_2O_2 by myeloperoxidase enzyme in the presence of HCl is converted into hypochlorite ($HOCl^-$), which kills bacteria.

e) **Deficiency of NADPH+H⁺-oxidase** leads to chronic bacterial infection.

E. **<u>Regulation of Pentose phosphate pathway</u>:**

Glucose-6-phosphate dehydrogenase is the key enzyme of pentose phosphate pathway. It is stimulated by insulin and NADP⁺ and inhibited by NADPH, H⁺ and acetyl CoA.

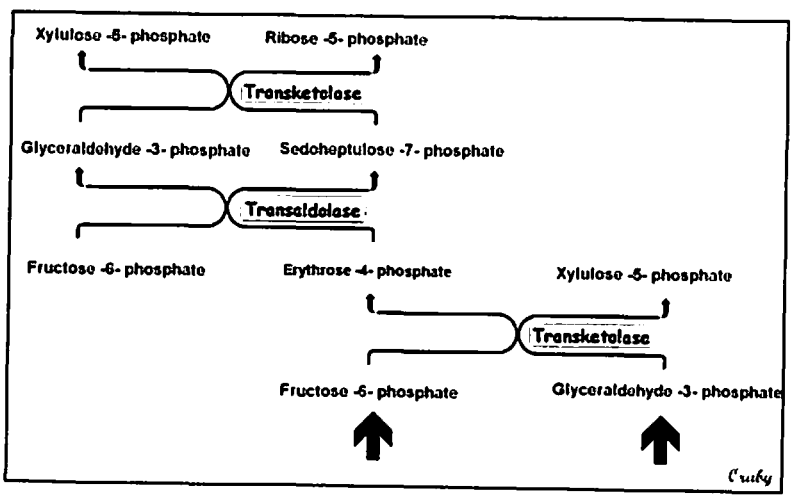


F. **Differences between pentose phosphate pathway (PPP) and glycolysis:**

	PPP	Glycolysis
Location	In certain cells	In all cells
Oxidation of glucose	Oxidation occurs in the first reaction.	Phosphorylation occurs first then oxidation
Coenzyme	$NADP^+$	NAD^+
Energy	No energy production	2 or 8 ATP
CO_2	Produced	Not produced
Pentoses	Produced	Not produced

G. **Pentose phosphate pathway in skeletal muscles:**

1. Skeletal muscles are poor in glucose-6-phosphate dehydrogenase enzyme, but they contain transketolase and transaldolase enzymes.
2. Skeletal muscles obtain their pentose requirement by reversible reactions of pentose phosphate pathway, using fructose-6-phosphate and glyceraldehyde-3-p and the enzymes transketolase and transaldolase.



H. **Defects of pentose phosphate pathway:**

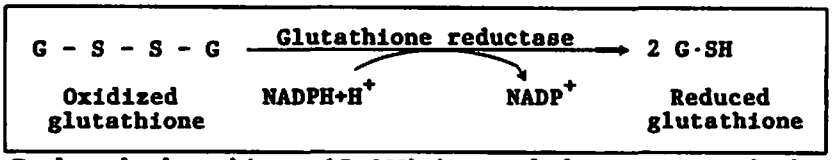
Favism (Deficiency of glucose-6-phosphate dehydrogenase enzyme):

1. Definition:

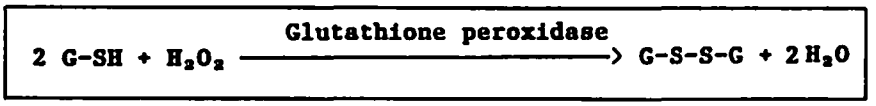
It is type of a hemolytic anemia (excessive destruction of RBCs) results after ingestion of fava beans and some other compounds. These compounds alter the structure of the enzyme → ↓ G-6-P dehydrogenase enzyme activity.

2. Mechanism:

a) Deficiency of glucose-6-P dehydrogenase ➔ Decreased NADPH,H⁺ production (which is essential to reduce glutathione in RBCs).



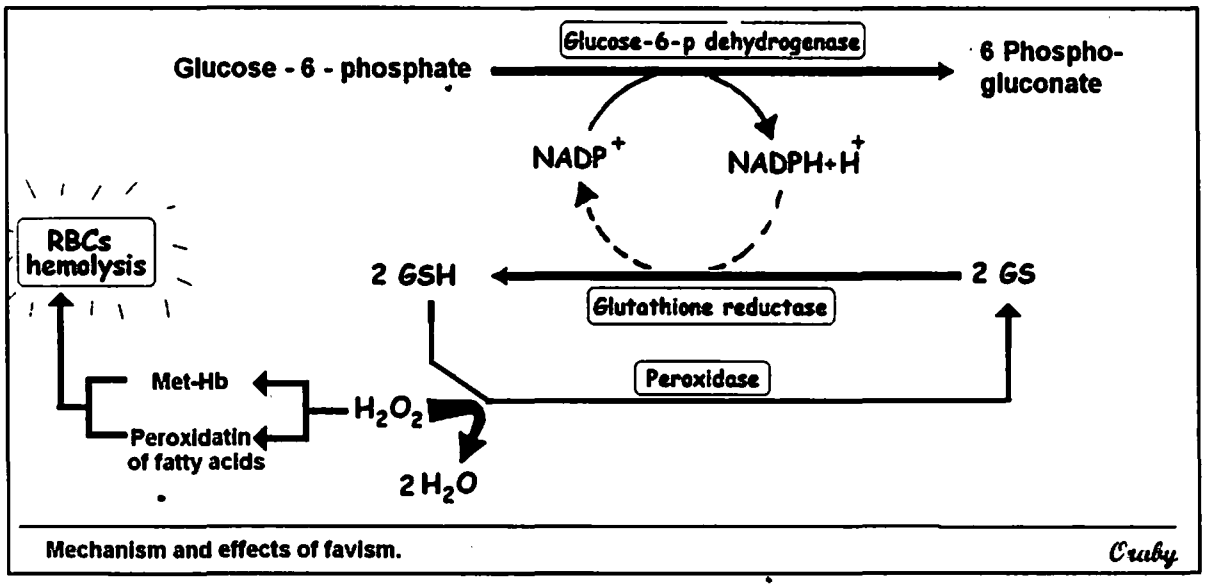
b) Reduced glutathione (G-SH) is needed to remove hydrogen peroxide (H_2O_2) which is toxic to the cell.



Deficiency of glucose-6-P DH → ↓ NADPH,H⁺ ➔ ↓ reduced glutathione ➔ Accumulation of H_2O_2 ➔ Hemolysis of RBCs

c) Effect of H_2O_2 on RBCs:

1) H_2O_2 causes peroxidation of fatty acids present in cell membrane ➔ Hemolysis.
2) H_2O_2 causes conversion of hemoglobin into met-hemoglobin. These toxic compounds increase the red cell membrane fragility.



Mechanism and effects of favism. *Cruby*

3. Signs and symptoms of favism:

a) Patients with enzyme deficiency show attacks of hemolytic anemia in the form of severe jaundice and decreased hemoglobin concentration when exposed to certain oxidizing agents such as:

1) **Special food** as fava beans.
2) **Antimalarial drugs:** as primaquine.
3) **Antibiotics** as streptomycin and sulfa.

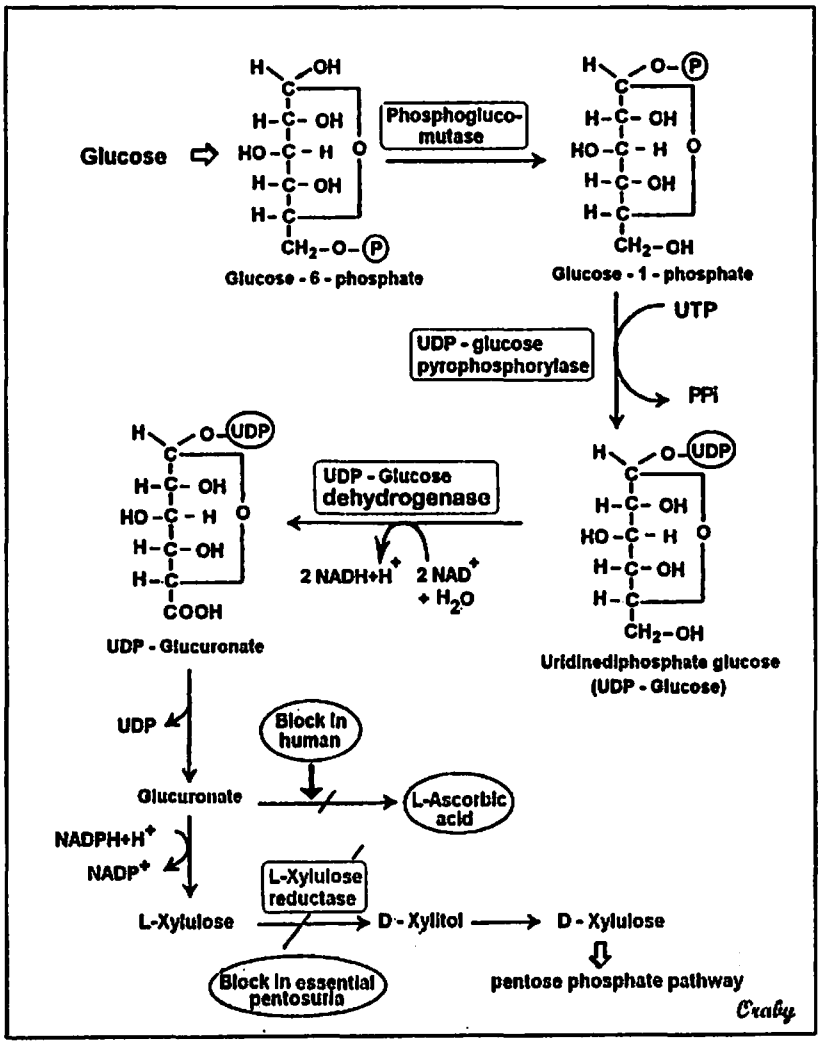
IV. *Uronic acid pathway:*

A. **Definition:** It is a minor pathway, in which glucose is converted into glucuronic acid.

B. **Location of the pathway:**

1. **Intracellular location:** cytosol.
2. **Organ location:** Mainly liver.

C. **Steps:**



D. **Functions (importance) of uronic acid pathway:**

This pathway produces **glucuronic acid**, which is important for:

1. **Synthesis of substrates:**
   a) Glycosaminoglycans.
   b) Vitamin C, L-ascorbic acid ***(not in human)***.

2. **Conjugation reactions:**
   UDP-glucuronic acid is used for conjugation with many body compounds to make them more soluble before excretion e.g. steroid hormones and bilirubin.
3. **Detoxification reactions:**
   UDP-glucuronic acid is used for conjugation with toxic compounds to make them less toxic e.g. phenols (see chapter of xenobiotics, part I).

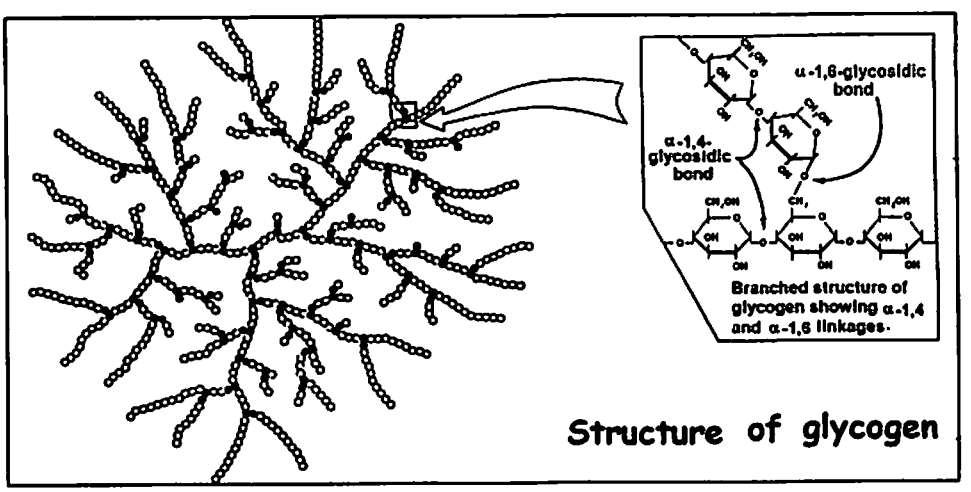
E. **Fate of glucuronic acid** (see diagram):
UDP-glucuronate is converted to glucuronate then → L-xylulose → D-xylitol → D-xylulose → which then joins pentose phosphate pathway to be completely oxidized.

F. **Defects of uronic acid pathway = Essential pentosuria:**
1. It is ***benign*** rare hereditary disease due to failure of conversion of L-xylulose into D-xylulose (due to **deficiency of L-xylulose reductase).**
2. L-xylulose will accumulate and be excreted in urine. Subjects excrete 1 to 4 grams of L-xylulose in the urine each day.

Glycogen Metabolism

1. *Structure of glycogen:*

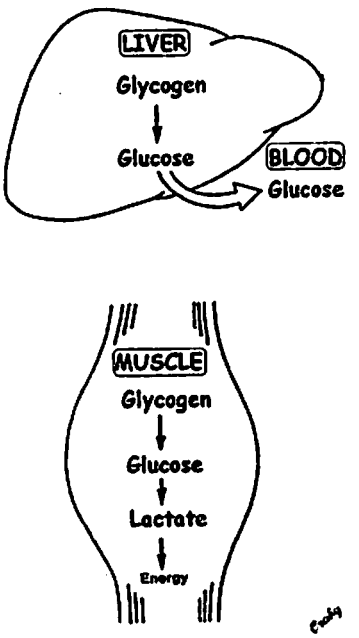
A. Glycogen is homopolysaccharide formed of branched α D glucose units (α 1,4 and α 1,6).
B. The main glycosidic bond is α1-4-linkage. Only at the branching point, the chain is attached by α1-6 linkage.
C. Each branch is made of 12-14 glucose units.



II. ***Location of glycogen:*** Glycogen is present mainly in cytosol of liver and muscles.

A. **Liver glycogen** is about 120 grams (about 6 % of liver weight).

B. **Muscle glycogen** is about 350 grams (about 1 % of total muscles weight).



Location and function of liver and muscle glycogen.

III. ***Functions of glycogen:***

A. **Liver glycogen:** It maintains normal blood glucose concentration especially during the early stage of fast (between meals). After **12-18 hours** fasting, liver glycogen is depleted.

B. **Muscle glycogen:** It acts as a source of energy within the muscle itself especially during muscle contractions.

IV. ***Synthesis of glycogen (glycogenesis):***

A. **Definition:** It is the formation of glycogen in liver and muscles.

B. **Substrates for glycogen synthesis:**

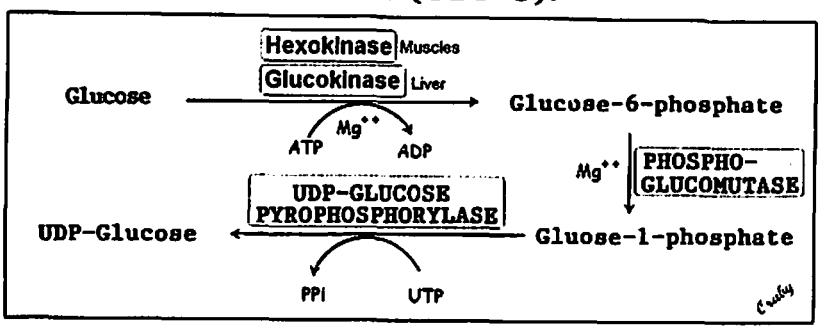
1. **In liver:**
   a) Blood glucose.
   b) Other hexoses: fructose and galactose.
   c) Non-carbohydrate sources: (gluconeogenesis) e.g. glycerol and lactate. These are converted first to glucose, then to glycogen.
2. **In muscles:**
   a) Blood glucose only.

C. **Steps:**

Glucose molecules are the first activated to uridine diphosphate glucose **(UDP-G)**. Then these UDP-G molecules are added to a glycogen primer to form glycogen.

$$\text{Glucose} \Rightarrow \text{UDP-Glucose} \xrightarrow[\text{Glycogen synthase, Branching enzyme}]{\text{Glycogen primer}} \text{Glycogen}$$

1. Formation of UDP-Glucose (UDP-G):



- **Note: Glucose is converted into glucose-6-phosphate by glucokinase in liver and hexokinase in muscles.**

2. Formation of glycogen:

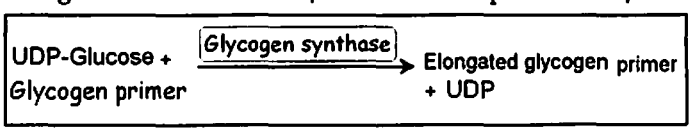
a) UDP-Glucose reacts with glycogen primer, which may be:

1) **Few molecules of glucose** linked together by α1-4 linkage.

2) **A protein called glycogenin.** UDP-G molecules react with -OH of tyrosine of that protein to initiate glycogen synthesis.

b) Glycogen synthase enzyme:

By the action of glycogen synthase (key enzyme of glycogenesis), UDP-G molecules are added to glycogen primer causing elongation of the α1-4 branches up to 12-14 glucose units.



c) Branching enzyme:

It transfers parts of the elongated chains (5-8 glucose residues) to the next chain forming a new α1-6 glycosidic bond. The new branches are elongated by the glycogen synthase and the process is repeated.

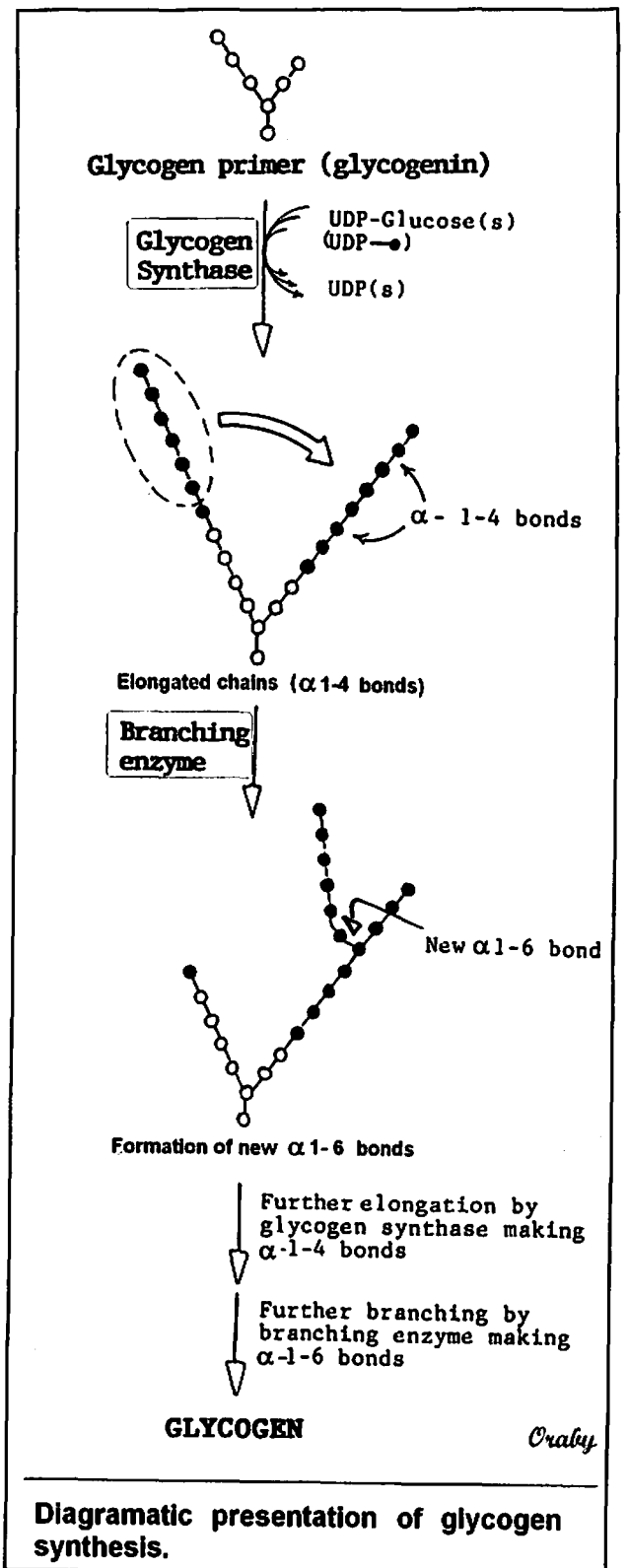
V. *Breakdown of glycogen (Glycogenolysis):*

A. Definition:

It is the breakdown of glycogen into glucose (in liver) and lactic acid (in muscles).

B. Steps:

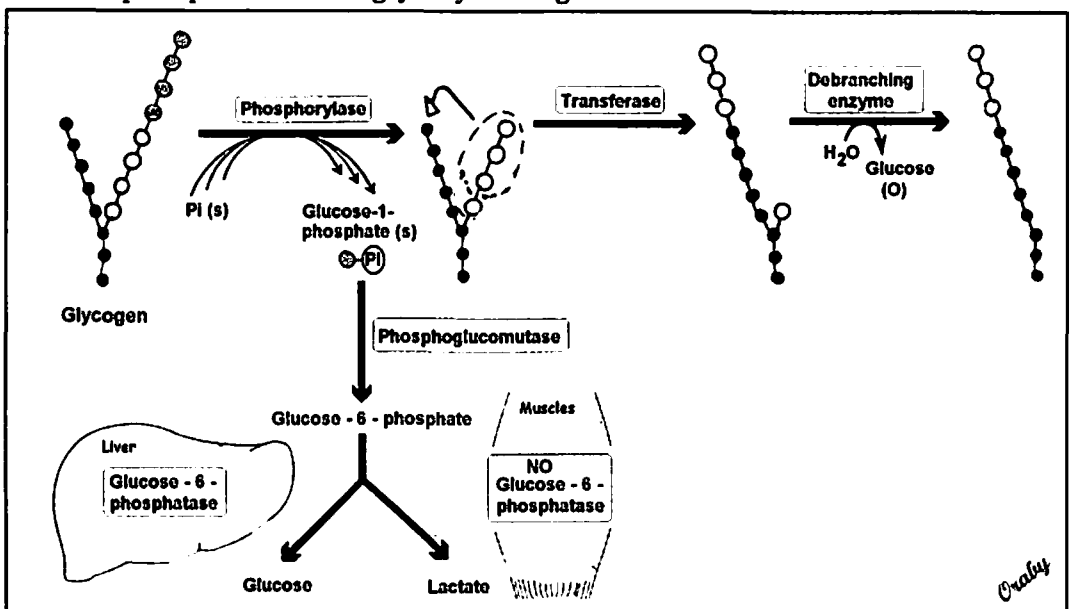
1. **Phosphorylase** (the key enzyme of glycogenolysis) acts on α1-4 bonds, breaking it down by phosphorolysis (i.e. breaking down by addition of inorganic phosphate "Pi"). Therefore, it removes glucose units in the form of glucose-1-phosphate.



Diagramatic presentation of glycogen synthesis.

2. Phosphorylase enzyme acts on the branches containing more than 4 glucosyl units.
3. When the branch contains 4 glucose units, 3 of them are transferred to a next branch by transferase enzyme, leaving the last one.

4. The last glucose unit that is attached to the original branch by α1-6 bond is removed by debranching enzyme by hydrolysis (i.e. breaking the bond down by addition of H_2O).
5. Glucose-1-phosphate molecules are converted to glucose-6-phosphate, by mutase enzyme.
6. **Fate of glucose-6-phosphate:**
   a) **In liver:** glucose-6-phosphate is converted to glucose by glucose-6-phosphatase.
   b) **In muscles:** there is <u>no</u> glucose-6-phosphatase, so glucose-6-phosphate enters glycolysis to give lactate.



Mechanism of glycogenolysis in liver and muscles.

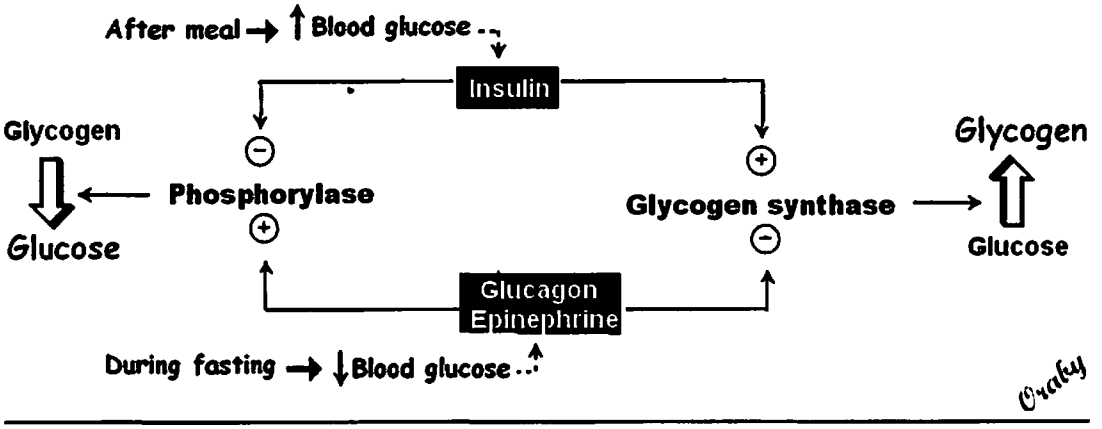
VI. *Regulation of glycogenesis and glycogenolysis:*

A. There is a **coordinated** regulation of glycogenesis and glycogenolysis i.e. conditions leading to stimulation of glycogenolysis, inhibiting at the same time glycogenesis and vise versa.

B. <u>**During fasting,**</u> glycogenolysis is stimulated and glycogenesis is inhibited. This provides blood glucose from liver glycogen.

C. <u>**After meal,**</u> part of absorbed glucose (40%) goes to general circulation to be utilized. The remaining (60%) is converted into glycogen in liver. So after meal, glycogenesis is stimulated and glycogenolysis is inhibited.

D. <u>**The principle enzymes**</u> controlling glycogen metabolism are glycogen **synthase** and **phosphorylase**. These are regulated as follows:



Summary of hormonal regulation of glycogenesis and glycogenolysis.

1. During fasting:

Glycogenolysis is stimulated by:

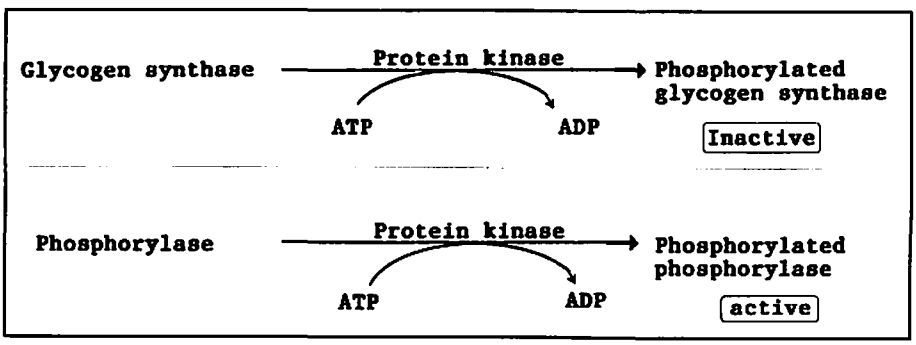
* Cyclic AMP and protein kinase
* Calcium ions and calmodulin protein

a) Blood glucose level tends to be decreased. This stimulates secretion of epinephrine, nor-epinephrine and glucagon hormones .

b) These hormones stimulate adenylate cyclase enzyme, which converts ATP into cyclic AMP (cAMP).

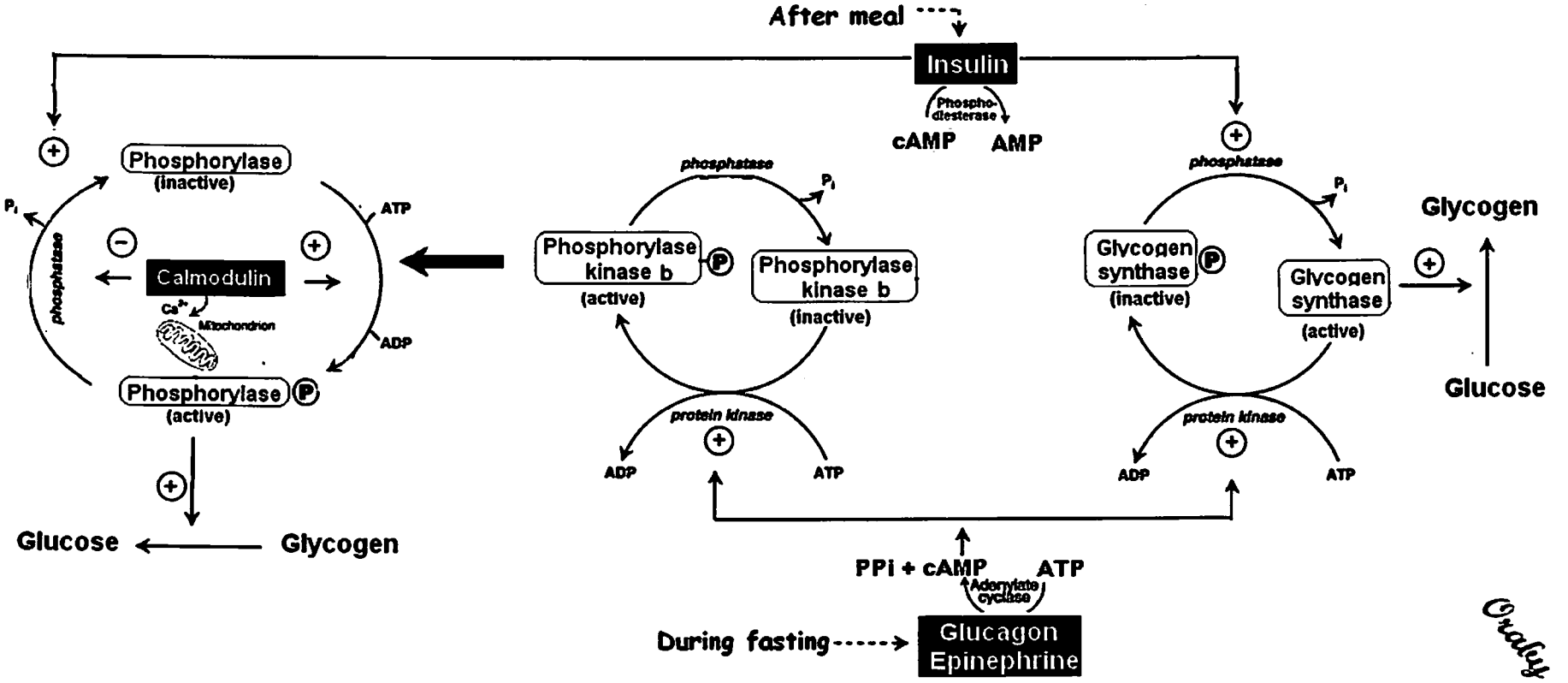
c) cAMP stimulates protein kinase enzyme .

d) Protein kinase enzyme stimulates phosphorylation of both glycogen phosphorylase and glycogen synthase.



e) As a result, glycogenolysis proceeds causing increase of blood glucose. At the same time , glycogenesis will be inhibited.

f) Epinephrine and nor epinephrine stimulate mobilization of calcium ions (Ca^{2+}) from mitochondria to cytosol. Ca^{2+} ions then combine with a protein called calmodulin causing its conformational changes and activating it.



Hormonal regulation of glycogenesis and glycogenolysis.

g) **The active calmodulin causes** phosphorylation of both glycogen phosphorylase and synthase (like protein kinase). This leads to stimulation of glycogenolysis.

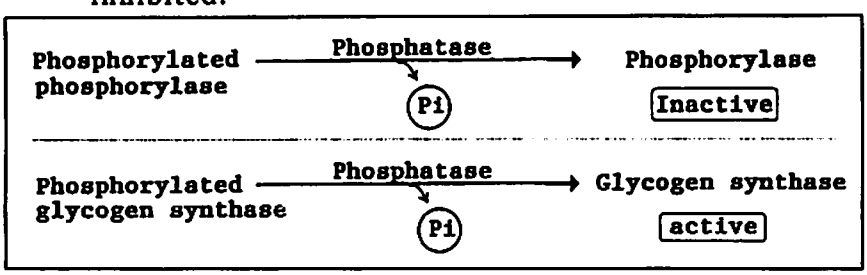
2. **After meal:**

a) Blood glucose level tends to be increased. This stimulates secretion of insulin hormone.

b) **Insulin causes the following:**

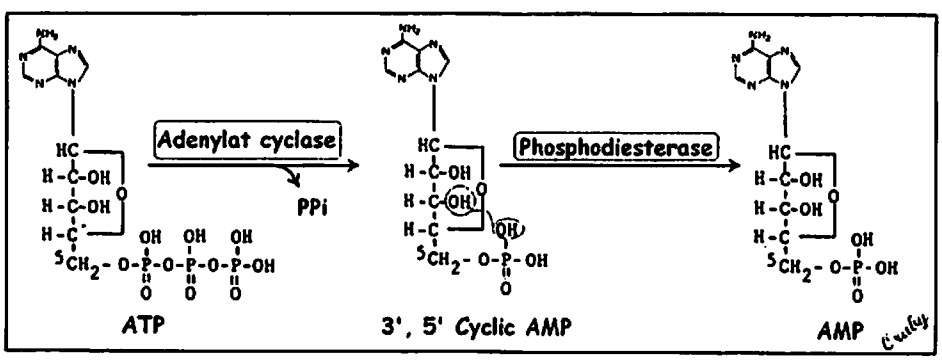
1) Stimulation of phosphodiesterase enzyme, which converts, cAMP into AMP i.e. abolishes the stimulatory effect of cAMP.

2) Stimulation of phosphatase enzyme, which removes phosphate from phosphorylase (inhibiting it) and glycogen synthase (stimulating it). As a result glycogenesis will proceed and glycogenolysis will be inhibited.

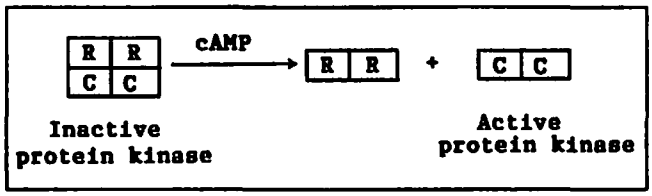


Notes:

1. **cAMP** is an intracellular compound , formed from ATP by an enzyme called : **adenylate cyclase** and destroyed by an enzyme called **phosphodiesterase** enzyme.



2. **Protein kinase** is an enzyme causing phosphorylation of substrate using ATP as a source of phosphate. **This enzyme is composed of 4 subunits:** 2 regulatory **(R)** and 2 catalytic **(C)**. The whole protein kinase is inactive, but binding of cAMP with it removes **(R)** subunits leaving the active catalytic subunit.



VII. *Differences between liver glycogen and muscle glycogen:*

	Liver glycogen	Muscle glycogen
Sources:	1-Blood glucose. 2-Other hexoses: e.g. fructose. 3-Non-carbohydrate sources : e.g. lactate	Blood glucose only.
Amount:	120 grams maximum	350 gram maximum
Concentration :	6%	1%
Functions:	It maintains normal blood glucose concentration between meals	private source of energy for muscles only
End product:	Glucose	Lactate (due to absence of glucose-6- phosphatase).
Effect of hormone:		
Insulin:	Stimulates glycogenesis	Same
Epinephrine:	Stimulates glycogenolysis	Same
Glucagon:	Stimulates glycogenolysis	**No effect**

VIII. *Glycogen storage diseases:*

A. Definition:

These are group of inherited disorders characterized by deposition of abnormal type or quantity of glycogen in the tissues.

B. Causes:

They are mainly due to deficiency of one of **enzymes of glycogen metabolism** e.g. glucose-6-phosphatase, debranching enzyme etc.

C. Types: (8 types):

1. Type one (I): Von Gierk`s disease:

a) It is due to deficiency of **glucose-6-phosphatase.**

b) It is the commonest type and characterized by:

1) Accumulation of large amount of glycogen in liver. This leads to **disturbance of liver functions.**

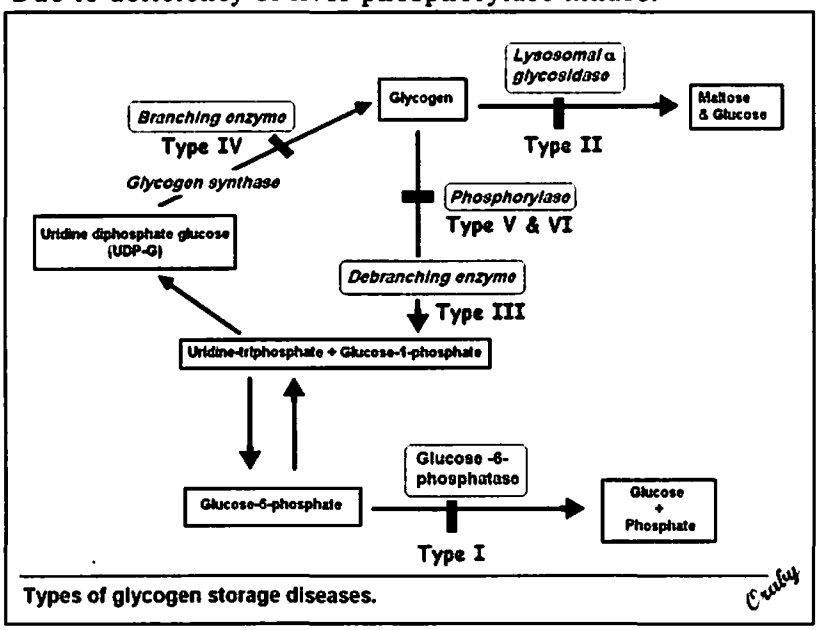
2) **Enlargment of liver** (hepatomegally).

3) **Fasting hypoglycemia.**

4) **Ketosis** and hyperlipidemia.

5) **Hyperuricemia (gout):** ↓ glucose-6-phosphatase ➔ ↑ Glucose-6-phosphate ➔ ↑ Pentose phosphate pathway ➔ ↑ Ribose production ➔ ↑ Uric acid ➔ Gout.

2. **Type two (II): Pompe`s disease:**
   a) Glycogen accumulates in lysosomes of all tissue cells.
   b) Normally, 1-3% of cellular glycogen is hydrolyzed by lysozomal glucosidase (acid maltase). Absence of this enzyme in lysosomes results in accumulation of glycogen in all tissues including heart. Death occurs during first year due to **heart failure.**
3. **Type three (III): Limit dextrosis (Cori`s disease):**
   a) Due to deficiency of **debranching enzymes** in liver, muscles and heart.
   b) Glycogen has many short branches.
4. **Type four (IV): Amylopectinosis:**
   a) It is due to absence of **branching enzyme.**
   b) Glycogen formed has no or few branches.
   c) Death due to cardiac or liver failure in first year of life.
5. **Type five (V): McArdle`s syndrome:**
   a) Due to deficiency of **muscle phosphorylase.**
   b) Glycogen is accumulated in muscles, and does not breakdown during exercise. This leads to decreased energy.
   c) **Muscle cramps** after short period of exercise.
6. **Type six (VI): Her`s disease :**
   a) Due to deficiency of **liver phosphorylase.**
   b) Glycogen is accumulated in liver.
   c) Fasting hypoglycemia.
7. **Type seven (VII): Tarui`s disease:**
   a) Due to deficiency of **phosphofructokinase** in muscle and RBCs.
   b) Hemolytic anemia and muscle cramps.
8. **Type eight (VIII):**
   a) Due to deficiency of liver phosphorylase kinase.



Types of glycogen storage diseases.

Gluconeogenesis

I. *Definition:*

Gluconeogenesis is the formation of glucose from non-carbohydrate sources. **These sources include:**

1. Lactate.
2. Pyruvate.
3. Glycerol.
4. Some amino acids.
5. Propionate (in ruminants only).

II. *Functions of gluconeogenesis:*

A. **Gluconeogenesis supplies the body with glucose:**

1. Glucose is the only source of energy for nervous tissues, RBCs and skeletal muscles during exercises.
2. Glucose is the precursor of milk sugar (lactose) in mammary gland.
3. Glucose is important during low carbohydrate diet or when liver glycogen is depleted (liver glycogen is depleted after 12-18 hours).

B. **Gluconeogenesis clears the blood** from the waste products of other tissues as lactate (produced by muscles and RBCs).

III. *Location of gluconeogenesis::*

A. **Intracellular location:** cytosol and mitochondria.

B. **Organ location:**

1. Liver (90%).
2. Kidney (10%).

IV. *Steps:*

The steps of gluconeogenesis are mainly the reversal of glycolysis, except for the three irreversible kinases which are replaced by the following enzymes:

Glycolysis	Gluconeogenesis
1. Glucokinase	1. Glucose-6-phosphatase
2. Phosphofructokinase- 1	2. Fructose 1,6 bisphosphatase
3. Pyruvate kinase	3. Pyruvate carboxylase
	4. Phosphoenol pyruvate carboxykinase

A. **Fructose 1,6 bisphosphate to fructose-6-phosphate:**

This reaction is catalyzed by the enzyme fructose 1,6 bisphosphatase.

B. **Glucose-6-phosphate to glucose:**

This reaction is catalyzed by the enzyme glucose-6-phosphatase.

C. **Pyruvate to phosphoenol pyruvate:**

1. This conversion is done by dicarboxylic acid shuttle and needs 2 enzymes:
   a) Pyruvate carboxylase: present in mitochondria.
   b) Phosphoenol pyruvate carboxykinase: present in cytosol.

2. Pyruvate should pass first from cytosol to mitochondria by special transporter.

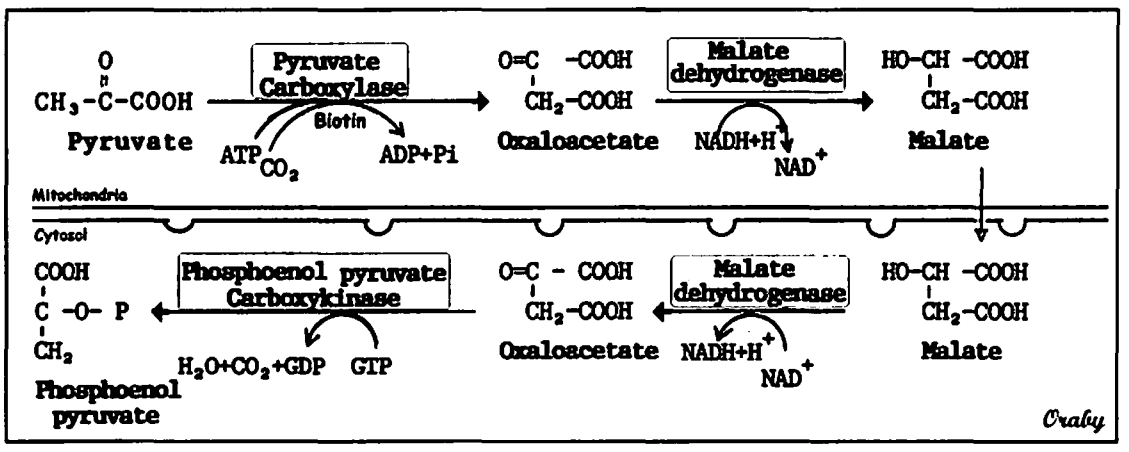
3. Pyruvate is then converted into oxaloacetate by pyruvate carboxylase (in the presence of biotin, CO_2 and ATP).

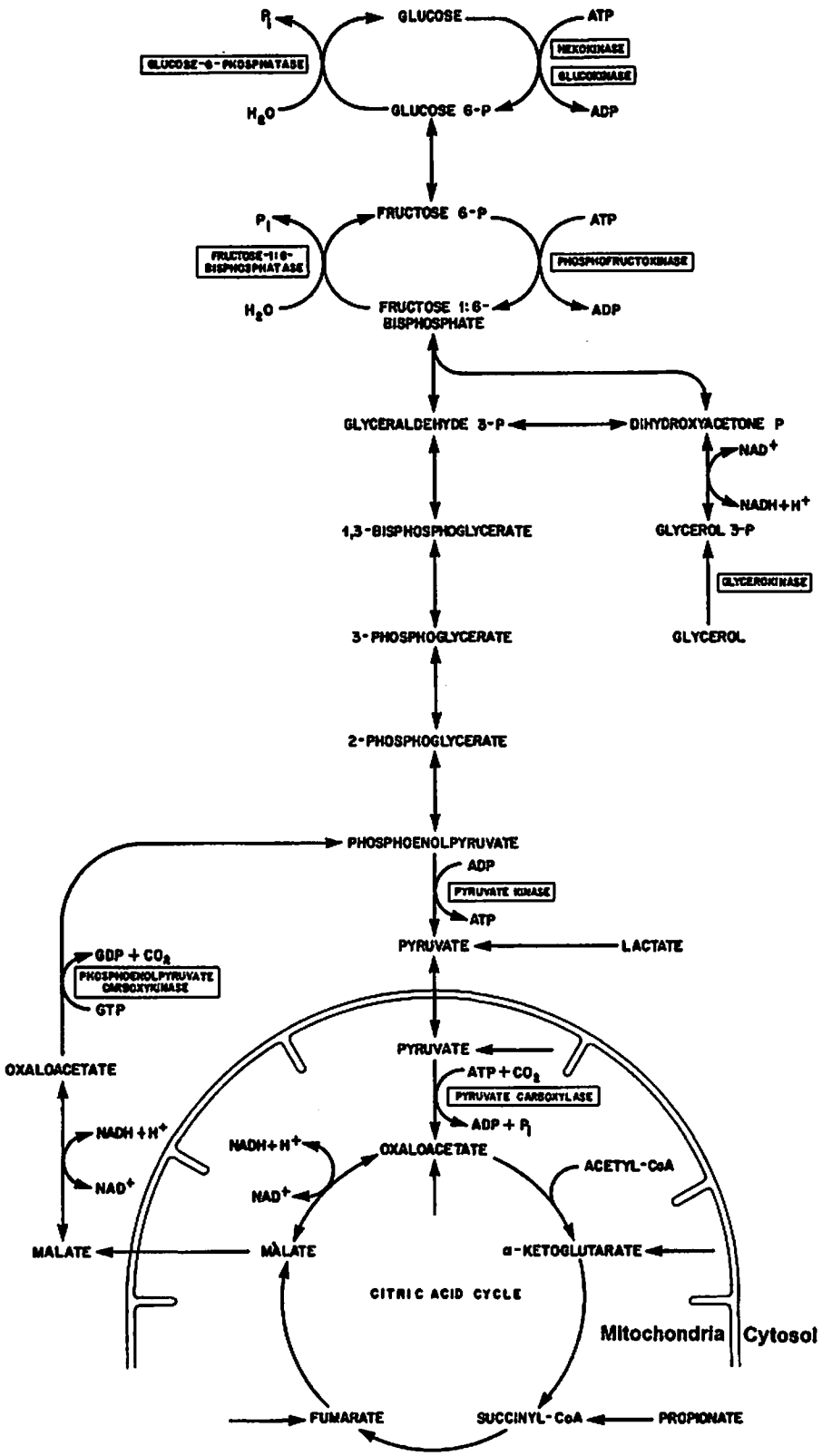
4. The mitochondrial membrane is impermeable to oxaloacetate. So oxaloacetate is converted to malate by malate dehydrogenase.

5. Malate is transported to cytosol, where it is converted again into oxaloacetate (by cytosolic malate dehydrogenase).

6. Oxaloacetate is converted into phosphoenol pyruvate by phosphoenolpyruvate carboxykinase (PEP).

Note: Pyruvate never goes in the course of citric acid pathway to reach malate, because this pathway needs insulin and other factors, which are deficient during gluconeogenesis.

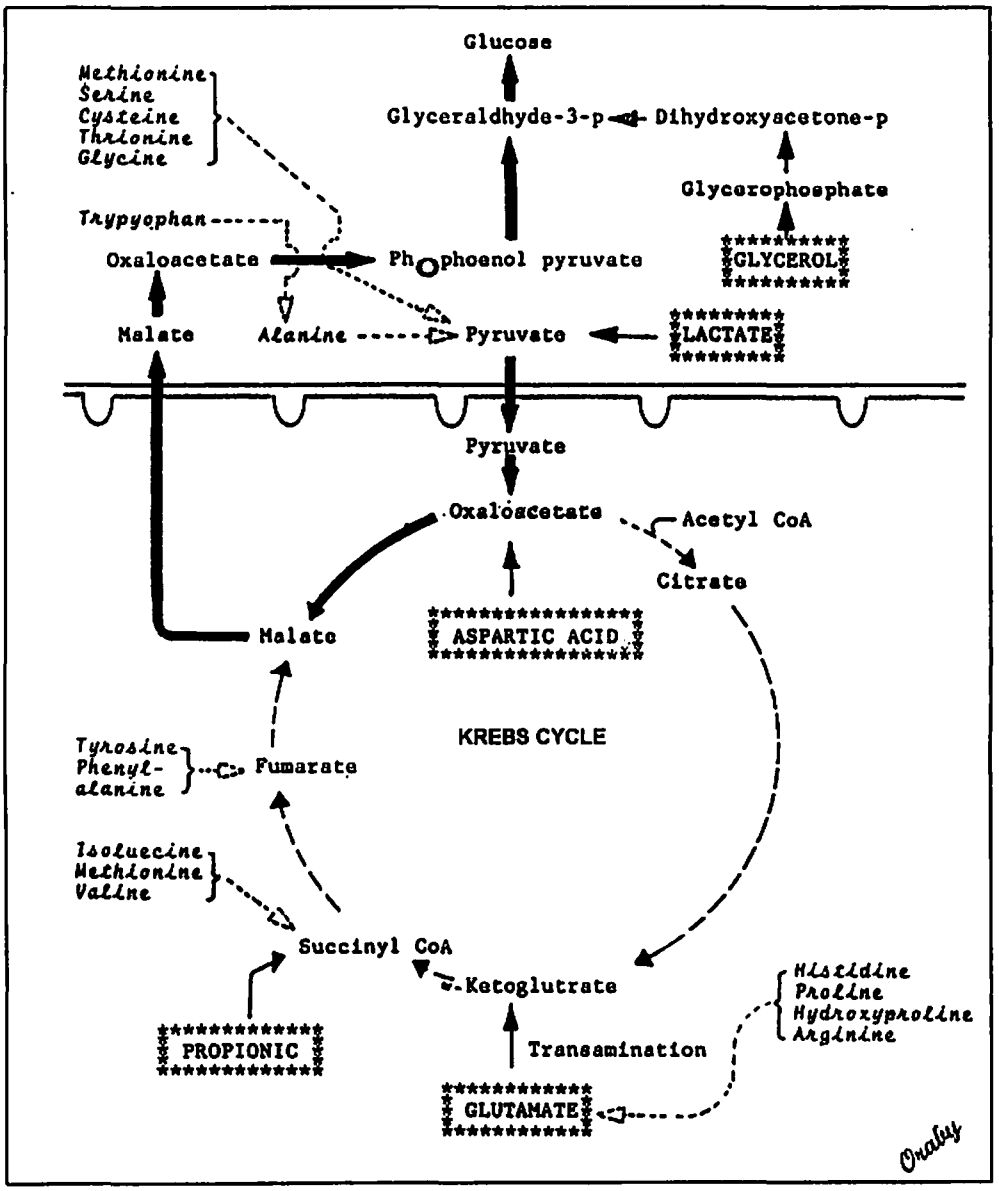




Major pathways in gluconeogenesis in the liver. Entry points of glucogenic amino acids after transamination are indicated by unlabeled arrows.

V. *Pathways for different sources of gluconeogenesis:*

Any substance that can join common pathway of gluconeogenesis is considered glycogenic.



A. Gluconeogenesis from lactate:

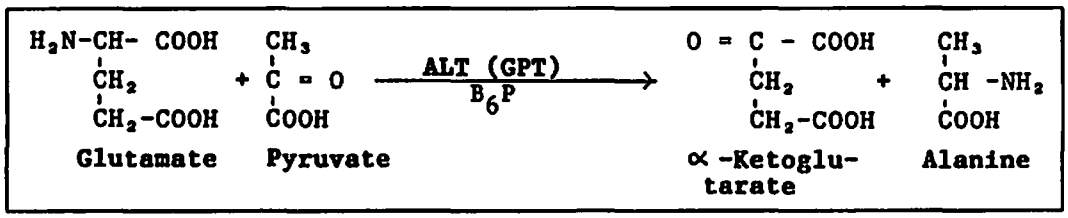
1. Lactate is converted into pyruvate by lactate dehydrogenase:

$$\text{Lactate} + NAD^+ \xrightarrow{\text{Lactate dehydrogenase}} \text{Pyruvate} + NADH + H^+$$

2. pyruvate then joins common pathway to give glucose (as indicated in the diagram).

B. Gluconeogenesis from glutamate:

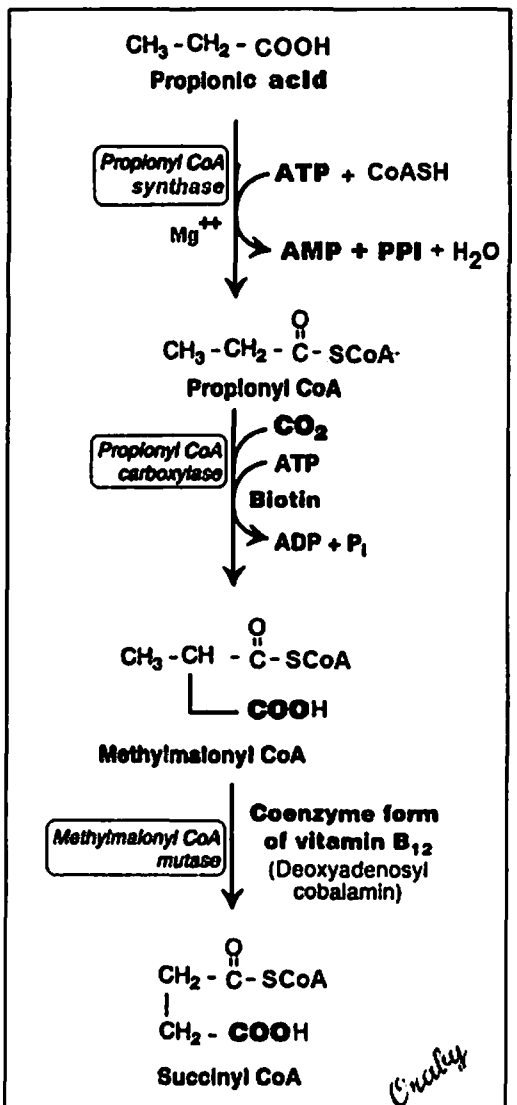
1. Glutamate is converted into α-ketoglutarate by transamination reaction:



2. α -Ketoglutarate is converted into malate as follows :
   α -Ketoglutarate → Succinyl CoA → Fumarate → Malate.
3. Malate then passes out the mitochondria and join common pathway to glucose (as indicated in the diagram).

C. Gluconeogenesis from propionic acid:

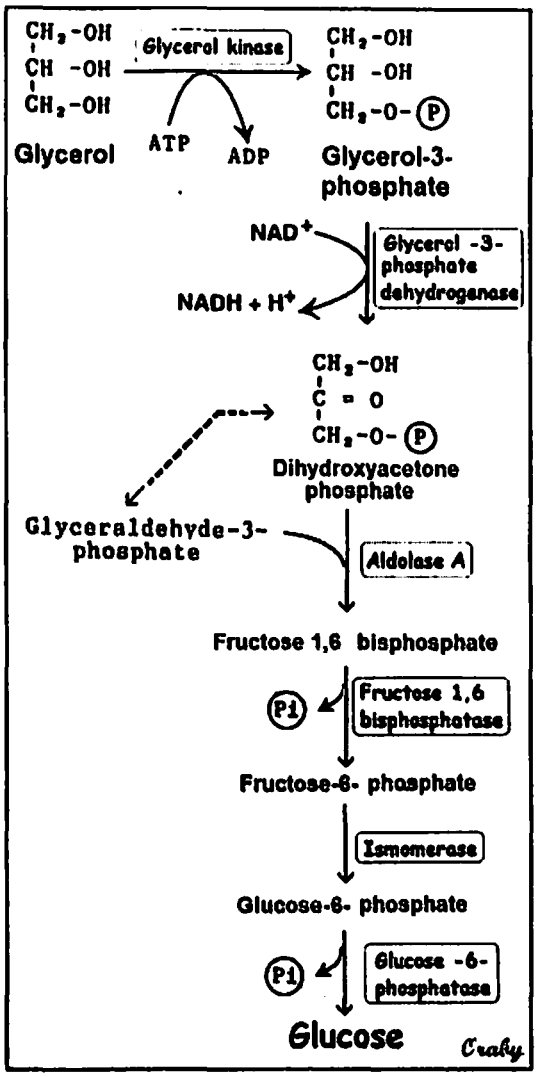
1. This occurs only in **ruminants** and not in human.
2. Propionic acid is converted into succinyl CoA as shown in the figure.
3. Succinyl CoA is then converted into malate:
   Succinyl CoA → Fumarate → Malate.
4. Malate passes out the mitochondria and join common pathway to glucose (as indicated in the diagram).



Gluconeogenesis from propionic acid.

D. Gluconeogenesis from glycerol:

1. Glycerol is mobilized from adipose tissue during fasting. Two molecules of glycerol are used to form glucose:



Note:

Two important cycles are related to gluconeogenesis: Cori cycle (see glycolysis, part II) and alanine glucose cycle (see protein metabolism, part III).

VI. *Energy cost of gluconeogenesis:*

Gluconeogenesis is an endergonic process (anabolic). For conversion of two molecules of pyruvate to one molecule of glucose, 4 molecules of ATP, 2 molecules of GTP and two molecules of NADH, H^+ are utilized as follows:

A. **ATP and GTP:**

Two pyruvate ➔ Two oxaloacetate (-2ATP)

Two Oxaloacetate ➔ Two phosphoenolpyruvate (-2 GTP)

Two 3 phosphoglycerate ➔ Two 1,3 Bisphosphoglycerate (-2 ATP)

B. **$NADH,H^+$:**

Two 1,3 Bisphosphoglycerate ➔ Two glyceraldhyde-3-phosphate (-2 NADH+H^+)

VII. *Regulation of gluconeogenesis:*

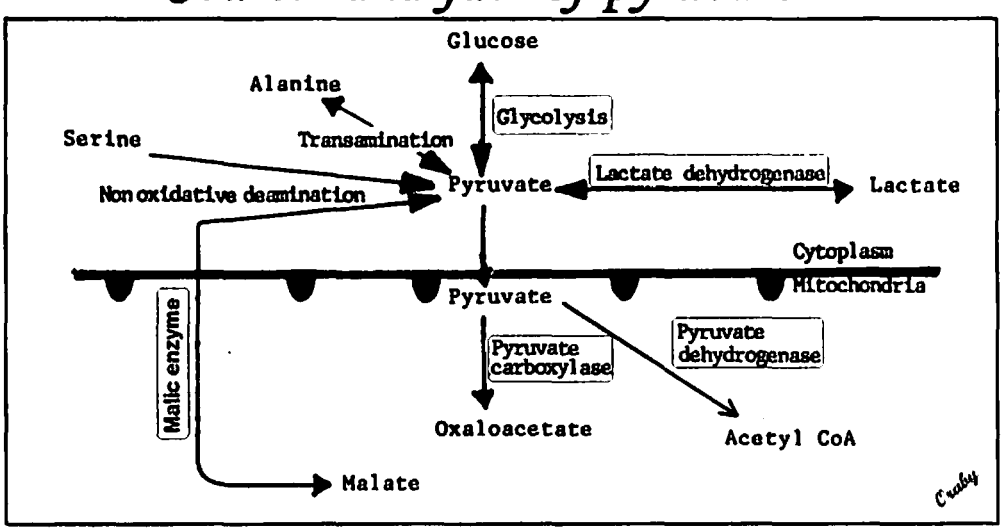
A. Hormonal regulation:

1. **Glucocorticoids e.g. *cortisol*:** stimulate gluconeogenesis by the following mechanisms:
   a) They **Induce: (stimulate)** the synthesis of gluconeogenesis enzymes which are: pyruvate carboxylase, phosphoenolpyruvate carboxykinase, fructose 1,6 bisphosphatase and glucose-6-phosphatase.
   b) Glucocorticoids stimulate protein catabolism by tissues → ↑ glycogenic amino acids available for gluconeogenesis.
2. **Glucagon**: Stimulates gluconeogenesis by lowering the level of fructose 2,6 bisphosphate (see regulation of glycolysis).
3. **Insulin:** Inhibits gluconeogenesis. It acts as repressor (inhibitor) for synthesis of enzymes of gluconeogenesis: pyruvate carboxylase etc.

B. Acetyl CoA and ATP:

1. Stimulate gluconeogenesis by inhibiting glycolysis (through inhibiting phosphofructokinase-1) and stimulate gluconeogenesis (by stimulating fructose 1,6 bisphosphatase).
2. Acetyl CoA also stimulates pyruvate carboxylase (gluconeogenesis) and inhibit pyruvate dehydrogenase (oxidation).

Sources and fate of pyruvate



A. Sources:

1. **Glucose oxidation:** glycolysis.
2. **Lactate:** by lactate dehydrogenase.
3. **Malate:** by malic enzyme.

4. **Alanine:** by transamination.
5. **Serine:** by non-oxidative deamination.
6. **Other amino acids:** methionine, cysteine, threonine and glycine.

B. **Fate:**

1. **Glucose formation:** gluconeogenesis.
2. **Lactate formation:** by lactate dehydrogenase.
3. **Malate formation:** by malic enzyme.
4. **Alanine formation:** by transamination.
5. **Oxaloacetate formation:** by pyruvate carboxylase.

Metabolism of monosaccharides

Galactose metabolism

I. ***Importance of galactose:*** In the form of UDP-galactose:

A. Synthesis of lactose (=milk sugar).

B. Synthesis of glycolipids (cerebrosides).

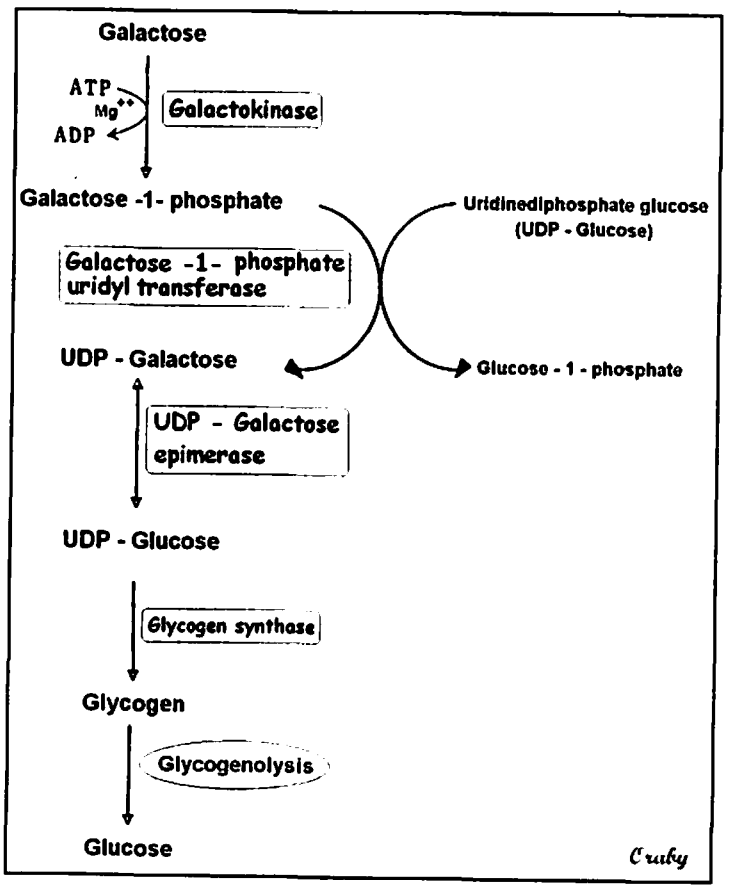
C. Synthesis of glycoproteins and proteoglycans.

D. Synthesis of glycosaminoglycans.

II. ***Conversion of galactose into glucose:***

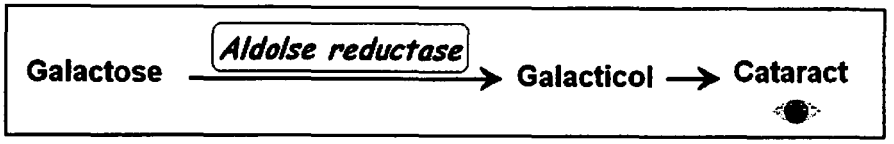
A. **Site:** Liver.

B. **Steps:**



C. **Galactosemia:**

1. **Definition:** It is increase blood galactose concentration due to inability to metabolize galactose.
2. **Causes:** Inherited enzyme deficiency of:
   a) Galactokinase.
   b) Galactose-1-P uridyl transferase.
   c) Epimerase.
3. **Effect:**
   a) **Cataract** (=opacity of eye lens):
   Galactose in the eye is reduced by an enzyme called aldose reductase into galacticol, which accumulates causing cataract.



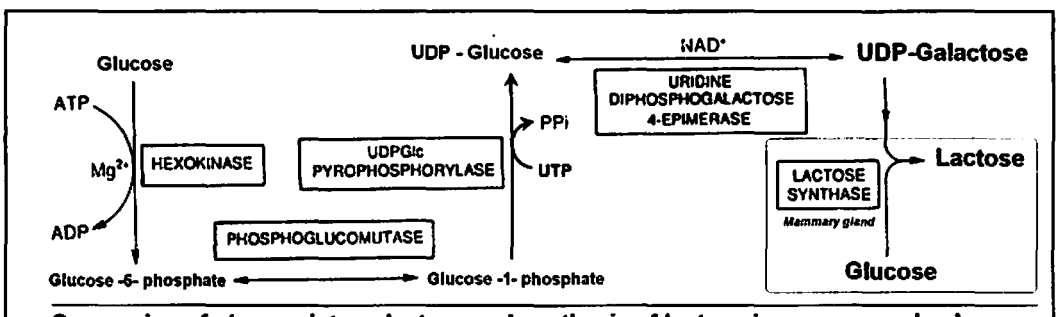
   b) Liver failure.
   c) Mental retardation.
   d) **Galactosuria**: excretion of galactose in urine.

III. *Conversion of glucose into galactose in mammary gland and lactose synthesis:*

Lactose is a disaccharide formed of β -galactose attached to α-glucose by β1-4 bonds. It is called milk sugar.

A. **Steps:**

1. Glucose is first converted into UDP-galactose.



Conversion of glucose into galactose and synthesis of lactose in mammary gland.

2. UPD-Galactose reacts with a molecule of glucose in the presence of lactose synthase enzyme to form lactose.

Fructose metabolism

I. Dietary Sources of fructose:

A. **Sucrose:** (table sugar): Hydrolysis of sucrose ➔ glucose and fructose.

B. **Fructose as a monosaccharide** is present in honey and in many fruits and vegetables.

II. Importance of fructose:

A. **Energy production:** 15% of daily energy is derived from fructose.

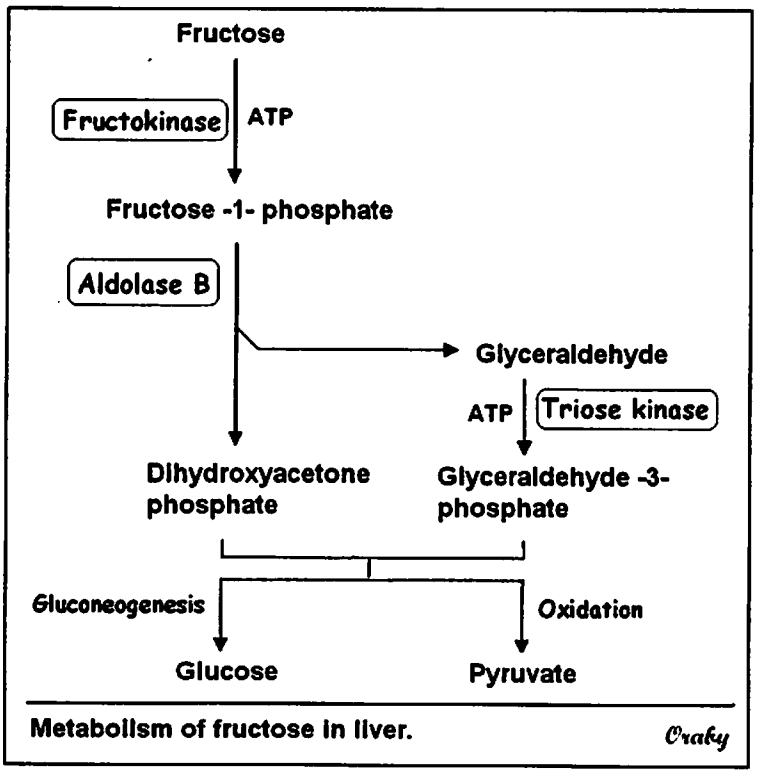
B. **Fructose is the major energy** source for spermatozoa in the seminal vesicle.

> **Note:** Entry of fructose to the tissue cells is not insulin dependent.

III. Metabolism of fructose:

A. **In the liver:**

1. **Liver contains fructokinase enzyme,** which phosphorylates fructose into fructose-1-phosphate.
2. Fructose-1-phosphate **by aldolase B enzyme** ➔ Dihydroxyacetone phosphate + glyceraldehyde.
3. Glyceraldehyde ➔ glyceraldehydes 3- phosphate.

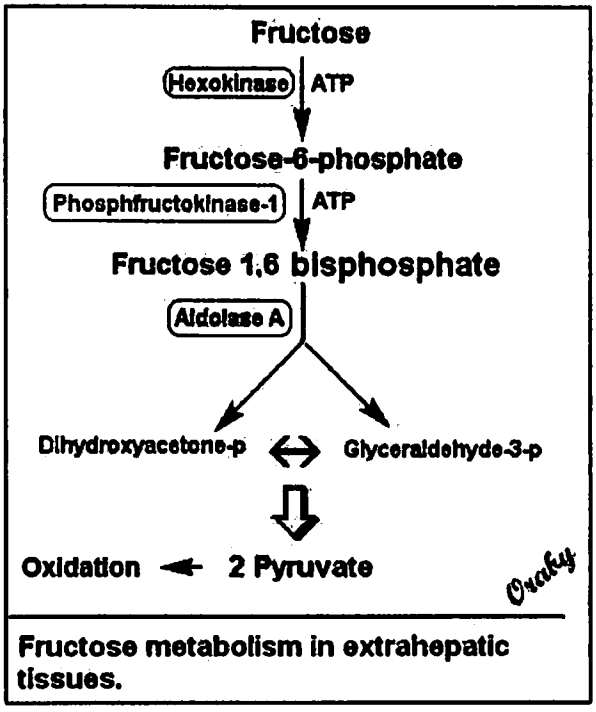


Metabolism of fructose in liver. Oraby

4. Glyceraldehyde-3-p + Dihydroxyacetone p may undergo:
   a) Glucose formation (gluconeogenesis): main pathway.
   b) Oxidation to pyruvate (glycolysis).

B. In extra-hepatic tissues:

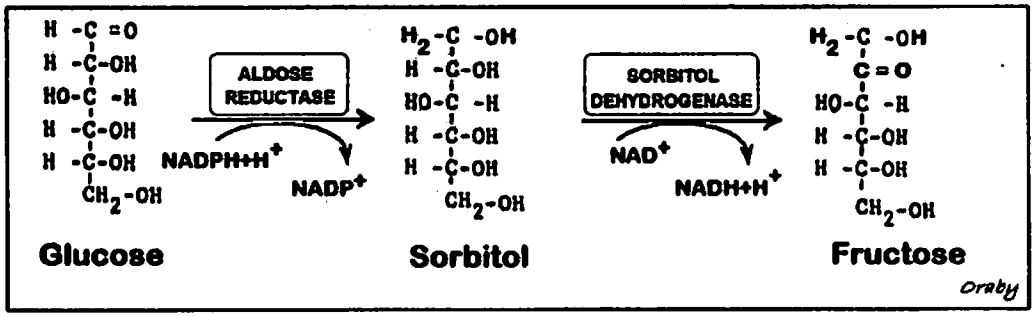
1. Because fructokinase is not available in muscles and adipose tissue, fructose is metabolized by hexokinase and other enzymes into pyruvate ➔ oxidation.



Fructose metabolism in extrahepatic tissues.

C. In the testis (seminal vesicle), lens, peripheral nerves and renal glomeruli:

1. Glucose is converted into fructose through sorbitol formation:



2. Fructose is the main **nutrient for sperms.**
3. Deficiency of fructose in semen correlates with male infertility.

IV. *Genetic disorders of fructose metabolism:*

A. **Essential fructosuria:**

1. **Cause:** Due to deficiency of ***fructokinase*** enzyme.
2. **Effect:** Not serious condition. The excess accumulated fructose is lost in urine (fructosuria).

B. **Hereditary fructose intolerance:**

1. **Cause:** Due to deficiency of ***aldolase-B*** enzyme. This leads to accumulation of fructose-1-phosphate.
2. **Effect:** the accumulation of fructose-1-phosphate leads to:
   a) Damage of liver and kidney tissues → Liver and kidney failure.
   b) Inhibition of phosphorylase enzyme. This leads to inhibition of glycogenolysis and hypoglycemia.

Blood glucose

I. *Plasma glucose levels:*

A. **Fasting level** = 65-110 mg/dl (3.6-6.1 mmol/L)

B. **One hour after carbohydrate meal** = 120-150 mg/dl (6.7-8.3 mmol/L).

C. **Two hours after carbohydrate meal (PP)**: 65-140 mg/dl (3.6-7.8 mmol/L).

Note: For glucose 1 millimole/L (mmol/L) = 18 mg/dl.

II. *Sources of blood glucose:*

A. **Dietary carbohydrate:** absorbed glucose following digestion of carbohydrate e.g. starch.

B. **Liver glycogen:** through glycogenolysis. Liver glycogen can supply body with glucose for about 18 hours of fasting.

C. **Amino acids and other metabolites:** (gluconeogenesis): liver and kidney can convert these substrates into glucose.

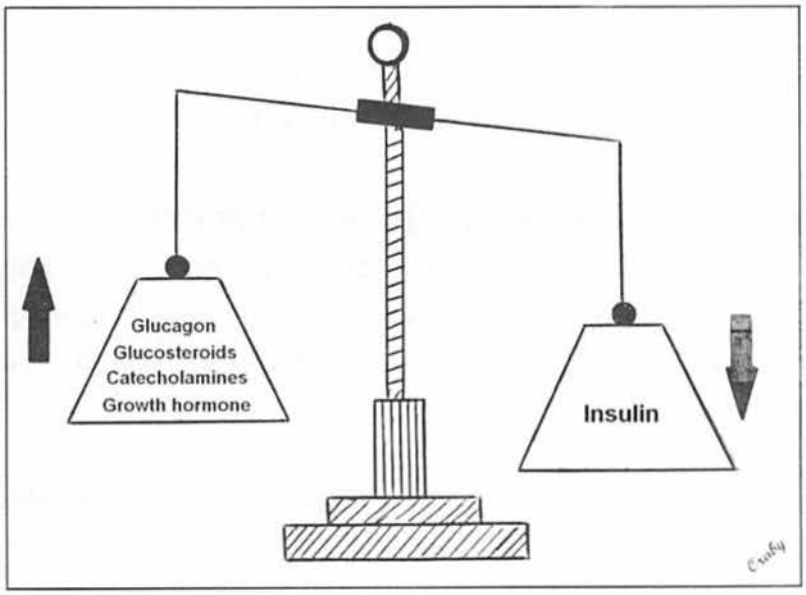
III. *Regulation of blood glucose level:*

A. Blood glucose level is maintained within narrow range 65-150 mg/dl. This is because:

1. Hypoglycemia (low blood glucose level) causes impairment of cerebral function, as brain is very dependent on blood glucose for its energy supply.

2. Hyperglycemia (high blood glucose level) can also cause cerebral dysfunction by its effect on extra cellular osmolality (discussed later).
3. Regulation of blood glucose level can be achieved by hormonal, hepatic and renal mechanisms.

B. Hormonal regulation:



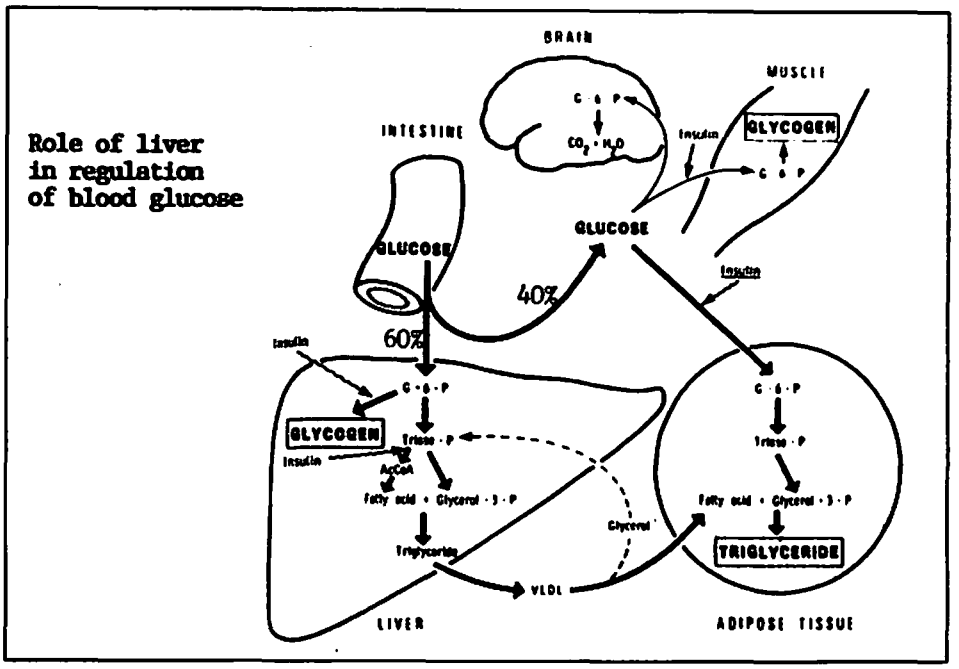
1. **Insulin** is a hormone secreted by beta cells of islets of Langerhans of pancreas (see chapter of hormones). It is the only hormone which reduces blood glucose level through:
   a) **Transfer** of glucose inside the cells.
   b) **Stimulation** of glucose oxidation.
   c) **Stimulation** of glycogen storage (glycogenesis).
   d) **Inhibition** of glycogen breakdown (glycogenolysis).
   e) **Inhibition** of conversion of amino acids and other metabolites into glucose (gluconeogenesis).
   f) **Stimulation** of conversion of glucose into fat (lipogenesis).
2. **Glucagon:** is a hormone that secreted by alpha cells of islets of Langerhans of pancreas. It tends to increase blood glucose level through:
   a) Stimulation of glycogen breakdown (glycogenolysis): by stimulating adenylate cyclase enzyme.
   b) Stimulation of gluconeogenesis.
3. **Catecholamines (epinephrine and norepinephrine):** They are secreted from suprarenal medulla. They tend to increase blood glucose level even if there is hyperglycemia as in

case of stress. They increase blood glucose level through stimulation of glycogenolysis.

4. **Glucocorticoids as cortisol:** They are a group of steroid hormones secreted by suprarenal cortex. They tend to increase blood glucose level through stimulation of gluconeogenesis.
5. **Growth hormone:** It is a hormone secreted from anterior pituitary gland. It antagonizes insulin hormone and tends to increase blood glucose level through:
   a) Inhibition of glucose uptake by cells.
   b) Blocks the insulin action at cell membranes.

C. **Hepatic regulation (role of liver):** Liver plays a very important role in keeping blood glucose level within range. Simply during fasting, liver adds glucose to the blood (glycogenolysis). After carbohydrate meal, it takes up about 60% of the glucose load (glycogenesis).

1. **During fasting:**
   a) Liver adds glucose to blood by glycogenolysis and gluconeogenesis.
   b) It can convert fatty acids (acetyl CoA) released from adipose tissue to ketone bodies, which can be used by other tissues, including brain when glucose is in short supply.
2. **After carbohydrate meal:** Glucose is transported in portal blood to the liver.
   a) About **40%** passes to the blood stimulating insulin secretion and taken up by brain, muscles, adipose tissue and liver.
   b) The remaining **60%** of glucose is taken up by liver, and converted into glucose-6-phosphate, by **glucokinase enzyme:**
      1) This enzyme has **low affinity** for glucose relative to hexokinase found in most tissues.
      2) Glucokinase is stimulated **(induced) by insulin** secreted in response to hyperglycemia. For this reason, less glucose is taken up by liver during fasting.
      3) Glucose-6-phosphate will undergo one of the following **fate in the liver:**
         - Conversion to glycogen (glycogenesis) which is stimulated by insulin.
         - Conversion to fatty acids (lipogenesis). It is also stimulated by insulin.



D. **Renal regulation**: (role of kidney):

1. Circulating blood glucose is filtered in glomerular filtrate and reabsorbed again by certain tubular enzymes. This saves blood glucose from loss in urine .
2. If blood glucose level exceeds certain limits (average 180 mg/dl), glucose will increase in glomerular filtrate and exceeds the capacity of tubular enzymes to reabsorb it. Thus glucose appears in urine (glycosuria). This limit is called renal threshold for glucose reabsorption.

Definition of renal threshold for glucose reabsorption: It is the blood glucose level above which glucose appears in urine. (Average **180 mg/dl**).

3. **Low renal threshold:** Where glucose appears in urine even when blood glucose level is low (may be just 100 mg/dl). **This occurs in:**
   a) Normally in some persons due to defective tubular enzymes for glucose reabsorption (diabetes innocence).
   b) In 20% of pregnant females.
4. **High renal Threshold:** Where glucose appears in urine in concentration above the average "renal threshold" (which may reach 220 mg/dl or more). This occurs in:
   a) Elderly people due to reduced glomerular filtration rate.
   b) In cases of diabetes associated with renal damage.

Variations in blood glucose

Hypoglycemia

I. Definition:

A. It is the decrease of blood glucose concentration below normal fasting average concentration: **less than 65 mg/dl. Symptoms** of hypoglycemia appear if blood glucose concentration becomes less than**45 mg/dl.**

II. Effect of hypoglycemia:

A. Hypoglycemia causes cerebral dysfunction (as brain and nervous tissue are dependent on glucose as a source of energy).

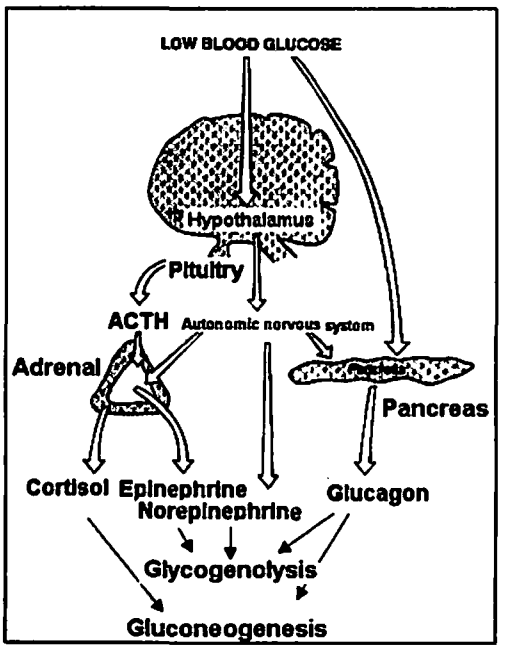
B. If hypoglycemia is severe and prolonged, it causes coma (hypoglycemic coma) and then death.

III. Mechanism that corrects hypoglycemia: Hypoglycemia activates:

A. **α Cells of islets of Langerhans** ➔ Glucagon ➔ Glycogenolysis ➔ ↑ Blood glucose.

B. **Receptors in hypothalamus:** This stimulates:

1. Secretion of **epinephrine** (mediated by autonomic nervous system) ➔ ↑ Glycogenolysis ➔ ↑ Blood glucose.
2. Secretion of **anterior pituitary** hormones:
   a) **Growth hormone:** insulin antagonist.
   b) **ACTH** ➔ Stimulation of suprarenal cortex ➔ Glucocorticoids ➔ ↑ Gluconeogenesis ➔ ↑ Blood glucose.

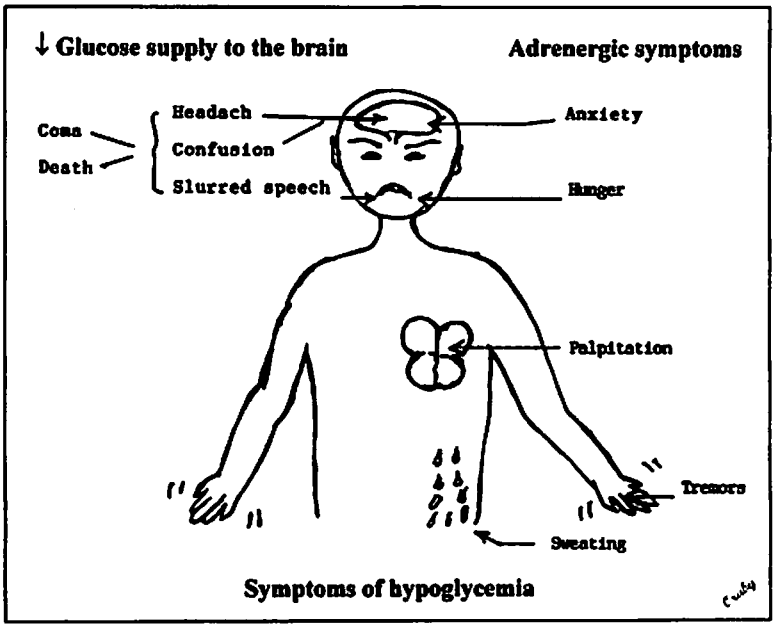


Note:

1) **Glucagon and epinephrine are most important in the acute, short-term regulation of blood glucose levels.**

2) **Glucocorticoids and growth hormone play a role in long-term regulation of blood glucose levels.**

IV. *Symptoms of hypoglycemia:*



A. Adrenergic symptoms (that is, symptoms mediated by elevated epinephrine) ➔ *Anxiety, palpitation, tremors and sweating and hunger.*

B. Symptoms due to impaired glucose supply to the brain ➔ Impairment of brain function ➔ *Headache, confusion, slurred speech, coma and death.*

V. *Types of hypoglycemia:* It may be divided into 2 types, fasting and stimulative.

A. Fasting hypoglycemia:

1. It is the inability to maintain normal glucose concentration in the fasting state.
2. It is usually due to organ diseases.
3. **Causes:**
   a) **Pancreatic disease:** disorders affecting β-cells of islets of Langerhans e.g. ***insulinoma*** (=Pancreatic β cell tumor) ➔ ↑↑ insulin secretion.
   b) **Liver diseases:** leads to ↓ glycogenolysis and ↓ gluconeogenesis:
      1) Hepatocellular damage e.g. hepatic carcinoma
      2) Glycogen storage diseases.
      3) Prolonged starvation: due to depletion of liver glycogen.

c) **Adrenocortical diseases:** ➔ lead to decreased secretion of epinephrine (↓ glycogenolysis) and glucocorticoids (↓ gluconeogenesis).

B. Stimulative (reactive) hypoglycemia:

1. It occurs due to some stimuli usually after taking a meal.
2. **Causes:**

a) **Drugs and poisons:**

1) **Therapeutic insulin administration:** over dosage of insulin during treatment of diabetes can lead to hypoglycemia.
2) **Over dosage of sulphonylurea:** it is an oral hypoglycemic drug that stimulates β-cells to produce insulin.
3) **Liver poisons:** as chloroform and phosphorus.
4) **Alcohol (ethanol) ingestion:**
   - Excessive alcohol ingestion is a common cause of hypoglycemia.
   - Metabolism of alcohol ➔ ↑ $NADH+H^+$ levels in the liver ➔ Inhibits gluconeogenesis ➔ Hypoglycemia.

b) **Postgastrectomy:** i.e. after partial removal of stomach. Glucose is absorbed rapidly which leads to stimulation of excessive insulin secretion.

c) **Leucine hypersensitivity of infancy and childhood:** Leucine amino acid in the diet stimulates β-cells to secrete insulin. In sensitive infants, there is excessive insulin release.

d) **Inborn errors of metabolism:** as in:

1) **Galactosemia:** due to deficiency of galactose-1-phosphate uridyl transferase.
2) **Hereditary fructose intolerance:** due to deficiency of aldolase B enzyme ➔ Inhibition of phosphorylase ➔ Inhibition of glycogenolysis.

Types and causes of hypoglycemia:

Types	Organ or cause	Examples
Fasting hypoglycemia	**Pancreas**	Insulinoma.
	Other endocrine glands	Adrenocortical hypofunction.
	Liver	Prolonged starvation. Hepatocellular damage. Glycogen storage diseases.
Stimulative Hypoglycemia	**Drugs and poisons**	↑dose of insulin or ↑dose of sulphonylurea. Liver poisons: chloroform, Alcohol (ethanol).
	Postgastrectomy **Leucine hypersensitivity**	
	Inborn errors of metabolism	• Galactosemia • Hereditary fructose intolerance.

Hyperglycemia

I. ***Definition:***

It is the rise of blood glucose above normal average level.

II. ***Causes:***

1. **Diabetes mellitus**: *Most common cause* (discussed later).
2. In patients receiving intravenous fluid containing glucose.
3. Temporarily in severe stress.
4. After cerebro-vascular accidents.
5. Disturbance in hyperglycaemic hormones.

Diabetes mellitus

I. ***Definition:***

A. It is an endocrinc disease caused by a relative or absolute **deficiency of insulin** hormone.

B. It is characterized by a **chronic hyperglycemia.**

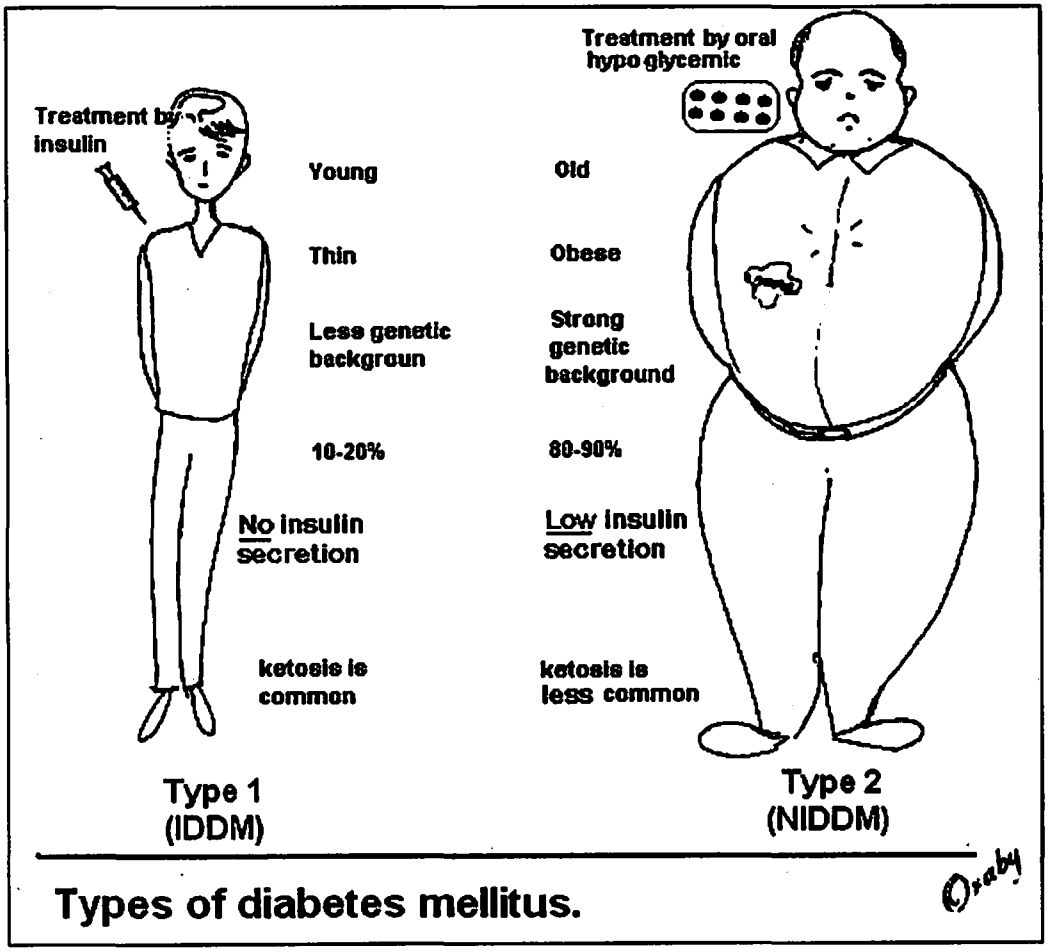
C. **Glycosuria** (presence of glucose in urine) is usually present.

Note: The word DIABETES means increase urine volume.

II. ***Types of diabetes:*** 2 Types; insulin dependent and non-insulin dependent diabetes mellitus.

Comparison of two types of diabetes mellitus

	Type 1 Insulin-dependent diabetes mellitus (IDDM)	Type 2 Non-insulin-dependent diabetes mellitus (NIDDM)
Synonym (other names)	juvenile-onset diabetes	adult-onset diabetes
Age of onset	Usually during childhood or puberty **(young)**	Frequently after age 35 **(old).**
Nutritional status at time of onset of disease	Frequently undernourished **(thin)**	Obesity usually present
Prevalence	10-20% of diagnosed diabetics	80-90% of diagnosed diabetics
Genetic predisposition	Moderate	Very strong
Defect or deficiency	β-cells destruction **(no insulin).**	Inability of β-cells to produce appropriate quantities of insulin or insulin resistance.
Ketosis	Common	Rare
Plasma insulin	Low to absent	Normal to high
Acute complications	Ketoacidosis	Hyperosmolar coma
Oral hypoglycemic drugs	No response	Responsive
Treatment with insulin	Always necessary	usually not required

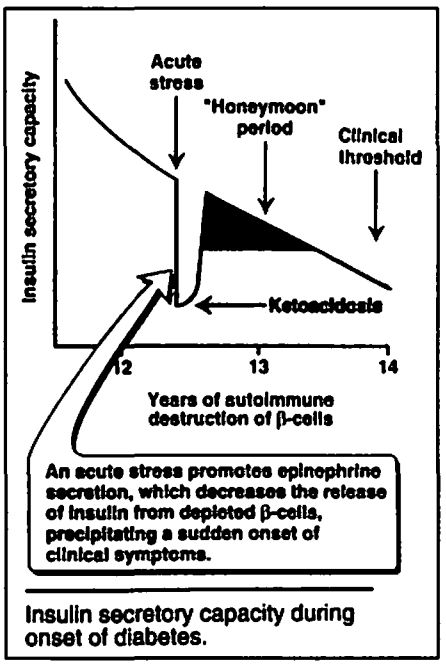


Types of diabetes mellitus.

A. **Insulin-dependent (type I) diabetes mellitus (IDDM):** This type is less common (10-20%) and characterized by an **absolute deficiency** of insulin caused by:

1) **Destruction of β-cells** of the pancreas.
2) This destruction may be due to **viral infection** or formation of **antibodies against β cells.**

- Patients of this type are usually
  - **young** and
  - **develop ketosis.**



Insulin secretory capacity during onset of diabetes.

B. **Non-Insulin dependent (type II) diabetes mellitus (NIDDM):** This type is the most common form of the disease (80-90%) and characterized by a relative deficiency of insulin caused by:

1) **Dysfunction of β-cells** of the pancreas.
2) **Resistance to insulin** action at cellular level. This may explain why some patients of NIDDM show normal or even elevated plasma insulin.

- Patients of this type are usually
  - **Old** and
  - **Obese** at the onset of diabetes.
  - Strong **genetic** background

III. ***Biochemical disturbance of diabetes mellitus:*** insulin deficiency causes disturbance in carbohydrate, lipids, protein, vitamins and minerals metabolism.

A. **Carbohydrate metabolism:** insulin deficiency leads to decrease glucose uptake by tissues, ↓ glucose oxidation, ↑ gluconeogenesis and ↑ Glycogenolysis. This leads to:

1. ↓ **Intracellular glucose** ➔ hunger pain ➔ **Polyphagia (excessive eating).**
2. ↑ **Blood glucose (hyperglycemia),** this causes ↑ **Plasma osmolality** ➔ dehydration:
   a) Dehydration of brain cells ➔ Hyperglycemic, hyperosmolal coma.

b) Dehydration of body cells → Sense of thirst ➔ Excessive drinking **(polydepsia).**

c) Symptoms and sign of dehydration.

3. **Glycosuria:** If blood glucose level exceeds renal threshold ➔ Glucose appears in urine **(glycosuria).** This will leads to ↑ osmotic diuresis that causes:
   a) Excessive and frequent urination **(polyurea).**
   b) Loss of water soluble vitamins e.g. B_1.
   c) Loss of minerals e.g. Na^+ and K^+.

B. **Protein metabolism:** insulin deficiency leads to increased protein breakdown and stimulation of gluconeogenesis. i.e. conversion of amino acids to glucose. This results in :

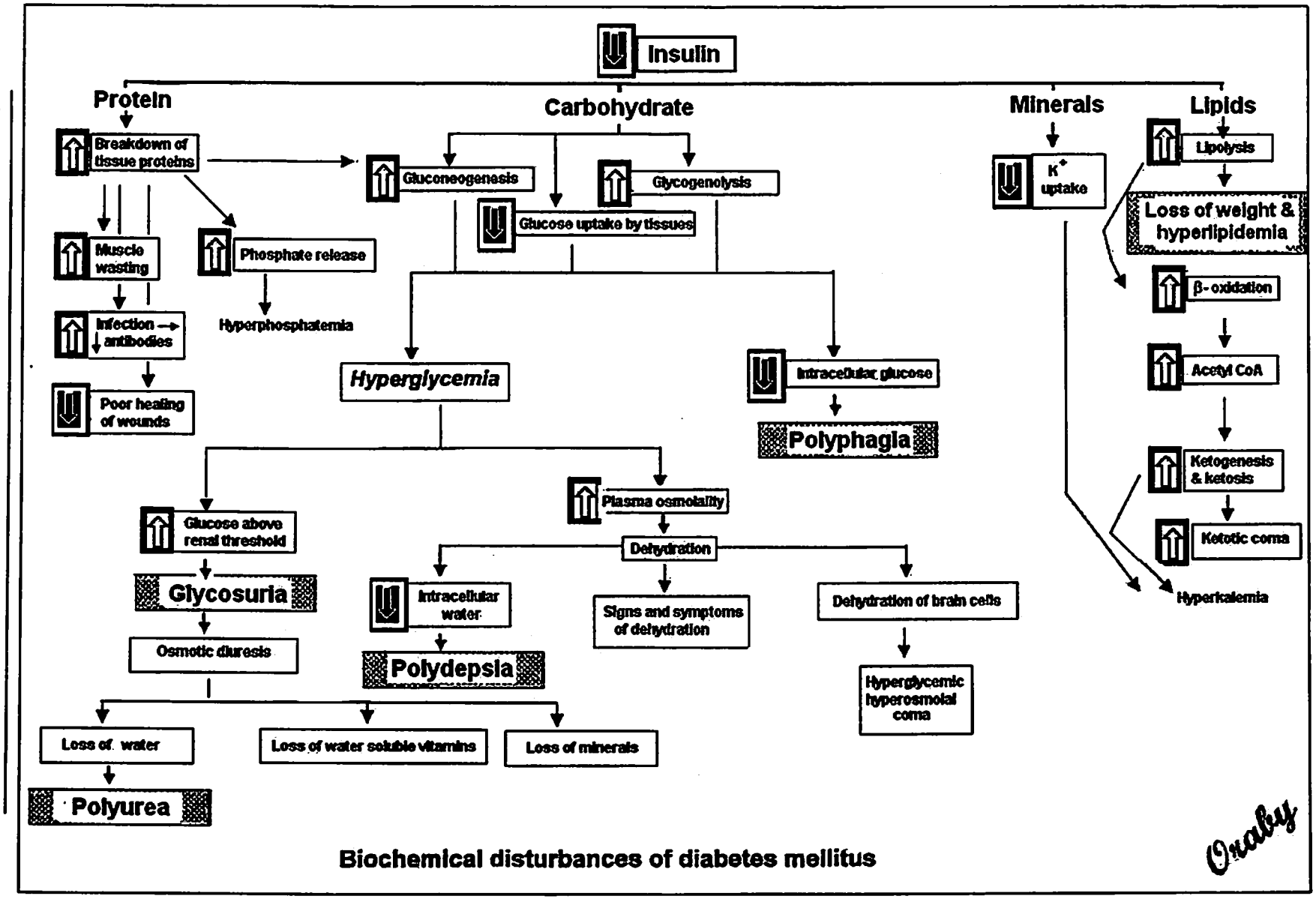
1. Phosphate release ➔ **hyperphosphatemia.**
2. Excessive breakdown of tissue proteins causing **muscle wasting.**
3. Decreased **antibody formation** causing low resistance and infection.
4. **Poor healing of wounds.**

C. **Lipid metabolism:** Insulin deficiency leads to excessive lipolysis in adipose tissue ➔ mobilization of free fatty acids and glycerol to the blood ➔ Then to the liver and other tissues. This leads to:

1. Loss of weight.
2. Hyperlipidemia ➔ Atherosclerosis.
3. Fatty liver (Re-estrification of fatty acids again with glycerol to form triacylglycerol in liver).
4. Excessive ketone bodies formation ➔ Ketonemia ➔ Ketosis. This leads to:
   a) Coma: ketotic coma.
   b) Hyperkalemia.

D. **Microangiopathy:**

1. It is a degeneration that affects small blood vessels as capillaries especially those of the kidneys and retina of the eyes.
2. Thus two of common **chronic complications** of diabetes mellitus are **renal failure** and **blindness.**


Insulin
Protein
Carbohydrate
Minerals
Lipids
Breakdown of tissue proteins
Gluconeogenesis
Glycogenolysis
Glucose uptake by tissues
K+ uptake
Lipolysis
Loss of weight & hyperlipidemia
Muscle wasting
Phosphate release
Hyperphosphatemia
Infection → ↓ antibodies
Poor healing of wounds
Hyperglycemia
Intracellular glucose
Polyphagia
β- oxidation
Acetyl CoA
Ketogenesis & ketosis
Ketotic coma
Hyperkalemia
Glucose above renal threshold
Glycosuria
Osmotic diuresis
Plasma osmolality
Dehydration
Intracellular water
Polydepsia
Signs and symptoms of dehydration
Dehydration of brain cells
Hyperglycemic hyperosmolal coma
Loss of water
Loss of water soluble vitamins
Loss of minerals
Polyurea
Biochemical disturbances of diabetes mellitus

IV. *Diagnosis of diabetes mellitus:*

Symptoms:
1- Fasting plasma glucose.
2- Postprandial plasma glucose.
No symptoms:
1- OGTT

A. **If a patient has symptoms** of diabetes mellitus, the following tests may be done:

1. ***Measurement of fasting plasma glucose level*** (8-12 hours after last meal):
   a) If concentration is 65-110 mg/dl → **Normal blood glucose.**
   b) If concentration is 110-126 mg/dl → **Impaired glucose utilization.**
   c) If concentration is > 126 mg/dl → **Diabetes mellitus.**
2. ***Measurement of two-hours postprandial plasma glucose level:*** (2 hours after last meal):
   a) If concentration is 65-140 mg/dl → **Normal blood glucose.**
   b) If concentration is 140-200 mg/dl→**Impaired glucose utilization.**
   c) If concentration is > 200 mg/dl → **Diabetes mellitus.**

	Glucose concentration (mg/dl)	
	Fasting	**Two hours after last meal**
Normal	65-110 mg/dl	65-140 mg/dl
Impaired glucose tolerance	110-126 mg/dl	140-200 mg/dl
Diabetes mellitus	More than 126 mg/dl	More than 200 mg/dl

Summary of diagnostic criteria for patients with symptoms of diabetes mellitus (WHO criteria).

B. **If patient has no symptoms:** ***Oral glucose tolerance test (OGTT) is done:***

1. **Indication of the test:** There are only two indications for performing OGTT:
   a) People with no or mild symptoms of diabetes mellitus.
   b) To determine the renal threshold for glucose.
2. **Procedure:**
   a) Fasting plasma sample is taken after overnight fast (8-12 hours).
   b) Then patient is given 75 grams glucose orally.
   c) Plasma and urine glucose concentrations are determined in fasting and at 30 minutes intervals for the next 3 hours after glucose ingestion. Then draw a curve.
3. **Explanation of normal glucose tolerance curve:**
   a) **Fasting level:** 65-110 mg/dl.

b) The plasma glucose level rises and reaches the maximum in one hour (120-150 mg/dl). Normally the maximum value equals 1.5 fasting value. ***The ascending limb of the curve represents glucose absorption.***

Thus any cause impairing glucose absorption as diarrhea will make the glucose level reaches the maximum in much slower rate (more than one hour).

c) **The plasma glucose returns to fasting level after 2 hours:**

1) ***The descending limb represents glucose utilization by the tissues in response to insulin secretion.*** In diabetes mellitus, glucose returns to fasting level in much slower rate (more than 2 hours).
2) Insulin secretion continues till the plasma glucose level becomes less than normal fasting level. Then insulin secretion stops and plasma glucose level returns once more to normal fasting level. This is called **hypoglycemic response or insulin overshoot.** The return to normal level is due to secretion of glucagon and epinephrine in response to low plasma glucose level.

d) Normally, **all urine samples contain no glucose** as all plasma glucose levels are below the normal threshold.

4. **WHO criteria for diagnosis of diabetes mellitus using oral glucose tolerance test :**

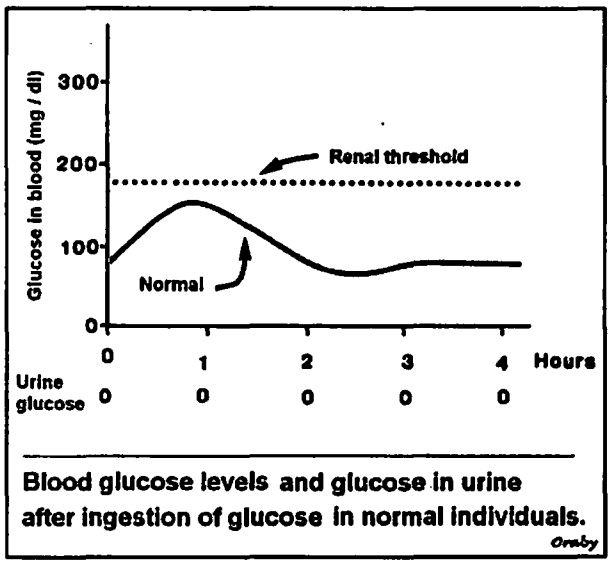
a) fasting plasma glucose is greater than 126 mg/dl (7 mmol/L) .
b) At least, one of the intermediate (30, 60, 90 min) plasma specimens has plasma glucose greater than 200 mg/dl.
c) At any urine specimen, if the rate of glomerular filtration of glucose exceeds that of tubular reabsorption in the kidney (renal threshold), glucose will appear in urine.

5. **Glucose tolerance curve of renal glycosuria:**

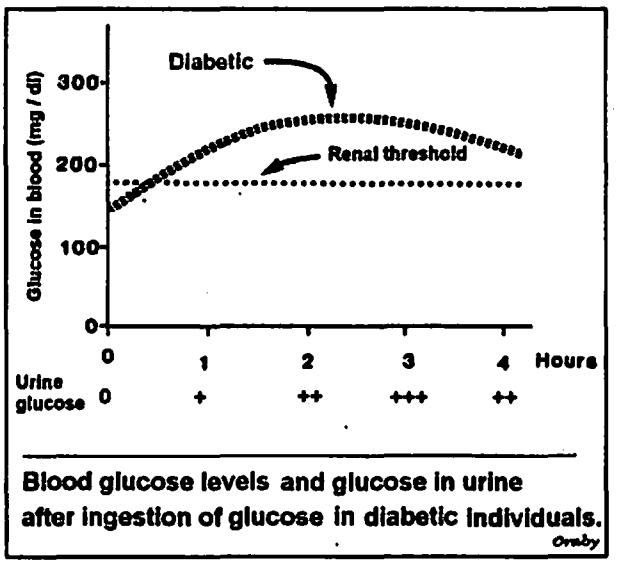
a) The curve is normal, indicating normal glucose utilization and no insulin deficiency.

b) Abnormalities lie in appearance of glucose in some or all urine specimen.

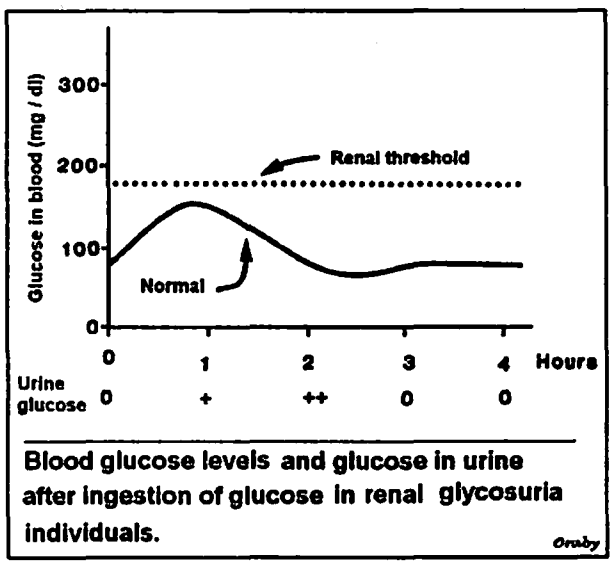
6. **Hypoglycemic glucose tolerance curve:** Fasting level is below normal, maximum rise is below normal and returns to fasting level are very rapid.



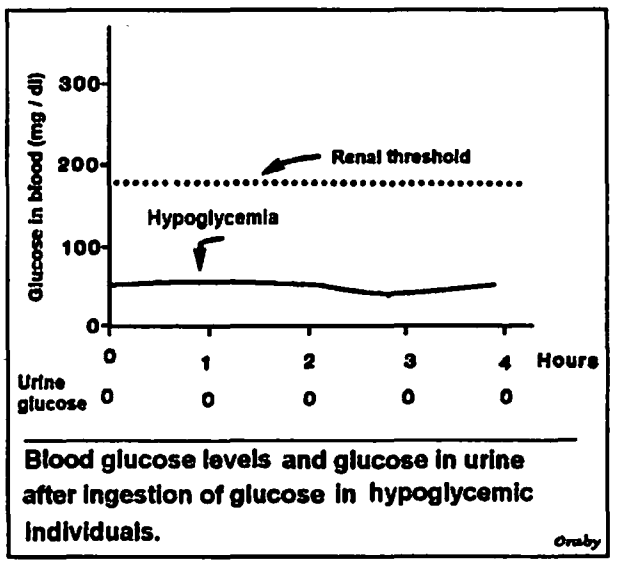
Blood glucose levels and glucose in urine after ingestion of glucose in normal individuals.



Blood glucose levels and glucose in urine after ingestion of glucose in diabetic individuals.



Blood glucose levels and glucose in urine after ingestion of glucose in renal glycosuria individuals.



Blood glucose levels and glucose in urine after ingestion of glucose in hypoglycemic individuals.

C. Investigations for follow up of diabetes mellitus:

1. Glucose in urine (glycosuria):

a) Glucose appears in urine if blood glucose level exceeds renal threshold (180 mg/dl).

b) Glycosuria ***is not used for diagnosis of diabetes mellitus***, as there are other causes of glycosuria.

c) Glycosuria is used rather for follow up (monitoring) of effectiveness of treatment of diabetes.

2. Glycated proteins: all proteins including hemoglobin and albumin undergo slow, non-enzymatic, covalent formation of glycated protein (protein + glucose). The rate of formation of such glycoprotein is related to the concentration of glucose over the life span of that protein.

a) **Glycated hemoglobin:** (formerly named glycosylated hemoglobin):

1) Hemoglobin A is the major hemoglobin in adults(97%) .

2) Normally, about 5-8% of hemoglobin A reacts non-enzymatically with glucose to form a derivative known as **glycated hemoglobin** or HBA_{1c}. This glycated hemoglobin remains over the life span of RBCs (120 days).

3) The concentration of glycated hemoglobin is directly proportional to the glucose level over the life span of RBCs.

4) In diabetes mellitus with uncontrolled hyperglycemia, glycated hemoglobin will be higher than normal (may reach 12% or more of the total hemoglobin A).

5) **Importance of the test:** This test is used as an index of diabetic control over 2-3 months. It correlates with the mean plasma glucose concentration during this period. The higher the percentage, the poorer the mean diabetic control.

b) **Glycated albumin (fructosamine):**

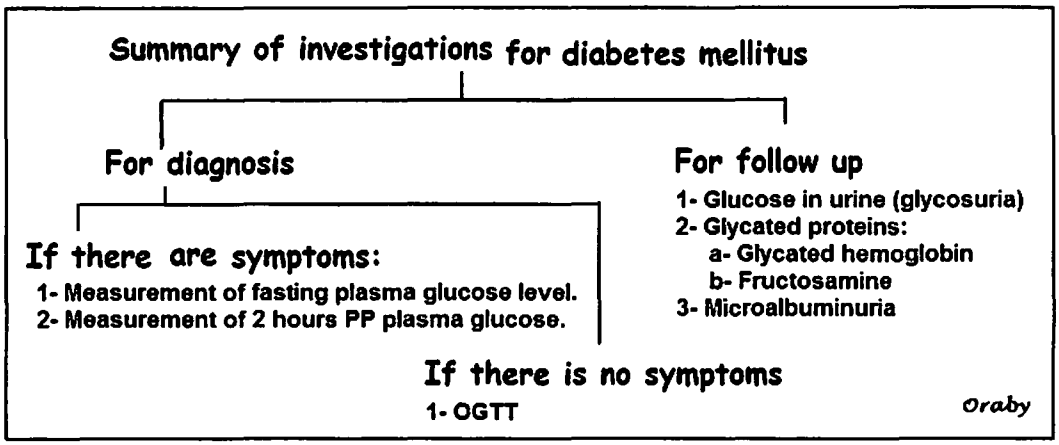
1) Plasma albumin has shorter life span (about 3 weeks) than hemoglobin (2 months).

2) Albumin is also glycated non-enzymatically and can be measured as fructosamine.

3) **Importance of the test:** This test is used as an index of diabetic control over 2-3 weeks (life span of albumin).

This is very useful in short-term evaluation of glucose control (2 weeks) as in pregnancy.

3. **Microalbuminuria:**
   a) **Normally**, proteins (mostly albumin) are excreted in urine in trace amounts (less than 20-30 mg/L).
   b) Albuminuria means excretion of protein in amount more than 200 mg/L.
   c) Excretion of very small amount of protein or albumin (30-200 mg/L) is called microalbumnuria.
      1) It indicates early affection of kidney as in diabetes mellitus.
      2) It cannot be detected by ordinary methods (as heat coagulation test) and needs special techniques for its detection.



V. ***Coma in diabetes mellitus:*** Three types of coma may result in advanced uncontrolled diabetes. These are:

A. **Ketotic coma:** due to acidosis (ketosis).

B. **Hyperglycemic, hyperosmolar, nonketotic coma:** due to hyperglycemia, increased osmolality and dehydration of brain cells. Here, there is neither ketosis nor fall in blood pH.

C. **Lactic acidosis:** due to hyperlactatemia. This type may occur in some patients who receive an oral hypoglycemic drug called: Phenformin (cidophage).

Note: Insulin increases the transfer of potassium and inorganic phosphate into cells. In case of treatment of diabetic coma with intravenous insulin and glucose, it may be fatal if potassium is not given at the same time due to hypokalemia ➔ Cardiac affection.

V. *Types of diabetes:*

A. **Diabetes mellitus:** caused by defective insulin action.

B. **Diabetes insipidus:** hereditary disease caused by defective action of antidiuretic hormone.

C. **Diabetes innocence (renal diabetes):** caused by low renal threshold for glucose reabsorption.

D. **Bronze diabetes:** It is a hereditary disease caused by excessive absorption of iron and its precipitation in tissues as :

1. **Skin:** causing its bronze discoloration.
2. **Pancreas:** causing diabetes mellitus.
3. **Liver:** causing hepatic cirrhosis.

VII. *Glycosuria:*

A. **Definition:** It is the presence of glucose in urine in amount detectable by ordinary methods.

B. **Causes (types):**

1. ***Diabetes mellitus:*** It is the most common cause of glycosuria. It accounts about 90% of all cases of glycosuria.
2. **Excess excretion of diabetogenic hormones:** as growth hormone, glucocorticoids, epinephrine and glucagon.
3. **Diabetes innocence (renal glycosuria):** due to abnormal low renal threshold for glucose absorption. It may be inherited as an autosomal dominate trait.
4. **Pregnancy glycosuria:**
   a) Occurs in 20% of all pregnancies.
   b) It is due to increase of glomerular filtration rate by about 50% during pregnancy.

C. **Detection of glycosuria:**

1. **Glucose oxidase method:**
   It is a specific test, because the enzyme does not react except with glucose. This is done by urine strips containing this enzyme.
2. **Benedict's test:**
   Benedict's reagent gives colored precipitate with urine contains glucose. It is non-specific semi-quantitative test because it gives positive results with other substances than glucose as vitamin C.

Summary of Carbohydrate pathways

	Glycolysis	Pyruvate to Acetyl CoA	Krebs' cycle	Pentose PP	Uronic acid pathway	Glycogenesis	Glycogenolysis	Gluconeo-genesis
Definition	Glucose to pyruvate or lactate	Pyruvate to Acetyl CoA	Acetyl CoA to H_2O, CO_2 and Energy	is a process that generates NADPH and pentoses	is a process that generates glucuronic acid	Synthesis of glycogen	Breakdown of glycogen	Formation of glucose from non carbohydrate sources
Location	Cytosol	Mitochondria	Mitochondria	Cytosol Liver, RBCs ...	Cytosol Liver	Cytosol Liver& muscle	Cytosol Liver& muscle	Cytosol & mito Liver& kidney
Steps	See	See	See	See	See	See	See	See
Energy	No O_2: 2 ATP O_2: 6 or 8 ATP	3 ATP	12 ATP	NO energy	NO energy	NO energy	NO energy	Consume energy
Functions	Energy DAHP → Fat 2,3 BPG 3 p glyc → serine Pyruv → alanine	Energy	• Energy • Catabolic functions • Anabolic Functions	Production of: • Pentoses • NADPH+H$^+$	Production of: Glucuronic acid	Maintains blood glucose concentration in <u>short</u> term fasting		Maintains blood glucose concentration in <u>long</u> term fasting
Key enzymes	Glucokinase Hexokinase PFK PK	Pyruvate dehydrogenase	Citrate synthase Isocitrate dehydrogenase α ketoglutarate dehydrogenase	Gluocose -6-phosphate dehydrogenase	UDP glucose dehydrogenase	Glycogen synthase	Phosphorylase	G-6-phosphatase F 1,6 bisphosphatase PC PEPK Kinase
Regulation	Hormonal Energy Substrate	Phosphorylation dephosphoryla-tion	ATP, NADH+H$^+$ Acetyl CoA Long acyl CoA	NADP$^+$, insulin	-	Phosphorylation dephosphoryla-tion		Glucocorticoids Insulin Glucagon
Diseases	PK – HA HK –HA Lactic acidosis	-	-	Favism	Essential pentosuria	Von Gierk's disease		-

DAHP= Dihydroxy acetone phosphate – 2,3 BPG = 2,3 Bisphosphoglycerate – 3 p gly = 3 phosphoglycerate – PFK = Phosphofructokinase – PK = Pyruvate kinase – HA= Hemolytic anemia – HK= hexokinase – PC= Pyruvate carboxylase, PEPK kinase= Phosphoenol pyruvate carboxy kinase